

Ordered Chiral Geometry

Volume I of the integrated monograph: two associative directions, aligned
locality, three bars, and bar–cobar reconstruction

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Abstract

The weak Jacobi form $\phi_{0,1}$, the weight-five Igusa product, the Gritsenko–Nikulin Borcherds Lie superalgebra, the primitive reduced $K_3 \times E$ partition function, and the mixed holomorphic–topological boundary theory are assembled here as one typed construction. The work begins from the exact Jacobi coefficients and constructs the automorphic product, root algebra, proper-height Chevalley coalgebra, local–wrapped current object, Hong quantum supergroup, finite-window Hall and three-Calabi–Yau models, root-diagonal Fredholm family, and reduced inverse-square scalar. The common root grading proves their compatibility. A universal property characterizes this realization, while a parity-refined PBW theorem gives the exact criterion by which a geometric Hall/BPS source identifies with it. The subsequent books develop ordered and associative chiral algebras, the formal Hamiltonian open–closed theorem, Calabi–Yau Hall and quantum-group assembly, and the universal BV–Einstein envelope. Every scalar is the value of a specified trace, Euler character, determinant, or automorphic section of a previously constructed object.

The theorem spine

The work is organized by the order in which geometry creates algebra:

$$\begin{aligned} \text{collision} &\longrightarrow \text{ordered fusion} \longrightarrow \text{centre} \longrightarrow \text{Hall extension} \\ &\longrightarrow \text{quantum double} \longrightarrow \text{character}. \end{aligned}$$

Each arrow is constructed in a fixed category. Each inverse limit uses a proper positive filtration. Each physical interpretation is the image of a displayed open–closed, Hall, current, trace, or determinant morphism. The central examples are calculated in generators and relations before their abstract generalization.

Notation and normalization

The ground field has characteristic zero. Stable categories are presentable and idempotent-complete. Tensor products preserve colimits separately. Completions are separated inverse limits of finite weight or proper-height quotients. The Igusa normalization is

$$\Delta_S^\theta = 64D_S, \quad \chi_{10} = D_S^2,$$

with D_S monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

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The five coupled constructions

The governing puzzle is the coexistence of two associative directions on a factorisable object over a curve. Ordered fusion acts along a transverse topological line, whereas associative chiral composition acts through collision geometry on the curve. Their products have different carriers and their bar constructions agree only after a mixed distributive law has been specified. The aligned-decomposition idempotent isolates the sector on which that law is geometrically meaningful.

Five constructions meet along this interface. Ordered chiral geometry supplies the two products and their three distinct bars. Mixed holomorphic–topological Deligne theory supplies the open–closed comparison. Igusa–Borcherds theory supplies the root-graded denominator. Calabi–Yau Hall geometry supplies extension correspondences and quantum doubles. The BV–Einstein construction supplies the deformation and trace carriers. No scalar shadow alone determines or constructs these structured objects: every passage is made by a displayed bar–cobar, open–closed, Hall, current, trace, or determinant morphism under its stated hypotheses.

The first construction is developed in detail below. The remaining parts record its comparison maps with the other four carriers, retaining separate completion, parity, descent, and anomaly hypotheses.

Book entrance: Ordered and Associative Chiral Geometry

Collision on a complex curve, order on a transverse line, exchange in a plane, and the three corresponding bar constructions.

BOOK I

The invariant object

Two directions of locality

1.1 THE LOCAL EXPERIMENT

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \rightarrow z_2$ and $t_1 \rightarrow t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2 . The second composes operators in the order selected by the orientation of I .

Principle 1.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X . Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

1.2 CARRIER AND OPERATION

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_* \mathfrak{g}$$

in the chiral pseudo-tensor category of right D -modules [6]. The D -module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\mathrm{gr}_m^F \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \quad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

The Ran coefficient category

2.1 THE FIXED SIX-FUNCTOR SETTING

Fix a presentable stable category $\mathcal{D}_X$ of right D -modules on $\mathrm{Ran}(X)$ with the following operations:

- (i) the pointwise tensor product $\otimes^!$;
- (ii) chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;
- (iii) base change and projection formulae for the disjoint-union correspondences;
- (iv) geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D -module setting of chiral Koszul duality has these operations [33]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \xleftarrow{j} (\mathrm{Ran}(X) \times \mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F \otimes^{\mathrm{ch}} G = u_! j^!(F \boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 2.1 (Aligned-decomposition projector). For a finite set I , let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A \sqcup B$ and let

$$a_I : \mathrm{Dec}(I) \longrightarrow \mathrm{Dec}(I) \times \mathrm{Dec}(I), \quad d \longmapsto (d, d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!} a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 2.2 (Aligned sector in the finite decomposition model). *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I , and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1, d_2 \in \mathrm{Dec}(I)} \mathcal{M}_{d_1, d_2}.$$

The projection

$$e_I : \mathcal{M}_I \longrightarrow \mathcal{M}_I, \quad e_I|_{\mathcal{M}_{d_1, d_2}} = \begin{cases} \text{id}, & d_1 = d_2, \\ 0, & d_1 \neq d_2, \end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\text{al}} = \bigoplus_{d \in \text{Dec}(I)} \mathcal{M}_{d, d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\text{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1, d_2, d_3} \mathcal{M}_{d_1, d_2, d_3} \quad \text{onto} \quad \bigoplus_d \mathcal{M}_{d, d, d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. $\square$

Theorem 2.3 (Global aligned projector). *In the fixed six-functor Ran category, the common-decomposition inclusion is a finite clopen inclusion on every finite-set chart. The functors*

$$e_I = a_{I,!} a_I^!$$

therefore assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and Beck–Chevalley transports this equality through every Ran transition map.

Proof. The decomposition index is a finite discrete groupoid. Equality of two decomposition indices selects a union of connected components, so the diagonal inclusion is clopen and its extension-by-zero followed by restriction is an idempotent. A surjection of finite sets pulls equal decompositions to equal decompositions, which gives naturality. For three indices, both iterated projectors select the same clopen union $d_1 = d_2 = d_3$. The six-functor base-change isomorphisms identify the corresponding pull–push composites. $\square$

The aligned image carries the canonical interchange obtained by applying the projection formula to one common disjoint-union correspondence. Locality states that transverse multiplication is a morphism for this factorization coproduct. A strong interchange occurs on the full mixed object exactly when every off-aligned summand vanishes.

2.2 FACTORIZATION COALGEBRAS

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\text{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\text{Fact}_D(X) \subset \text{CoCAlg}_{\text{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [33, 45].

Proposition 2.4 (Pointwise closure). *The componentwise tensor product of two factorization objects is a factorization object. Thus $\text{Fact}_D(X)$ is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G . $\square$

The two noncommutative chiral objects

3.1 TRANSVERSE ORDER

A transversely ordered chiral algebra is an E_1 -algebra in the pointwise coefficient category. Its multiplication composes two operators that occupy the same chiral support and are ordered in an independent real direction. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 3.1 (Transversely ordered chiral algebra). A transversely ordered chiral algebra is an augmented object

$$A^\perp \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

3.2 ASSOCIATIVITY INSIDE CHIRAL CONVOLUTION

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 3.2 (Associative chiral algebra). An associative chiral algebra is an augmented algebra

$$A_{\text{Ass}}^{\text{ch}} \in \text{Alg}_{E_1}^{\text{aug}}(\mathcal{D}_X, \star^{\text{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\text{ch}} \in \text{Alg}_{\text{Lie}}(\mathcal{D}_X, \star^{\text{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^\perp : A \otimes^! A \rightarrow A$ composes transverse intervals. The operation $m^{\text{ch}} : A \star^{\text{ch}} A \rightarrow A$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 3.3 (Independence of the two associative structures). *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_\perp : M \otimes^! M \rightarrow M$ defines a transversely ordered algebra on $\mathbf{1}_\perp \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\text{ch}} : M \star^{\text{ch}} M \rightarrow M$ defines an associative chiral algebra on $\mathbf{1}_{\text{ch}} \oplus M$,*

while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. $\square$

Definition 3.4 (Mixed associative chiral datum). A mixed associative chiral datum is a factorization object A equipped with

- (i) a pointwise multiplication $m^\perp : A \otimes^\perp A \rightarrow A$ and a unit map $\eta_\perp : \mathbf{1}_\perp \rightarrow A$;
- (ii) a chiral-convolution multiplication $m^{\text{ch}} : A \star^{\text{ch}} A \rightarrow A$ and a unit map $\eta_{\text{ch}} : \mathbf{1}_{\text{ch}} \rightarrow A$;
- (iii) an aligned mixed distributive law identifying the two composites on $\text{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

3.3 FORMAL-DISC REALIZATION

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state–field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 3.5 (Formal-disc realizations). *Let V be a translation-equivariant state–field system with vacuum and injective state–field map.*

- (i) *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*
- (ii) *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S -locality and the hexagon compatibility, then V is a quantum vertex algebra.*
- (iii) *A translation-equivariant associative chiral-convolution algebra produces such a state–field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li’s reconstruction theorem for quasi-compatible fields [55]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [25, 56]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. $\square$

3.4 THREE SEPARATION LAWS

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The m th normal symbol of a pole is recovered by the m th normal-jet map. An oriented

line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

Transversely ordered chiral algebras

Definition 4.1 (Ordered chiral algebra). An ordered chiral algebra on X is an augmented E_1 -algebra object

$$\mathcal{A} \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

Equivalently, $\mathcal{A}$ is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1, I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1, I_2} \otimes^! \Delta_{I_1, I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 4.2 (Locality recognition). *To give an ordered chiral algebra is equivalent to giving a factorization family A_I , an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1 -algebra in the symmetric monoidal category $\text{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\text{Fact}_D(X)$. A morphism in $\text{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. $\square$

4.1 PRODUCT FIELD THEORIES

Let $\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R}) \simeq \text{Alg}_{E_1}(\text{HolFact}(X^{\text{an}}))$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\text{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 4.3 (Holomorphic–topological recognition). *Let $\mathbb{F}$ be a product-generated factorization algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathbb{F}$ to a transverse point is an E_1 -algebra in holomorphic factorization algebras on X . A regular-holonomic algebraization produces an ordered chiral algebra on X .*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent E_1 structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\text{OrdCh}_X = \text{E}_1 \otimes \text{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 5.1 (Operadic exponential law). *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\text{Alg}_{P \otimes Q}(\mathcal{V}) \simeq \text{Alg}_P(\text{Alg}_Q(\mathcal{V})).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\text{E}_1 \otimes \text{Ch}_X$ classifies E_1 -algebra objects in Ch_X -algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [60]. The second statement is the specialization $P = \text{E}_1$ and $Q = \text{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. $\square$

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \dots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

5.1 MIXED FORESTS

Fix a cofibrant dg-operad $Q_X \rightarrow \text{Ch}_X$. Apply the functorial W -construction to $\text{E}_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\text{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\text{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 5.2 (Two-color forest differential). *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction*

square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy

$$d_{\perp}^2 = 0, \quad d_{\text{ch}}^2 = 0, \quad d_{\perp}d_{\text{ch}} + d_{\text{ch}}d_{\perp} = 0.$$

Thus $d = d_{\text{int}} + d_{\perp} + d_{\text{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^2$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\text{ch}}^2 = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1 , proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. $\square$

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

BOOK II

Bar, cobar, and the boundary state

Three bar constructions

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 6.1 (Transverse, associative-chiral, and Lie-chiral bars). For augmented objects of the indicated types define

$$\mathrm{Bar}^\perp(\mathcal{A}^\perp) := \mathbf{I} \otimes_{\mathcal{A}^\perp}^{\mathbb{L},!} \mathbf{I},$$

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}}) := \mathbf{I} \star^{\mathrm{ch}}_{\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}}} \mathbf{I},$$

and

$$\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) := C_*^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[\mathbf{I}])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^!$, the second with $\star^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 6.2 (Typed bar theorem). *The three bars have the following chain models.*

- (i) $\mathrm{Bar}^\perp(\mathcal{A}^\perp)$ is the tensor coalgebra $T_{\otimes^!}^c(s\tilde{\mathcal{A}})$ with coderivation induced by $m^\perp$.
- (ii) $\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}})$ is the tensor coalgebra $T_{\star^{\mathrm{ch}}}^c(s\tilde{\mathcal{A}})$ with coderivation induced by m^{ch} .
- (iii) $\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}})$ is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. $\square$

Definition 6.3 (PBW-admissible chiral Lie algebra). A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}} U^{\mathrm{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 6.4 (Filtered chiral PBW comparison). *Let $\mathfrak{g}$ be a chiral Lie algebra in a complete filtered chiral category. Assume:*

- (i) its chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$ exists;
- (ii) the PBW filtration is exhaustive, complete, and separated;
- (iii) the chiral PBW morphism

$$\text{Sym}^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{gr } U^{\text{ch}}(\mathfrak{g})$$

is an isomorphism;

- (iv) the induced filtrations on the two bar complexes are strongly convergent.

Then antisymmetrization induces a filtered quasi-isomorphism

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(U^{\text{ch}}(\mathfrak{g})) \simeq \text{Bar}_{\text{Lie}}^{\text{ch}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\text{Sym}^{\text{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is the PBW lane of chiral Koszul duality [6, 33]. $\square$

Definition 6.5 (Bar-admissible mixed datum). A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 6.6 (Mixed Fubini theorem). *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\left| p \mapsto \left| q \mapsto B_{p,q}(A) \right|_{\text{ch}} \right|_{\perp} \simeq \left| \text{diag } B_{\bullet,\bullet}(A) \right| \simeq \left| q \mapsto \left| p \mapsto B_{p,q}(A) \right|_{\perp} \right|_{\text{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. $\square$

The notation used in the remainder of this part is

$$\text{Bar}_X^{\text{ord}} = \text{Bar}^{\perp}.$$

It always refers to transverse order.

Collision maps on finite powers of the curve

At a binary collision, choosing the transverse order selects one of two branches. At a triple collision, two successive choices must select the same total order. These statements have a realization by maps of schemes and by morphisms of differential modules. They give an explicit finite-arity compatibility for the pointwise ordered bar.

7.1 THE BINARY COEFFICIENT AND ITS TWO RESTRICTIONS

Work over $\mathbb{C}$ with cohomological grading. Put $R = \mathbb{C}[x]$ and $K = R[x^{-1}]$, where x is a normal coordinate to the diagonal. Glue two copies of $\text{Spec } R$ along $\text{Spec } K$. The ordered Čech algebra is

$$D^0 = R \oplus R, \quad D^1 = K, \quad d(a, b) = b - a.$$

Its products are

$$(a, b)(c, e) = (ac, be), \quad (a, b)v = av, \quad v(c, e) = ve, \quad vv' = 0.$$

The equality $be - ac = (b - a)e + a(e - c)$ proves the only nonzero Leibniz identity. Associativity follows from the two endpoint actions. Thus $H^0(D) = R$ and $H^1(D) = K/R$.

Proposition 7.1 (Two collision restrictions). *The two chart restrictions induce distinct algebra maps*

$$D \otimes_R^{\mathbb{L}} \mathbb{C} \simeq \mathbb{C} \times \mathbb{C} \longrightarrow \mathbb{C}.$$

They are the two coordinate projections. The algebra D is not formal as a unital associative algebra with central R -action.

Proof. The terms of D are flat over R . Tensoring its two-term complex with $R/(x)$ therefore computes the derived fibre. The localization term vanishes, leaving $\mathbb{C}^2$ with componentwise product. Endpoint restriction gives the asserted projections.

A bounded flat model of the cohomology algebra is $E = R \oplus L$, where $L = [R \rightarrow K]$ occupies degrees zero and one and has square-zero multiplication. The quotient $K \rightarrow K/R$ gives an algebra quasi-isomorphism $E \rightarrow H^*(D)$. Its derived fibre is $\mathbb{C}[\epsilon]/(\epsilon^2)$ in degree zero. This algebra has a nonzero nilpotent, whereas $\mathbb{C} \times \mathbb{C}$ has none. Derived scalar extension preserves algebra quasi-isomorphisms: it can be computed by the finite free scalar resolution $[R \xrightarrow{x} R]$. Consequently an R -algebra zigzag from D to $H^*(D)$ would give an impossible isomorphism of these fibre algebras. $\square$

In particular, the two restrictions are data of the derived coefficient algebra. The dual-number fibre has only one unital $\mathbb{C}$ -algebra map to $\mathbb{C}$.

7.2 ORDER CHARTS AND DIFFERENTIAL MODULES

Let X be a smooth separated finite-type complex algebraic curve. For a nonempty finite set I , set $S_I = X^I$ and let $\text{Ord}(I)$ be its set of total orders. For a nonempty set $J \subset \text{Ord}(I)$, let $D(J)$ be the set of unordered label pairs whose order varies in J . Define the open subset

$$U_J = S_I \setminus \bigcup_{\{i,j\} \in D(J)} \Delta_{ij}.$$

Take a copy of S_I for each total order and identify two copies on $U_{\{\sigma,\tau\}}$ by the coordinate identity. Denote the result by Y_I and its projection to S_I by p_I . The identity

$$D(\{\sigma, \tau, \rho\}) = D(\{\sigma, \tau\}) \cup D(\{\sigma, \rho\})$$

proves the gluing cocycle. Thus Y_I is a smooth, possibly nonseparated scheme. The map p_I is an étale local isomorphism. For $X = \mathbb{A}^1$, these are the order charts of [2, §§4.1–4.2].

Write $j_J : U_J \hookrightarrow S_I$ and $\mathcal{R}_J = j_{J,*} \mathcal{O}_{U_J}$. The diagonals are effective Cartier divisors because X is smooth of dimension one and separated. Locally $\mathcal{R}_J$ is therefore a flat localization of $\mathcal{O}_{S_I}$. Vector fields act by derivations of that localization.

Let

$$\Omega_J = \mathbb{C}[t_\sigma, dt_\sigma \mid \sigma \in J] / (\sum t_\sigma - 1, \sum dt_\sigma), \quad d(t_\sigma) = dt_\sigma.$$

The variables t_σ have degree zero and dt_σ degree one. Define the sheaf of commutative dg algebras

$$C_I = \{(c_J) : c_J \in \mathcal{R}_J \otimes_{\mathbb{C}} \Omega_J, \quad c_J|_K = c_K|_{U_J} \text{ for } \emptyset \neq K \subset J\}.$$

The left side of each face equation sets the variables outside K to zero. The differential acts on simplex forms. Vector fields on S_I act on $\mathcal{R}_J$ and kill all simplex variables. These actions commute with d , satisfy the flat-connection identities, and act by derivations of multiplication. Thus C_I is a commutative dg algebra in left D_{S_I} -modules with their tensor product over $\mathcal{O}_{S_I}$.

This is a model of $R p_{I,*} \mathcal{O}_{Y_I}$ as a complex of $\mathcal{O}_{S_I}$ -modules with flat connection. Here $R p_*$ denotes derived sheaf pushforward of the structure sheaf, with the connection just specified. Indeed, the finite chart Čech resolution has terms

$$\check{C}_I^r = \bigoplus_{\sigma_0 < \dots < \sigma_r} \mathcal{R}_{\{\sigma_0, \dots, \sigma_r\}}.$$

An auxiliary ordering is used only in this formula. The augmented resolution contracts on each stalk by inserting a chart containing that stalk. Each intersection maps affinely to S_I , so its higher pushforwards vanish. Polynomial integration from the compatible simplex forms to this finite complex is a quasi-isomorphism, as in the Thom–Sullivan comparison [2, Appendix A.3]. It commutes with the coefficient derivations. Integration is used as a comparison of complexes, not as a strictly multiplicative map to the ordered cup product.

To express the construction in right differential modules, put $C_I^r = \omega_{S_I} \otimes C_I$. Our pointwise convention is

$$M \otimes^! N = (M \otimes_{\mathcal{O}_S}^L N) \otimes_{\mathcal{O}_S} \omega_S^{-1}, \quad \mathbf{1}_S = \omega_S.$$

For $f : S \rightarrow T$, write

$$f^\dagger M = \omega_S \otimes Lf^*(\omega_T^{-1} \otimes M)$$

with the induced flat connection. This specifies the unshifted inverse-image convention; no dimension shift is implicit. The transfer-module description gives this inverse image and its chain rule [31, §28.5, equation (21)].

7.3 MERGING ORDERED BLOCKS

Let $p : I \rightarrow J$ be a surjection. Choose a total order τ_j on each fibre $p^{-1}(j)$. If σ orders J , let $e_{p,\tau}(\sigma)$ be the order on I obtained by listing the fibres in order σ and each fibre in order τ_j . Let $\delta_p : S_J \hookrightarrow S_I$ repeat the coordinate of each block.

Theorem 7.2 (Ordered collision morphism). *The chart maps $S_J \xrightarrow{\delta_p} S_I$ indexed by $\sigma \mapsto e_{p,\tau}(\sigma)$ glue to a morphism*

$$\iota_{p,\tau} : Y_J \rightarrow Y_I, \quad p_I \iota_{p,\tau} = \delta_p p_J.$$

They induce a morphism of commutative algebra objects in right D_{S_J} -modules,

$$\kappa_{p,\tau} : \delta_p^\dagger C_I^r \rightarrow C_J^r.$$

For nested surjections, these morphisms compose according to concatenation of the chosen fibre orders. Identity surjections give identity morphisms. Bijections act by relabelling.

Proof. Two expanded orders have identical restrictions inside every fibre. Their varying pairs are therefore cross-fibre pairs only. A pair i, i' in different fibres varies precisely when the corresponding pair $p(i), p(i')$ varies in the two block orders. Hence

$$\delta_p^{-1} U_{e_{p,\tau}(L)} = U_L \quad (\emptyset \neq L \subset \text{Ord}(J)).$$

This proves both the gluing assertion and the existence of the coefficient localization maps. It also proves compatibility on every overlap, including intersections of more than two charts.

In the simplex model, restrict c to the component indexed by $e_{p,\tau}(L)$, identify $t_{e_{p,\tau}(\sigma)}$ with t_σ , and pull back its regular coefficients by δ_p . This gives a map $\delta_p^* C_I \rightarrow C_J$ of commutative dg algebras with connection. No within-fibre denominator is inverted on a retained component. In coordinates the connection identity is

$$\partial_{Z_J} \kappa_{p,\tau}(c) = \kappa_{p,\tau} \left(\sum_{i \in p^{-1}(j)} \partial_{z_i} c \right).$$

The coordinate identity is the chain rule for the geometric map, so it glues under changes of coordinates. Precomposing with the derived-to-underived pullback map gives $L\delta_p^* C_I \rightarrow C_J$. Equivalently, apply derived pullback to a flat replacement before the displayed map. Tensoring with ω_{S_J} gives (7.2).

Suppose $q : J \rightarrow K$ has fibre orders v . The induced order on $(qp)^{-1}(k)$ first lists the j 's in order v_k , then lists $p^{-1}(j)$ in order τ_j . Both composites use the same coordinate diagonal and the same injection of total-order charts. Their simplex restrictions and coefficient pullbacks are consequently equal. Derived pullback transports this equality through its canonical composition isomorphism. Unary identities and relabelling follow from the same formulas. $\square$

On $\mathbb{A}^1$, define $q_{ii'} = \sum_{i <_\sigma i'} t_\sigma$ componentwise and $\omega_{ii'} = dq_{ii'} / (z_i - z_{i'})$. On a component where this denominator is unavailable the numerator is zero, and the component is defined to be zero. Thus $\omega_{ii'} \in C_I^1$ is a closed coefficient. Its class is nonzero: on the edge joining two orders differing by an

adjacent exchange, integration gives $\pm(z_i - z_{i'})^{-1}$. An exact element would instead give a difference of polynomial endpoint values. The collision morphism satisfies

$$\kappa_{p,\tau}(\omega_{ii'}) = \begin{cases} 0, & p(i) = p(i'), \\ \omega_{p(i),p(i')}, & p(i) \neq p(i'). \end{cases}$$

In the first case, restriction to the retained face sets $dq_{ii'} = 0$ before coordinate substitution.

Example 7.3 (Three labels). Merge 1, 2 to a , keep 3 as b , and choose $1 < 2$ in the merged fibre. The two retained orders are 123 and 312. The classes ω_{13}, ω_{23} both map to ω_{ab} , whereas ω_{12} maps to zero. Merging $a < b$ next and merging $2 < 3$ first, then $1 < (23)$, both select the chart 123. At total collision their common map is the projection $\mathbb{C}^6 \rightarrow \mathbb{C}$ at 123. Choosing $2 < 1$ instead selects a different branch.

7.4 ORDERED STATES AND THE BAR COMPATIBILITY

Let V_I have basis $u_i, i \in I$, in cohomological degree zero and weight one. Use the free associative algebra of finite words $T(V_I)$ with zero differential and its word-length augmentation. Define

$$\mathcal{A}_I = C_I \otimes_{\mathbb{C}} T(V_I).$$

Its state generators are horizontal. Products concatenate words, and coefficients commute with states with the graded tensor sign. This makes $\omega_{S_I} \otimes \mathcal{A}_I$ an augmented associative algebra over C_I^r in pointwise right differential modules. The augmentation is to C_I^r .

For an ordered surjection define the state map by

$$u_{i_1} \cdots u_{i_m} \mapsto u_{p(i_1)} \cdots u_{p(i_m)}.$$

Together with $\kappa_{p,\tau}$ it gives an augmented algebra morphism $\alpha_{p,\tau} : L\delta_p^* \mathcal{A}_I \rightarrow \mathcal{A}_J$ in left differential modules, and hence its right-module version. In particular, $u_1 u_2$ becomes u_a^2 when 1, 2 merge. The target retains the resulting composite state as an actual word.

Theorem 7.4 (Collision and the relative ordered bar). *Form the relative reduced bar*

$$B_{C_I} \mathcal{A}_I = \bigoplus_{r \geq 0} (s \overline{\mathcal{A}_I})^{\otimes_{C_I} r}, \quad |s| = -1.$$

Ordered collision induces a morphism

$$L\delta_p^* B_{C_I} \mathcal{A}_I \longrightarrow B_{C_J} \mathcal{A}_J$$

of coalgebras with differential and connection. It commutes with every adjacent bar face and every word cut. These maps have the nested-partition coherence of Theorem 7.2.

Proof. Write $[a_1 | \cdots | a_r] = sa_1 \otimes \cdots \otimes sa_r$ and use $s(ca) = (-1)^{|c|} csa$. The bar differential is determined by

$$b_1(sa) = -s(da), \quad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

Insert these degree-one maps with the tensor sign. Two contractions on disjoint input pairs cancel. The overlapping contribution is

$$b_2^2[a|b|c] = (-1)^{|b|} s((ab)c - a(bc)) = 0.$$

The mixed internal terms give the Leibniz identity, and the remaining terms give $d^2 = 0$. Thus the total differential squares to zero.

Send a bar word to the word of its images under $\alpha_{p,\tau}$. This map has degree zero, commutes with the internal differential, and preserves products and degrees. It therefore commutes with the two displayed cogenerator maps, including their signs. It also commutes with a cut because it does not alter the number or order of bar letters. The coefficient chain rule gives the connection identity. Derived extension of scalars gives the stated map; all bar constructions can be computed by the free state-word model just displayed. Composition reduces to composition on each coefficient and each state letter. $\square$

For the three-label merger above, the nonzero calculation is

$$\alpha b[u_1|u_2|u_3] = [u_a^2|u_b] - [u_a|u_a u_b] = b\alpha[u_1|u_2|u_3].$$

Its next boundary is $[u_a^2 u_b] - [u_a(u_a u_b)] = 0$. Multiplying the input by ω_{13} transports its coefficient to ω_{ab} , with the coefficient differential and tensor sign retained. Thus collision acts simultaneously on the actual derived coefficients and the ordered composite states.

For each weight the relative bar is finite free over C_I : state words have positive weights and there are only finitely many ordered cuts of a word of a fixed weight. The relative augmentation dual is consequently

$$\prod_{w \geq 0} \text{Hom}_{C_I}(B_{C_I} \mathcal{A}_I(w), C_I),$$

where the product is taken in each cohomological degree. The differential is $dF = d_{C_I} F - (-1)^{|F|} Fb$ and the product uses deconcatenation. The left bar resolution is $P_I = \mathcal{A}_I \otimes_{C_I} B_{C_I} \mathcal{A}_I$. Its additional leading face is

$$a_o \otimes [a_1 | \cdots | a_r] \mapsto -(-1)^{|a_o|} a_o a_1 \otimes [a_2 | \cdots | a_r].$$

The degree-minus-one insertion $h(a_o \otimes w) = -1 \otimes [\bar{a}_o | w]$ satisfies $dh + hd = 1 - \eta\epsilon$ on length zero and $dh + hd = 1$ on positive lengths. Adjacent terms cancel, leaving the leading augmentation-ideal term and its scalar complement. The internal terms cancel by $d\bar{a}_o = \overline{da}_o$. The differential of a free state-word basis vector uses shorter bar words. A map into an acyclic module is therefore null-homotopic by choosing primitives in increasing bar length. This proves the semifree resolution property. Its coefficient dual computes $R \text{Hom}_{\mathcal{A}_I}(C_I, C_I)$ in dg modules over the coefficient algebra. This is a relative algebraic Hom, not the Hom of horizontal sections. Deconcatenation makes P_I a right bar comodule. A coefficient functional F lifts to $(1 \otimes F)\Delta_{P_I}$, with the tensor sign. Cutting twice identifies convolution with composition of these lifts. Their postcomposition with the augmentation recovers F , so the identification also preserves the derived product. Dualizing a collision also changes its coefficient algebra. Put $Q_I = B_{C_I} \mathcal{A}_I$ and

$$Q_I^\vee(w) = \text{Hom}_{C_I}(Q_I(w), C_I).$$

For the ordered surjection (p, τ) , set $\mathcal{B} = L\delta_p^* C_I$ and use $\kappa_{p,\tau} : \mathcal{B} \rightarrow C_J$ in left differential modules. Its scalar-extension functor on C_I -modules is

$$E_{p,\tau}(M) = C_J \otimes_{\mathcal{B}}^{\mathbb{L}} L\delta_p^* M.$$

The bar collision map has the C_J -linear form

$$\gamma_{p,\tau,w} : E_{p,\tau}(Q_I(w)) \longrightarrow Q_J(w).$$

Proposition 7.5 (Dual collision after scalar extension). *Transposition gives the morphism of complexes*

$$\begin{aligned} d_{p,\tau,w}^\vee &: \operatorname{Hom}_{C_J}(Q_J(w), C_J) \\ &\longrightarrow C_J \otimes_B^{\mathbb{L}} L\delta_p^* \operatorname{Hom}_{C_I}(Q_I(w), C_I). \end{aligned}$$

These maps preserve the connections. Their degreewise products over weight define a unital morphism of convolution dg algebras

$$\prod_{w \geq 0} Q_J^\vee(w) \longrightarrow \prod_{w \geq 0} E_{p,\tau}(Q_I^\vee(w)).$$

For nested ordered surjections $(p, \tau) : I \rightarrow J$ and $(q, v) : J \rightarrow K$, use the concatenated order on each fibre of $q \circ p$. Under the canonical identification $E_{q,v} E_{p,\tau} \simeq E_{q \circ p, v \circ \tau}$,

$$d_{q \circ p, v \circ \tau, w}^\vee = E_{q,v}(d_{p,\tau,w}^\vee) \circ d_{q,v,w}^\vee.$$

Proof. At weight w , $Q_I(w)$ is graded free on finitely many state words and their ordered cuts. The bar-length filtration has at most w steps. Its differential preserves the coefficient part and sends a state-word basis vector to shorter bar words. It is therefore a finite semifree module, hence perfect over C_I ; the displayed Hom computes its derived dual.

For a single shifted free module, the evaluation map identifies the dual after scalar extension with the scalar extension of the dual. The identification is natural, commutes with differentials, and persists under finite sums, shifts, and mapping cones. Applying the finite semifree filtration gives

$$\operatorname{RHom}_{C_J}(E_{p,\tau}(Q_I(w)), C_J) \simeq E_{p,\tau}(Q_I^\vee(w)).$$

In the fixed free models the map in (7.5) sends F to $F \circ \gamma_{p,\tau,w}$, followed by this identification. Since γ is a degree-zero chain map, $d(F\gamma) = (dF)\gamma$. The connection on Hom is the commutator connection, so horizontality of γ proves the connection assertion as well.

The cut map at weights a, b has target $Q_I(a) \otimes_{C_I} Q_I(b)$ and source $Q_I(a + b)$. The coalgebra identity for γ consequently gives $d^\vee(F * G) = d^\vee(F) * d^\vee(G)$ with the tensor signs. In weight zero the map is the identity of C_J , so it preserves the unit. At output weight w , convolution uses only the finitely many pairs $a + b = w$. Thus these maps form algebra maps on every finite weight quotient and then on their inverse limit, represented in the stated models by degreewise products. The target extends coefficients separately at each weight before forming that product.

For nested collisions, associativity of derived scalar extension and the identity for the coefficient maps give the displayed identification of the E functors. The bar maps obey $\gamma_{qp} = \gamma_q \circ E_q(\gamma_p)$ by Theorem 7.4. Precomposition reverses this equality. Naturality of the finite-module dual identification then gives the stated composition formula. Every step occurs in a fixed weight, so the same equality holds after completion. $\square$

The scalar extension is necessary even for the binary coefficient fibre. Take $B = \mathbb{C}^2$, let $C = \mathbb{C}$ carry the B -action through the first projection, and put $e_2 = (0, 1)$. Every B -linear map $f : C \rightarrow B$ satisfies $e_2 f(1) = f(e_2 \cdot 1) = 0$. Its image lies in the first idempotent summand, so it cannot send 1 to $1_B = (1, 1)$. There is therefore no unital B -linear reverse map $C \rightarrow B$. After extension, $C \otimes_B^{\mathbb{L}} B \simeq C$, and the weight-zero dual collision map is the identity, as required.

Proposition 7.6 (Completed reconstruction and collision). *Let $Q_I = B_{C_I} \mathcal{A}_I$, with positive-weight reduced part $\overline{Q}_I$, and let t have degree one. The cobar differential on tensor words in $t\overline{Q}_I$ is*

$$d_\Omega(tc) = -t(bc) + \sum_{\overline{\Delta}c = \sum c' \otimes c''} (-1)^{|c'|+1} (tc')(tc'').$$

It extends by Leibniz to a square-zero differential. The counit

$$\Omega_{C_I} Q_I \longrightarrow \mathcal{A}_I$$

sends $t(sa)$ to a and longer bar generators to zero. It is a quasi-isomorphism in each weight. The same holds after taking the derived inverse limit of the finite weight quotients. Ordered collision commutes with the counit and the cobar differential.

Proof. The square of the displayed differential vanishes by $b^2 = 0$, the coderivation identity, and coassociativity of cuts, respectively. On $t[a|b]$, the counit sends the internal and cut terms to $-(-1)^{|a|}ab$ and $(-1)^{|a|}ab$. They cancel. Single-letter generators commute with the internal differential; all longer generators have zero image under the chain-map identity.

Fix a weight w . Filter by the number of augmentation-ideal state letters before their multiplication. This number is at most w . State multiplication lowers it, whereas the cut and coefficient differentials preserve it. For a fixed sequence of m state letters in the associated graded, each of its $m - 1$ gaps is either inside one bar generator or between two cobar factors. The cut differential is the tensor differential of $m - 1$ two-term identity complexes. Indeed, after state letter j , let c_j count the preceding outside gaps. The splitting sign is $(-1)^{1 + \sum_{i \leq j} |a_i| - j + c_j}$. The degree before that gap in the interleaved letter-gap complex is $\sum_{i \leq j} |a_i| - (j - 1 - c_j)$, which has the same parity. For $m > 1$ a contraction on the first gap contracts this complex. For $m = 1$ it is the original state module, mapped identically to $\mathcal{A}_I(w)$. Filter the positive-weight target in step one. The finite filtration proves the weightwise quasi-isomorphism. Derived inverse limits preserve equivalences. On affine coefficient charts the transition projections are degreewise surjective, so the limit is represented by the degreewise product of weight complexes.

Collision preserves state products, weights, bar differentials, and cuts. Substitution in the cobar formula and the counit on generators therefore proves both commuting identities. The completed maps are the inverse limits of these finite-weight maps. $\square$

The completed target is $\widehat{\mathcal{A}}_I$, rather than the finite-word algebra $\mathcal{A}_I$. For example, $\sum_{m \geq 0} u_i^m$ is allowed in the completed target in degree zero.

7.5 THE COMPARISON WITH RAN COEFFICIENTS

Proposition 7.7 (The diagonal transition is not invertible). *The family C_I with the collision maps above is not a Cartesian family for the usual Ran diagonal transitions.*

Proof. At a point of S_I , partition the labels into equal-coordinate blocks B . Tensor the bounded flat Čech complex with the residue field. A term vanishes precisely when its overlap inverts a difference inside one such block. The surviving orders split into classes with identical restrictions to every B . Each class gives the cochain complex of a simplex and contributes one residue field in degree zero. Multiplication is coordinatewise. The derived fibre therefore has $\prod_B |B|!$ branches. For two coincident labels it is $\mathbb{C}^2$; for one label it is $\mathbb{C}$. The transition selects a projection between these fibres, which cannot be an equivalence. $\square$

Thus Theorem 7.4 constructs a relative pointwise bar on finite powers of X , with ordered diagonal maps. To compare it with $\text{Bar}^\perp$ on $\text{Fact}_D(X)$ requires a specified factorization object and a morphism carrying these coefficients, state words, and diagonal maps to that object. Chiral convolution requires its own multiplication morphism. The finite-power theorem supplies neither a Cartesian Ran replacement nor that multiplication.

A second obstruction concerns a proposed multiplication with the coefficient ω_{ij} . Suppose a_i, a_j, b_{ij} freely generate state summands over C_I , have degrees 0, 0, -1 , and are horizontal. The prescription $a_i a_j = \omega_{ij} b_{ij}$ cannot define a multiplication in differential modules with this connection. Indeed,

$$\partial_{z_i}(\omega_{ij} b_{ij}) = -\frac{dq_{ij}}{(z_i - z_j)^2} b_{ij} \neq 0,$$

as restriction to an adjacent-exchange edge shows, whereas $(\partial_{z_i} a_i) a_j + a_i (\partial_{z_i} a_j) = 0$. The prescription can define an associative dg coefficient algebra, but a differential-module realization must supply a different connection or additional state operations. The free-word algebra above has horizontal multiplication and its collision compatibility is proved on that stated carrier.

The internal ordered bar

Let A be an augmented ordered chiral algebra with augmentation ideal $\tilde{A}$. The simplicial two-sided bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Its realization is

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = |B_\bullet(\mathbf{1}, A, \mathbf{1})|.$$

The finite-power collision maps of Theorem 7.4 commute with the same adjacent faces and cuts. Their coefficient augmentation is relative to C_I ; identifying that construction with this Ran bar requires the comparison specified after Proposition 7.7.

Theorem 8.1 (Normalized bar). *In the fixed stable dg enhancement, Dold–Kan normalization identifies the geometric realization with the reduced tensor coalgebra. Explicitly, there is a natural equivalence*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq (T^c(s\tilde{A}), d_{\mathrm{int}} + b_m),$$

where b_m is the coderivation induced by the multiplication of A . The deconcatenation formula

$$\Delta_{\mathrm{dec}}[a_1 | \cdots | a_n] = \sum_{i=0}^n [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_n]$$

makes the bar a conilpotent coassociative coalgebra in $\mathrm{Fact}_D(X)$.

Proof. Normalization removes degeneracies containing the unit and leaves tensor powers of $s\tilde{A}$. The simplicial faces give the internal differential and adjacent multiplication. Associativity is precisely the equality that makes the multiplication coderivation square to zero. Deconcatenation is coassociative because a word cut twice has the same three pieces in either order. The bar differential is local with respect to every cut. $\square$

Theorem 8.2 (Double-coalgebra bar). *In every Ran enhancement in which geometric realization preserves chiral commutative coalgebras, $\mathrm{Bar}_X^{\mathrm{ord}}(A)$ carries the deconcatenation coproduct and the chiral factorization coproduct. Deconcatenation is a morphism of chiral coalgebras.*

Proof. Every simplicial operator is a morphism of factorization objects because the multiplication, unit, and augmentation satisfy the componentwise locality identity. The chiral coproduct therefore realizes levelwise. For a fixed decomposition $I = I_1 \sqcup I_2$, deconcatenation cuts the transverse word and factorization separates each letter into its I_1 - and I_2 -components. These two operations act on independent indices, so their two composites agree after the canonical symmetry. $\square$

8.1 THREE COPRODUCTS

The theorem system uses three coproducts with distinct geometric meanings:

Δ_{fact} : separation of chiral support,

Δ_{dec} : cutting an ordered bar word,

Δ_{Hopf} : fusion of representations.

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 8.3 (Coproduct separation). *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. $\square$

8.2 INTERVAL FACTORIZATION HOMOLOGY

For a right $\mathcal{A}$ -module R and a left $\mathcal{A}$ -module L , factorization homology of an interval with endpoint labels gives

$$\int_{([0,1], \{0,1\})} (\mathcal{A}; R, L) \simeq R \otimes_{\mathcal{A}}^{\mathbb{L}} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [4, 60]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

Cobar and twisting cochains

Let C be a coaugmented conilpotent coalgebra in $\text{Fact}_D(X)$. The cobar algebra is

$$\text{Cobar}_X^{\text{ord}}(C) = (T(s^{-1}\tilde{C}), d_{\text{int}} + d_{\Delta}).$$

A twisting cochain $\tau : C \rightarrow A$ has degree 1 and satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\tilde{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \text{Bar}_X^{\text{ord}}(A) \rightarrow A, \quad \tau_C : C \rightarrow \text{Cobar}_X^{\text{ord}}(C).$$

Theorem 9.1 (Bar–cobar adjunction). *There is a natural adjunction*

$$\text{Cobar}_X^{\text{ord}} : \text{Coalg}_{\text{conil}}(\text{Fact}_D(X)) \rightleftarrows \text{Alg}_{\text{E}_1}^{\text{aug}}(\text{Fact}_D(X)) : \text{Bar}_X^{\text{ord}}.$$

Its mapping space is the space of twisting cochains.

Proof. An algebra map from the tensor algebra $T(s^{-1}\tilde{C})$ is determined by its restriction to $s^{-1}\tilde{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\tilde{A})$. $\square$

Complete bar–cobar reconstruction

Let $A = \prod_{w \geq 0} A_w$ be complete for a positive weight grading, with $A_0 = \mathbf{1}$ and degreewise dualizable pieces. Let $F^p A = \prod_{w \geq p} A_w$. The invariant completion statement is the equivalence between nil-complete algebras and conilcomplete Koszul-dual coalgebras; this is the maximal equivalence locus of operadic Koszul duality [45]. The positive weight hypotheses below give an explicit sufficient criterion in the present dg setting.

Theorem 10.1 (Complete reconstruction). *Let $A = \prod_{w \geq 0} A_w$ be a positive complete augmented algebra with degreewise dualizable pieces. Assume products and inverse limits are exact on the weight towers used below. Let $C = \prod_{w \geq 0} C_w$ be a complete coaugmented coalgebra whose reduced coproduct is locally conilpotent in every weight, whose coradical filtration is exhaustive, and whose completion map to the inverse limit of its finite coradical quotients is an equivalence. Then the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is an equivalence, and the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C)$$

is an equivalence.

Proof. Filter both objects by total weight. On associated graded objects, the bar–cobar composite is the standard tensor resolution. Its contracting homotopy inserts the coaugmentation at the first admissible cut. The filtration is complete, separated, and bounded below in every total weight. The comparison spectral sequence therefore converges strongly and lifts the associated-graded equivalence. $\square$

10.1 QUADRATIC RECOGNITION

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\mathrm{q!}}$ is cogenerated by sV with correlations $s^2 R$. The canonical map $A^{\mathrm{q!}} \rightarrow \mathrm{Bar}(A)$ gives the Koszul complex.

Theorem 10.2 (Quadratic Koszul criterion). *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

- (i) $A^{\mathrm{q!}} \rightarrow \mathrm{Bar}(A)$ is a quasi-isomorphism;
- (ii) $\mathrm{Cobar}(A^{\mathrm{q!}}) \rightarrow A$ is a quasi-isomorphism;
- (iii) the left and right Koszul complexes have homology concentrated in weight zero;
- (iv) $\mathrm{Tor}_i^A(\mathbf{1}, \mathbf{1})$ is concentrated in internal weight i .

Proof. The universal twisting morphism $\mathcal{A}^{\text{q!}} \rightarrow \mathcal{A}$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [57, 68]. $\square$

Modules, comodules, and Morita theory

A twisting cochain $\tau : C \rightarrow A$ defines twisted tensor functors between A -modules and C -comodules. Put $I = \ker(A \rightarrow \mathbf{1})$ and $F^p A = \prod_{w \geq p} A_w$. Write $\text{Mod}_A^{\wedge I}$ for the full derived category of left A -modules M for which

$$M \longrightarrow \varprojlim_p (M \otimes_A^{\mathbb{L}} A / F^p A)$$

is an equivalence. Write

$$D_{\text{conil}, \text{cpl}}^{\text{co}}(C\text{-comod})$$

for the coderived category of coaugmented C -comodules whose coradical filtration is exhaustive and complete. The coderived localization kills coacyclic totalizations and is the natural unbounded target for coalgebra modules [67].

Theorem II.1 (Koszul module equivalence). *Let $C = \text{Bar}(A)$ and let A satisfy the positive completeness and exact-limit hypotheses of the complete reconstruction theorem. Assume the module and comodule weight towers are degreewise finite or satisfy the corresponding Mittag–Leffler exactness. Then the universal twisting cochain induces an equivalence*

$$\text{Mod}_A^{\wedge I} \simeq D_{\text{conil}, \text{cpl}}^{\text{co}}(C\text{-comod}).$$

For complete right and left modules, relative tensor products correspond to derived cotensor products of the transformed comodules.

Proof. The unit and counit are represented by two-sided twisted bar complexes. Their complete weight filtrations have associated graded complexes equal to the ordinary tensor contractions. In each total weight the filtration is finite; the degreewise-finite or Mittag–Leffler hypothesis removes derived-limit terms, so the spectral sequences converge strongly. Coacyclic totalizations vanish in the coderived target. The two-sided transform sends the diagonal A -bimodule to the diagonal C -bicomodule, and applying it to a two-sided resolution identifies derived tensor with derived cotensor. This is the complete algebra–coalgebra form of Koszul duality [60, 67]. $\square$

Let C be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \text{RHom}_C(b, b)$$

has ordered multiplication by composition.

Theorem 11.2 (Factorization Morita theorem). *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$\mathcal{C} \simeq \text{Mod}_{A_b}(\text{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. $\square$

Derived centre and bulk action

The ordered Hochschild cochains are

$$Z_{\text{ord}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A).$$

Theorem 12.1 (Higher Deligne centre). *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\text{ord}}(A)$ carries a canonical E_2 -algebra structure. It is the universal E_2 algebra acting centrally on A .*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [54, 61]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A -bimodule. $\square$

A boundary condition in a field theory gives a morphism

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [11].

Cyclic closure and modular sewing

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 13.1 (Perfect Calabi–Yau ordered algebra). A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra A together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(A) \longrightarrow \mathbf{1}[-d]$$

whose induced pairing $A \otimes A \rightarrow \mathbf{1}[-d]$ is nondegenerate.

Theorem 13.2 (Ribbon-graph realization of a Calabi–Yau algebra). *Let A be a homologically finite, smooth and proper cyclic A_∞ algebra with a nondegenerate pairing of degree d . Then the cyclic bar construction of A defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology $\mathrm{HH}_*(A)$.*

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The A_∞ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello’s construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [18]. $\square$

Remark 13.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

BOOK **III**

Exact noncommutative models

Free and square-zero pairs

Let $M \in \text{Fact}_D(X)$.

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \dots$$

is the free augmented ordered algebra.

Theorem 14.1 (Free–square-zero duality). *Assume exact tensor products and dualizable M . Then*

$$\text{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \quad \text{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\text{Bar}(A) = T^c(sM), \quad A^\dagger \simeq \widehat{T}(M^\vee[-1]).$$

Proof. For the tensor algebra, every word of positive length has a unique last letter. The standard extra degeneracy contracts every bar simplex with an internal cut, leaving the one-letter class. Dualization gives the square-zero algebra. For the square-zero algebra, the multiplication part of the bar differential vanishes, leaving the cofree tensor coalgebra; its continuous dual is the completed tensor algebra. $\square$

The quantum plane

Let

$$A_q = \mathbb{k}\langle x, y \rangle / (yx - qxy), \quad q \in \mathbb{k}^\times.$$

Give x and y weight one.

Theorem 15.1 (PBW basis of the quantum plane). *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q .*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal letters, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. $\square$

The quadratic dual is

$$A_q^\perp = \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 15.2 (Koszulness). *A_q is Koszul, and its Koszul complex is*

$$0 \rightarrow A_q \otimes \mathbb{k}(\xi\eta) \xrightarrow{d_2} A_q \otimes (\mathbb{k}\xi \oplus \mathbb{k}\eta) \xrightarrow{d_1} A_q \rightarrow \mathbb{k} \rightarrow 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = a y x - q a x y = 0$. The image of d_1 is the augmentation ideal (x, y) , so the complex is exact at A_q and has cokernel $\mathbb{k}$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j \geq 0} u_{ij} x^i y^j, \quad v = \sum_{i,j \geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b \geq 0} (q^b u_{a-1,b} + v_{a,b-1}) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j \geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $c y = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. $\square$

Theorem 15.3 (Yoneda algebra of the quantum plane). *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α . The augmentation module has*

$$\mathrm{Ext}_{A_q}^*(\mathbb{k}, \mathbb{k}) \cong \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^!$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\mathrm{Hom}_{A_q}(-, \mathbb{k})$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. $\square$

Tensoring A_q with any commutative factorization coefficient produces a concrete ordered chiral algebra with noncommutative transverse multiplication. Its bar, quadratic dual, and derived augmentation boundary are now explicit in generators and relations.

Quiver path algebras

Let Q be a finite quiver and $S = \bigoplus_{i \in Q_0} \mathbb{k} e_i$. Assign $M_a \in \text{Fact}_D(X)$ to each arrow $a : i \rightarrow j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \quad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots.$$

Words are composable paths.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\text{Ext}^1(S_1, S_2) \cong \mathbb{k}, \quad \text{Ext}^1(S_i, S_i) = 0, \quad \text{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 16.1 (Relative path-algebra bar). *For a tensor path algebra $T_S(M)$ over semisimple S ,*

$$S \otimes_{T_S(M)}^{\mathbb{L}} S \simeq S \oplus sM$$

as an S -bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S -bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. $\square$

Weyl and Clifford deformations

Let V be finite-dimensional and let $\omega : V \otimes V \rightarrow \mathbb{k}$ be alternating. The Weyl algebra is

$$W_{\hbar}(V, \omega) = T(V)[[\hbar]] / \langle uv - vu - \hbar\omega(u, v) \rangle.$$

For a purely odd space and symmetric ω , the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 17.1 (Filtered PBW theorem). *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra is $\text{Sym}(V)$ in the Weyl case and $\wedge(V)$ in the Clifford case.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. $\square$

The quadratic-linear-constant presentation has a curved Koszul dual. The constant term becomes the curvature functional

$$m_o : s^2 R \longrightarrow \mathbb{k}[2].$$

This construction is the algebraic origin of the curvature; geometric anomaly curvature enters through a separate comparison morphism.

Associative chiral algebras on the formal disc

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T . A state–field map

$$Y(-, z) : V \longrightarrow \text{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 18.1 (Associative convolution from weak associativity). *Suppose $Y(\mathbf{1}, z) = \text{id}$, $Y(a, z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a, z)] = \partial_z Y(a, z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a, z_1)Y(b, z_2)c = (z_1 - z_2)^N Y(Y(a, z_1 - z_2)b, z_2)c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. $\square$

Example 18.2 (Braided current fields). Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = \mathbf{1}$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1)a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)a_l(z_2)a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R -matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

Zamolodchikov–Faddeev exchange

Let V be finite-dimensional and let $R(u) \in \text{End}(V \otimes V) \otimes \mathbb{k}(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u)Z_2(v) = R_{12}(u-v)Z_2(v)Z_1(u).$$

Theorem 19.1 (Yang–Baxter confluence). *The exchange algebra has a consistent normal ordering for three generators precisely when*

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u)Z_2(v)Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. $\square$

For the rational permutation solution

$$R(u) = \mathbf{1} - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows from $P_{ij}^2 = \mathbf{1}$ and $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$.

The RTT Yangian of $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \quad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_{\hbar}(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)).$$

Proposition 20.1 (Coefficient relations). *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar(t_{kj}^{(r)}t_{il}^{(s)} - t_{kj}^{(s)}t_{il}^{(r)}), \quad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first modes form the enveloping algebra of the copy $\hbar\mathfrak{gl}_2$ inside the Yangian.

Proof. Multiply the generating-series relation by $u-v$ and compare the coefficient of $u^{-r-1}v^{-s-1}$. The two terms contributed by multiplication by $u-v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r=0$ gives the displayed first-mode relation. $\square$

Theorem 20.2 (Hopf structure). *The formulas*

$$\Delta(T(u)) = T(u) \dot{\otimes} T(u), \quad \varepsilon(T(u)) = 1, \quad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure in the u^{-1} -adic completion. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^2 \sum_{p=0}^r t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. $\square$

The evaluation map

$$\mathrm{ev}_a : T(u) \mapsto 1 + \frac{\hbar E}{u - a}, \quad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[[u^{-1}]]$ and gives the standard evaluation modules. The PBW theorem identifies the associated graded with $U(\mathfrak{gl}_2[t])$ [28, 62].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R -matrix.

The Hall algebra of the A_2 quiver

Let $Q : 1 \rightarrow 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \rightarrow S_2 \rightarrow M \rightarrow S_1 \rightarrow 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \quad [S_2] * [S_1] = [S_1 \oplus S_2], \quad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1 E_2 - v^{-1} E_2 E_1$$

is represented by the indecomposable extension.

Theorem 21.1 (Quantum Serre relation). *The Hall generators satisfy*

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0,$$

and the analogous relation with 1 and 2 exchanged.

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\text{rk } f$	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)(v^{-1} - (v + v^{-1}) + v) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust the dimension vector $(2, 1)$, proving the relation. This is Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [40, 70]. $\square$

Matrix traces and the necklace bracket

Let $F = \mathbb{k}\langle x, y \rangle$ and $F_{\text{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\text{cyc}} [a_1 \cdots a_n] = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y . Define

$$\{f, g\}_{\text{neck}} = \left[\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g \right]_{\text{cyc}}.$$

Theorem 22.1 (Trace realization). *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\text{Tr } f(X, Y), \text{Tr } g(X, Y)\} = \text{Tr} \{f, g\}_{\text{neck}}(X, Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. $\square$

In the stable commuting-matrix quotient, let

$$p_{ab} = \text{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1, b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

BOOK **IV**

The nonlocal ladder

Spectral exchange as an associative chiral operation

23.1 THE COEFFICIENT FIELD AND THE EXCHANGE OPERATOR

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \text{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 23.1 (Zamolodchikov–Faddeev algebra). The exchange algebra $\mathcal{Z}_R$ is generated by the components of a current $A(u) \in V \otimes \mathcal{Z}_R$ with relation

$$A_1(u)A_2(v) = R_{12}(u-v)A_2(v)A_1(u).$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 23.2 (Confluence is Yang–Baxter). *The quadratic rewriting system defined by (23.1) is confluent on words with pairwise distinct spectral labels precisely when (23.1) holds. In that case the normal ordered words form a K -basis of $\mathcal{Z}_R$.*

Proof. Every two-letter inversion is removed by one application of (23.1). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (23.1). Bergman’s diamond lemma then gives the normal-word basis. $\square$

23.2 A DIAGONAL EXCHANGE MODEL

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \quad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q, q_{21} = q^{-1}$.

Calculation 23.3 (Three-particle coefficient). For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent letters contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. $\square$

RTT and reflection algebras

24.1 THE RTT BIALGEBRA

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v).$$

Theorem 24.1 (RTT coproduct). *The formula*

$$\Delta(T(u)) = T_{13}(u)T_{12}(u), \quad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$.

Proof. Apply (24.1) in each tensor factor. The Yang–Baxter equation moves the three R -operators through the three matrix factors. The resulting identity is the RTT relation for $\Delta(T)$. Matrix multiplication is associative, hence the two iterated coproducts both equal $T_{14}T_{13}T_{12}$. $\square$

For the Yang matrix $R(u) = I - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient of $u^{-1}v^{-1}$ is

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

24.2 BOUNDARY EXCHANGE

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \text{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}.$$

Theorem 24.2 (Boundary confluence). *The relation (24.2) gives a coherent exchange law for three boundary insertions. The two reductions of every length-three overlap agree by the Yang–Baxter equation and its inverse.*

Proof. A boundary exchange consists of an incoming crossing, reflection, and outgoing crossing. The two reductions of three insertions sweep the same braid in the doubled half-plane. Algebraically, one inserts (24.2) three times and uses Yang–Baxter to pass the middle R -matrix through the outer pair. The remaining factors cancel in inverse pairs. $\square$

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R -matrix, while their products live in three different endomorphism objects.

Quantum vertex realization

25.1 TWO EXPANSION REGIONS

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T , state-field map Y , and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 25.1 (Rational quantum vertex datum). A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1)$$

for a sufficiently large N , and weak associativity in every triple of states.

Theorem 25.2 (Exchange coherence). *Suppose S is unitary,*

$$S^{21}(-z)S(z) = \mathbf{1},$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (25.1) define a braid-group action on every n -point field product. Weak associativity makes the action compatible with iterated operator products.

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. $\square$

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [55, 56].

25.2 THE YANGIAN FIELD

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S -locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u) \left(\mathbf{1} - \frac{\hbar P}{u} \right), \quad f(u)f(-u) = \left(\mathbf{1} - \frac{\hbar^2}{u^2} \right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Shuffle realization and the positive current algebra

26.1 PAIRWISE KERNEL

Let $K(z)$ be a rational function and put

$$\text{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

For $f \in \text{Sh}_m$ and $g \in \text{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \text{Sym} \left(f(z_1, \dots, z_m) g(z_{m+1}, \dots, z_{m+n}) \prod_{i \leq m < j} K(z_i - z_j) \right).$$

Theorem 26.1 (Associativity of the shuffle product). *The product (26.1) is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. $\square$

For the tripled Jordan quiver, one uses

$$K(z) = \prod_{a=1}^3 \frac{z + \epsilon_a}{z}, \quad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

Wheel conditions remove the forbidden triple residues. The resulting shuffle algebra is the positive current algebra that appears in the critical CoHA of $\mathbb{C}^3$ and in the positive affine Yangian of $\mathfrak{gl}_1$ [71, 72].

Calculation 26.2 (Degree one against degree one). For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

The antisymmetric part records the first current commutator. Its residues at $z_1 - z_2 = -\epsilon_a$ are the three elementary extension channels of the tripled quiver.

Proof. For two variables the symmetrization in the shuffle definition has two terms, which gives the displayed formula. Antisymmetrization exchanges the two chambers. The three numerator factors of the tripled-quiver kernel locate the three elementary shifted extension channels. $\square$

26.2 COPRODUCT AND DOUBLE

Choose a chamber expansion of $K(z)$ at infinity. Splitting the variables into two ordered groups and expanding every cross-kernel in that chamber defines a completed coproduct. A residue pairing between positive and negative shuffle halves gives a continuous Hopf pairing. The completed Drinfeld double is then a quasitriangular current algebra. Product, coproduct, and pairing are separate operations; each is visible in the shuffle formula that defines it.

The ladder in one diagram

The nonlocal structures form the chain

$$\begin{aligned} \text{ordered coefficient} &\longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S), \\ (V, Y, S) &\longrightarrow \text{Sh}^+ \longrightarrow D(\text{Sh}^+). \end{aligned}$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S -locality and weak associativity
Sh^+	extension shuffle	pairwise-kernel associativity
$D(\text{Sh}^+)$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

BOOK **V**

Field theory and master equations

A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \frac{1}{2}[A, A] \rangle.$$

Theorem 28.1 (Cotangent master equation). *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2 , the derivation identity for d_L , and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. $\square$

Proposition 28.2 (Coordinate form of the modular term). *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^c e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^c c^a c^b, \quad \Delta = - \sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \text{Tr}(\text{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^a c^i c^j$. Odd differentiation in c^a gives $f_{aj}^a c^j$, and the chosen sign in Δ converts this to $f_{ja}^a c^j = \text{Tr}(\text{ad}_{e_j}) c^j$. This is the displayed formula. $\square$

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 28.3 (Finite-dimensional quantum master equation). *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \quad \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta(e^{S_L/\hbar}) = 0.$$

The quantum observable differential

$$\partial_{S_L} = \{S_L, -\} + \hbar \Delta$$

satisfies $\partial_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \text{str}(\text{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar} \Delta(e^{S_L/\hbar}) = \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\partial_{S_L}^2 = \left\{ \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L, - \right\} = 0.$$

□

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{\circ,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathbf{I}$$

for a finite-dimensional nilpotent $\mathbf{I}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

Choose the standard interval polarization in which the $L[1]$ variables are fixed at one end and the cotangent variables at the other. Canonical quantization of the boundary linear Poisson algebra gives the associative zero-mode algebra $U(\mathbf{I})$. Gluing the two augmentation boundary states across an interval gives the derived tensor product

$$\mathbf{I} \otimes_{U(\mathbf{I})}^{\mathbb{L}} \mathbf{I} \simeq \text{Bar}(U(\mathbf{I})) \simeq C_*(\mathbf{I}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex produces the corresponding infinite-dimensional quantum factorization algebra.

BRST and ordered reduction

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 29.1 (BRST–bar bicomplex). *With the suspension sign convention,*

$$Q_{\text{Bar}}b + bQ_{\text{Bar}} = 0.$$

Hence $D = Q_{\text{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. $\square$

The filtration by bar length gives

$$E_1 \cong \text{Bar}(H^*(A, Q)) \implies H^*(\text{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

Stokes and algebraic identities

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E , and the assignments are compatible with restrictions to corners.

Theorem 30.1 (Stokes-to-master). *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \quad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S) \rho_{F|S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. $\square$

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

Two edge colors and two loop operators

Let $G^{\perp, \text{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-, -]_{\perp}$ and $[-, -]_{\text{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch} . Product orientation gives

$$\Delta_{\perp} \Delta_{\text{ch}} + \Delta_{\text{ch}} \Delta_{\perp} = 0.$$

Theorem 31.1 (Bicolored master equation). *Let $P^{\perp, \text{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_P$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_P^I$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_{\perp} + \frac{1}{2}[\Theta, \Theta]_{\text{ch}} + \hbar_{\perp} \Delta_{\perp} \Theta + \hbar_{\text{ch}} \Delta_{\text{ch}} \Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [34]. $\square$

Typed anomalies

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_o

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

BOOK VI

Calabi–Yau and automorphic interfaces

Calabi–Yau categories

Let C be a smooth linear category with a d -Calabi–Yau structure. Two universal consequences are available.

Theorem 33.1 (Cyclic and shifted outputs). *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2 - d)$ -shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [11]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $R\mathrm{Hom}(E, E)[1]$ with itself in degree $2 - d$ [10, 66]. $\square$

A geometric BV theory or an open factorization category supplies an ordered chiral algebra through the recognition theorem of Chapter 3. This is the bridge from Calabi–Yau geometry to the present volume.

Hall and BPS algebras

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [23]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [26, 71].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borchers–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H} \mathfrak{n}_+ = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta, \alpha}, \quad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+ / F^{>H} \mathfrak{n}_+.$$

Height additivity makes $F^{>H} \mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 35.1 (Bar–Chevalley identification). *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\text{Bar}_h^\wedge(U(\mathfrak{n}_+)) := \varprojlim_H \text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

Proof. For every H , the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\text{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1 - e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. $\square$

With the positive-root convention $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$, the denominator theorem reads

$$e^{-\rho} \chi C_{*,h}^\wedge(\mathfrak{n}_+) = D_\zeta(2\Omega) = 2^{-6} \Delta_\zeta(2\Omega)$$

[41, 43]. This identity is the exact meeting point of ordered bar theory and automorphic products.

Finite-height currents and an exact physical model

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\text{Cur}(\mathfrak{n}_+^{(H)}) = \mathbb{k}[\partial] \otimes \mathfrak{n}_+^{(H)}, \quad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H . The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta(a(-1)\mathbf{1}) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \quad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 36.1 (Height-complete current realization). *Let $\mathfrak{n} = \bigoplus_{\alpha > 0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} / \bigoplus_{h(\alpha) > H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\text{ch}}(\text{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. $\square$

Apply the finite BF model of Chapter 19 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \quad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K_3 \times E$ identification is the geometric comparison studied in the companion volumes.

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 37.1 (Three reconstruction fibres). *The following fibres contain distinct structured objects with the same scalar image.*

- (i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees $a, b, a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*
- (ii) *The superdimension-zero fibre contains both the zero super vector space and $\mathbb{k}^{1|1}$.*
- (iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ .*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 37.2 (Minimal PBW recognition). *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

Exact dependency theorem

Theorem 38.1 (Ordered-chiral closure). *Fix:*

- (i) *a six-functor Ran category and its factorization subcategory;*
- (ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*
- (iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. $\square$

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$. Associative chirality is an E_1 structure for $\star^{\text{ch}}$. Chiral Lie theory is a Lie structure for $\star^{\text{ch}}$. Their bars are $\text{Bar}^\perp$, $\text{Bar}_{\text{Ass}}^{\text{ch}}$, and $\text{Bar}_{\text{Lie}}^{\text{ch}}$. The aligned projector supplies the exact sector on which a mixed distributive law is defined. Every comparison in the theorem system factors through one of these typed maps.

The constructed operations form the two functorial rows

$$\text{HT boundary theory} \xrightarrow{\text{Obs}_\partial} \text{Alg}_{E_1}(\text{Fact}_D(X)) \xrightarrow{\text{Bar}^{\text{ord}}} \text{Coalg}_{\text{Econil}, \text{cpl}}(\text{Fact}_D(X)),$$

$$\text{Alg}_{E_1}(\text{Fact}_D(X)) \xrightarrow{Z_{\text{ord}}} \text{Alg}_{E_2}(\text{Fact}_D(X)).$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

BOOK VII

Technical appendices to the preceding
book

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \vec{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}[(ab)c - a(bc)],$$

which vanishes by associativity.

Rational R -matrix verification

Let P_{ij} permute tensor factors i and j . Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Concordance with the July 2026 Volume I draft

The present volume preserves the draft's five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative example ladder. It sharpens four interfaces:

- (i) a factorization carrier is paired explicitly with its operation;
- (ii) the Ran category is fixed before its tensor products are used;
- (iii) the block-ordered operad is introduced after a representing chiral operad is chosen;
- (iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

These refinements make every theorem a statement in a fixed category and every physical interpretation the image of a displayed comparison map.

The enveloping bar and the audit of the planar tables

The three bars of the typed bar theorem are computed here on enveloping algebras, and the computation settles two dimension formulas that circulated as bar cohomology.

43.1 CHEVALLEY COMPARISON

Theorem 43.1 (Bar–Chevalley comparison). *Let $\mathfrak{g}$ be a Lie algebra object in $\text{FactD}(X)$ for the pointwise product, K -flat, positively graded and complete, with every total weight admitting finitely many decompositions into positive weights, and suppose internal PBW holds for $U(\mathfrak{g})$. There is a natural quasi-isomorphism of conilpotent coalgebras in $\text{FactD}(X)$*

$$\text{Bar}^\perp(U(\mathfrak{g})) \xrightarrow{\sim} C_*(\mathfrak{g}),$$

compatible with the factorization coefficient and the PBW filtration.

Proof. Filter $U(\mathfrak{g})$ by PBW degree and both complexes by total word length. The associated graded enveloping algebra is $\text{Sym}(\mathfrak{g})$, and the bar of a symmetric algebra is the suspended symmetric coalgebra on $\mathfrak{g}$, which is the associated graded of Chevalley chains. The comparison is an isomorphism on the first page; finite decomposition gives a bounded exhaustive spectral sequence in each weight, and completeness promotes it to a quasi-isomorphism. Every step uses only internal tensor products and the bracket, so the map stays in $\text{FactD}(X)$. $\square$

Calculation 43.2 (Finite-dimensional simple case). For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$:

$$\text{Bar}^\perp U(\mathfrak{sl}_2) : 1 + t^3, \quad \text{Bar}^\perp U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \quad \text{Bar}^\perp U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The first two were recomputed by exact rational elimination on $\Lambda^\bullet \mathfrak{g}^*$ built from matrix units, with $d^2 = 0$ asserted on every cochain space rather than assumed, giving Betti numbers 1, 0, 0, 1 and 1, 0, 0, 1, 0, 0, 1. The Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$, so for $\mathfrak{sl}_2$ they are 1, 3, 3, 1 while the bar chain dimensions of an augmented algebra with $\dim \bar{A} = 3$ are 3^r .

43.2 THE VIRASORO ROW

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C} e_n, \quad [e_i, e_j] = (j - i)e_{i+j},$$

to which Theorem 43.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 43.3 (Pentagonal rigidity). *The bar homology of $U(L_1)$ satisfies*

$$\dim H^n(\text{Bar}^\perp U(L_1)) = 2 \quad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q(\text{Bar}^\perp U(L_1)) = \prod_{n \geq 1} (1 - q^n),$$

and (43.3) is accounted for degree by degree with no residual freedom.

Proof. By Theorem 43.1 the bar of $U(L_1)$ is $C_*(L_1)$, whose cohomology is Goncharova's: two dimensional in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (43.3) is the denominator product applied to L_1 , where every root space is one dimensional and even, so $\text{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_w^n$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

The cohomology was computed for $n = 1, 2, 3, 4$, occupying the weights $(1, 2), (5, 7), (12, 15), (22, 26)$, one dimension in each; the nonzero coefficients of $\prod_{n \geq 1} (1 - q^n)$ occupy 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs +, -, -, +, +, -, -, +, +.

43.3 THE PLANAR TABLES

Two formulas circulated as bar cohomology dimensions: $\dim H^n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \dots$ for the Virasoro family, with M the Motzkin numbers, and $\dim H^n = R(n+3) = 3, 6, 15, 36, 91, \dots$ for the $\mathfrak{sl}_2$ family, with R the Riordan numbers. The coincidence of the discriminant $\sqrt{1 - 2z - 3z^2}$ in the two generating functions was attributed to Drinfeld–Sokolov reduction.

Proposition 43.4 (The two sequences are one sequence). *For all $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \quad M(n+1) - M(n) = R(n+2) - R(n).$$

The Virasoro sequence is therefore $R(n+2) - R(n)$ and the $\mathfrak{sl}_2$ sequence is $R(n+3)$: both are elementary transforms of one sequence, and their generating functions are rational in the single algebraic function $\sqrt{1 - 2z - 3z^2}$ for that reason alone. No Drinfeld–Sokolov input is used or needed.

Proof. Riordan numbers count Motzkin paths with no level step at height zero. Splitting a Motzkin path of length n according to whether its first step is level gives $M(n) = R(n) + R(n+1)$; subtracting consecutive instances gives the second identity. Both were checked termwise for $n < 24$. $\square$

Theorem 43.5 (Carrier theorem for the planar series). *Neither sequence is the ordered bar homology of the algebra it names, and neither is a bar chain count.*

- (i) *For $\mathfrak{sl}_2$ the bar homology is 1, 0, 0, 1 by Calculation 43.2, and the bar chain dimensions are 3, 9, 27, 81, 243. The sequence 3, 6, 15, 36, 91 is neither.*
- (ii) *For the Virasoro family the bar homology is the constant 2 by Theorem 43.3. On degrees $n = 1, 2, 3$ the asserted values are 1, 2, 5; they agree with the truth at $n = 2$ and nowhere else.*

- (iii) *Independently of any presentation, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged.*

Consequently a chain-level identification of either planar series with an ordered bar requires a specified transverse multiplication together with a quasi-isomorphism from its planar complex to that bar, and by (i) and (ii) no such quasi-isomorphism exists for the algebras named.

Remark 43.6 (The type error beneath the numerical one). The bar homology of $U(L_1)$ is bigraded by cohomological degree and weight, and is finite in each degree only because Theorem 43.3 concentrates it in two weights. A formula assigning one integer to each degree, with no weight truncation named, does not specify an invariant of a graded object with infinite-dimensional graded pieces.

The obstruction constitution

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 44.1 (Where a mixed distributive law can live). *The sector on which a mixed distributive law between $\otimes^!$ and $\star^{\text{ch}}$ is geometrically defined is the common-decomposition sector, the image of the aligned idempotent e_{al} . Off that sector the two products admit no interchange, only the aligned retraction, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. The canonical comparison in the additive model is the aligned inclusion ι with retraction π , and $\pi\iota = \text{id}$ while $\iota\pi = e_{\text{al}} \neq \text{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e_{al} acts as the identity. $\square$

This locates a mixed law without constructing one. No example in the present corpus exhibits one; Zamolodchikov–Faddeev algebras are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction.

Theorem 44.2 (No quantum group from dimension loss). *The dimension of the kernel of $V^{\otimes n} \rightarrow \text{Sym}^n V$ determines no R -matrix, associator, or quantum group. That kernel carries no multiplication, coproduct, monodromy, or coherence equation.*

Proof. The kernel is a representation of Σ_n and nothing more. Two algebras with the same underlying graded vector space, hence the same kernel dimensions in every arity, may have different multiplications and inequivalent representation categories; a free algebra and a quadratic quotient with the same Hilbert series is an example. A quantum-group structure is a coproduct together with coherence data, none of which is a function of a dimension. $\square$

Theorem 44.3 (No modular object from topology alone). *The identity $\partial^2 = \circ$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 44.4 (No bulk from a boundary centre). *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. $\square$

Theorem 44.5 (No classification by central charge). *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 44.6 (No identification of anomalies of different types). *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borchers-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. $\square$

Theorem 44.7 (No unnamed comparison of bars). *An asserted equality $\text{Bar}^\perp(A) = \text{Bar}_{\text{Ass}}^{\text{ch}}(A)$ has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 44.8 (A scalar sequence is not homology data). *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. Calculation 43.2 exhibits an algebra whose bar chain dimensions are 3, 9, 27, 81, 243 and whose bar homology dimensions are 1, 0, 0, 1; Proposition 43.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. $\square$

Theorem 44.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

Book entrance: Mixed Holomorphic–Topological Deligne Theory

*Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization, and the all-arity
open–closed map.*

BOOK VIII

Finite Hamiltonian geometry

The formal symplectic plane

45.1 HAMILTONIANS AND THE POISSON BRACKET

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = dz_1 \wedge dz_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[[z_1, z_2]], \quad \{f, g\} = \partial_{z_1} f \partial_{z_2} g - \partial_{z_2} f \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p \mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p \mathfrak{h}, F^q \mathfrak{h}\} \subset F^{p+q} \mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\text{aff}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \text{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r . The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^d R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 45.1 (Finite Hamiltonian jet algebra). *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$ -module $\text{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^d (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω ; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. $\square$

45.2 UNIMODULARITY

Theorem 45.2 (Unimodularity of every finite jet). *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\mathrm{Tr}_{\mathfrak{h}_{\leq d}}(\mathrm{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^d R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$ -module R_r has trace zero. Linearity gives the result. $\square$

Corollary 45.3 (Canonical finite BV measure). *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. $\square$

The shifted cotangent theory

46.1 THE LIE ALGEBRA AND ITS FUNCTIONS

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\leq d} = \mathfrak{h}_{\leq d} \ltimes \mathfrak{h}_{\leq d}^\vee[-1],$$

where $\mathfrak{h}_{\leq d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 46.1 (Chevalley–Poisson identification). *There is a canonical isomorphism of commutative dg algebras*

$$C_{\text{CE}}^*(\mathfrak{g}_d) \cong C^*(\mathfrak{h}_{\leq d}, \text{Sym}(\mathfrak{h}_{\leq d})).$$

The differential on the right is the Chevalley differential for the adjoint action on $\text{Sym}(\mathfrak{h}_{\leq d})$. Under the identification $\text{Sym}(\mathfrak{h}_{\leq d}) = \mathcal{O}(\mathfrak{h}_{\leq d}^)$, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}_{\leq d}^\vee[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\leq d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}_{\leq d}^*$ and exterior ghosts for $\mathfrak{h}_{\leq d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. $\square$

46.2 THE BV ACTION

Choose dual bases e_a and e^a . Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2} \{S_d, S_d\} = 0$$

by Jacobi.

Theorem 46.2 (Exact finite quantum master equation). *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \text{Tr}(\text{ad } e_b) c^b = 0.$$

Consequently

$$\frac{1}{2} \{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d .

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. $\square$

The universal boundary brane

47.1 POLARIZATION

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_{\hbar}(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[[\hbar]] / \langle xy - yx - \hbar[x, y] \rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\text{sym}_{\hbar} : \text{Sym}(\mathfrak{h}_{\leq d})[[\hbar]] \xrightarrow{\sim} U_{\hbar}(\mathfrak{h}_{\leq d}).$$

Theorem 47.1 (Boundary quantization). *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_{\hbar}(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}_{\leq d}^*$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_{\hbar}(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. $\square$

BOOK **IX**

The exact open–closed equivalence

Hochschild cochains of the universal brane

48.1 THE CARTAN–EILENBERG MODEL

For every finite-dimensional Lie algebra $\mathfrak{l}$ there is a natural quasi-isomorphism

$$C_{\text{HH}}^\bullet(U\mathfrak{l}, U\mathfrak{l}) \simeq C^*(\mathfrak{l}, U\mathfrak{l}_{\text{ad}}).$$

The same statement holds for the Rees algebra after $\hbar$ -adic completion.

Proposition 48.1 (Filtered Hochschild model). *There is a filtered quasi-isomorphism*

$$C_{\text{HH}}^\bullet(U_\hbar(\mathfrak{h}_{\leq d}), U_\hbar(\mathfrak{h}_{\leq d})) \simeq C^*(\mathfrak{h}_{\leq d}, U_\hbar(\mathfrak{h}_{\leq d})_{\text{ad}}).$$

Its associated graded is

$$C^*(\mathfrak{h}_{\leq d}, \text{Sym}(\mathfrak{h}_{\leq d})).$$

Proof. Resolve the diagonal bimodule by the Chevalley–Eilenberg bimodule resolution

$$U_\hbar(\mathfrak{l}) \otimes \Lambda^\bullet \mathfrak{l} \otimes U_\hbar(\mathfrak{l}).$$

Applying bimodule homomorphisms into $U_\hbar(\mathfrak{l})$ gives the Chevalley complex with adjoint coefficients. PBW identifies the associated graded coefficient module with $\text{Sym } \mathfrak{l}$. $\square$

The Kontsevich–Duflo map

49.1 LINEAR POISSON FORMALITY

Let π_d be the linear Poisson bivector on $\mathfrak{h}_{\leq d}^*$. Kontsevich’s stable formality morphism, twisted by π_d , gives a filtered L_∞ quasi-isomorphism from polyvectors with differential $[\pi_d, -]$ to Hochschild cochains of the star algebra. Tamarkin’s refinement transports the full homotopy Gerstenhaber structure.

Theorem 49.1 (Finite mixed-HT Deligne equivalence). *For every $d \geq 2$ there is a filtered homotopy- E_2 quasi-isomorphism*

$$\Phi_d : C_{CE}^*(T^*[-1]\mathfrak{h}_{\leq d})[[\hbar]] \xrightarrow{\sim} C_{HH}^\bullet(U_\hbar(\mathfrak{h}_{\leq d}), U_\hbar(\mathfrak{h}_{\leq d})).$$

Its classical term is the Hochschild–Kostant–Rosenberg map. On invariant degree-zero functions it is the Duflo map

$$\text{Duf} = \text{sym}_\hbar \circ j^{1/2}(\hbar\partial), \quad j^{1/2}(x) = \det_{\mathfrak{h}_{\leq d}} \left(\frac{\sinh(\text{ad}_x/2)}{\text{ad}_x/2} \right).$$

Proof. Kontsevich formality identifies the dg Lie algebra of polyvectors on $\mathfrak{h}_{\leq d}^*$, twisted by the Maurer–Cartan element $\hbar\pi_d$, with the Hochschild dg Lie algebra of the corresponding star product. The star algebra of a linear Poisson structure is identified with the Rees enveloping algebra by PBW. Tamarkin formality lifts the equivalence from dg Lie algebras to homotopy E_2 algebras. The wheel graphs in the first Taylor coefficient sum to $j^{1/2}$, which gives Duflo on invariants. $\square$

49.2 THE BOUNDARY ACTION

The brace operad acts on the pair

$$(C_{HH}^\bullet(A, A), A)$$

for every associative algebra A . The closed color is the derived centre and the open color is the algebra itself.

Corollary 49.2 (All-arity open–closed structure). *Fix a stable formality morphism and a compatible Swiss-cheese refinement. Transport through Φ_d gives the pair*

$$(C_{CE}^*(T^*[-1]\mathfrak{h}_{\leq d})[[\hbar]], U_\hbar(\mathfrak{h}_{\leq d}))$$

a homotopy Swiss-cheese action associated with that refinement. Its binary closed bracket is the Poisson bracket, its open product is the enveloping product, and its first mixed operation is the quantized Hamiltonian action.

Proof. The brace operad acts on Hochschild cochains together with the algebra on which they act. The chosen homotopy- E_2 formality map transports the closed colour and every mixed brace to the Chevalley–Poisson model. The binary and first mixed operations are its first Taylor components. $\square$

The formal inverse limit

50.1 WEIGHT COMPLETION

Let $F^{>d}\mathfrak{h}$ be the Hamiltonians of polynomial degree greater than d . Then

$$\mathfrak{h} = \varprojlim_d \mathfrak{h}_{\leq d}.$$

Define

$$\widehat{U}_h(\mathfrak{h}) = \varprojlim_d U_h(\mathfrak{h}_{\leq d})$$

and complete every cochain complex in polynomial weight.

Theorem 50.1 (Continuous formal-Darboux equivalence). *A stable formality morphism can be chosen compatibly with the quotient maps $\mathfrak{h}_{\leq d+1} \rightarrow \mathfrak{h}_{\leq d}$. It induces a continuous homotopy- E_2 quasi-isomorphism*

$$\Phi_\infty : C_{\text{CE,cont}}^*(T^*[-1]\mathfrak{h})[[\hbar]] \xrightarrow{\sim} C_{\text{HH,cont}}^\bullet(\widehat{U}_h(\mathfrak{h}), \widehat{U}_h(\mathfrak{h})).$$

A compatible Swiss-cheese refinement gives the full formal mixed holomorphic-topological open-closed structure of the pointed Hamiltonian BF sector. Its cohomological Gerstenhaber structure is independent of the chosen chain-level refinement.

Proof. The Taylor coefficients of a stable formality morphism are universal finite sums of graphs. In any fixed polynomial weight, finitely many jets and graphs contribute. The maps therefore commute with the jet projections. The finite complexes form Mittag-Leffler towers in each weight, so inverse limit preserves the finite-level quasi-isomorphisms. Brace operations are defined by finite insertion in every weight and pass to the same limit. $\square$

Principle 50.2 (Universal brane principle). The boundary algebra $\widehat{U}_h(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. The bulk is its complete derived centre.

BOOK **X**

The finite-jet laboratory

The first three Hamiltonian jets

51.1 QUADRATIC HAMILTONIANS

Set

$$e = \frac{1}{2}z_1^2, \quad f = -\frac{1}{2}z_2^2, \quad h = -z_1z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \quad \{h, f\} = -2f, \quad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\leq 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_{\hbar}(\mathfrak{sl}_2)$.

51.2 CUBIC HAMILTONIANS

Let

$$v_j = z_1^{3-j} z_2^j, \quad 0 \leq j \leq 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \quad \{e, v_j\} = jv_{j-1}, \quad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 51.1 (The cubic jet algebra). *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\leq 3} \cong \mathfrak{sl}_2 \ltimes \text{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. $\square$

51.3 THE FIRST NONLINEAR LAYER

Write $w_k = z_1^{4-k} z_2^k$, $0 \leq k \leq 4$. In $\mathfrak{h}_{\leq 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i)w_{i+j-1},$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 51.2 (Two brackets).

$$\{z_1^3, z_2^3\} = 9z_1^2 z_2^2 = 9w_2, \quad \{z_1^2 z_2, z_1 z_2^2\} = 3z_1^2 z_2^2 = 3w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio 3 : 1.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1 z_2 \cdot 2z_1 z_2 - z_1^2 \cdot z_2^2 = 3z_1^2 z_2^2$. $\square$

Theorem 51.3 (The quartic jet presentation). *The Lie algebra $\mathfrak{h}_{\leq 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j , formula (51.3), and the induced $\mathfrak{sl}_2$ action on R_4 . The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (51.3). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. $\square$

Moyal quantization in coordinates

52.1 THE BIDIFFERENTIAL FORMULA

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^n (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f) (\partial_1^r \partial_2^{n-r} g).$$

Every polynomial pair gives a finite sum.

Theorem 52.1 (Quadratic exactness). *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

Thus the quadratic Moyal algebra is the Rees enveloping algebra of $\mathfrak{sl}_2$ together with the Heisenberg translations when linear Hamiltonians are retained.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. $\square$

Calculation 52.2 (Cubic correction). For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (52.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. $\square$

52.2 WEYL GENERATORS

Let q act by multiplication and let $p = \hbar \partial_q$. Then

$$[p, q] = \hbar.$$

Weyl ordering identifies polynomial symbols with differential operators. The Moyal formula is the pullback of operator composition under this identification.

Chevalley coordinates and the Poisson differential

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c_{ab} e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}_{\leq d}^*$. Then

$$d_{\text{CE}} c^c = -\frac{1}{2} f^c_{ab} c^a c^b, \quad d_{\text{CE}} x_a = f^c_{ab} c^b x_c.$$

The first equation is the Lie bracket. The second is the coadjoint action.

Proposition 53.1 (Hamiltonian interpretation). *For a polynomial $F(x)$,*

$$d_{\text{CE}} F = c^a \{x_a, F\}_{\text{lin}}.$$

Consequently $d_{\text{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c_{ab} x_c$. The Leibniz rule gives (53) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. $\square$

53.1 THE $\mathfrak{sl}_2$ INVARIANT

For the quadratic jet, identify the coordinate functions with e, f, h . The invariant

$$q = 2ef + \frac{1}{2}h^2$$

is d_{CE} -closed. Its Duflo image is $\Omega + 1/2$. The local open-closed map therefore sends the classical radius of the coadjoint orbit to the quantum Casimir with its exact zero-point energy.

Residue duality and the formal current algebra

54.1 THE CONTINUOUS DUAL

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \quad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\text{Res}_0(z_1^i z_2^j \rho_{ab}) = \delta_{ia} \delta_{jb}.$$

The continuous dual of $\mathbb{C}[[z_1, z_2]]$ is the direct sum of the ρ_{ab} . The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\text{aff}}$.

Proposition 54.1 (Coadjoint current formula). *For $f = z_1^i z_2^j$ and ρ_{ab} ,*

$$\text{ad}_f^*(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of f, g by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. $\square$

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^{\vee}[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

All-arity open–closed operations

55.1 HOCHSCHILD BRACES

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F . The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 55.1 (Brace realization of the universal centre). *For $A = U_{\hbar}(\mathfrak{h}_{\leq d})$, choose the Tamarkin–Kontsevich homotopy–Gerstenhaber refinement of Φ_d . It transports the homotopy Schouten operations of polyvector fields to the Hochschild brace algebra. The binary operation is the Gerstenhaber bracket, and the mixed operation with one boundary word is the quantized Hamiltonian action.*

Proof. Kontsevich formality is an algebra over a cofibrant resolution of the Gerstenhaber operad. Tamarkin’s formality identifies that resolution with the brace operad. Twisting by the linear Poisson Maurer–Cartan element produces the differential and preserves every brace operation. $\square$

55.2 THE FIRST MIXED IDENTITIES

Let F be a closed cochain and a, b boundary operators. The first Swiss-cheese identities are

$$F \cdot (ab) = (F_{(1)} \cdot a)(F_{(2)} \cdot b),$$

for the coproduct encoded by the brace diagonal, and

$$[F, G] \cdot a = F \cdot (G \cdot a) - (-1)^{|F||G|} G \cdot (F \cdot a).$$

For linear Hamiltonians these are the Leibniz rule and the commutator identity of derivations. Higher braces provide their coherent homotopies.

Corollary 55.2 (All-arity formal–Darboux theorem). *The inverse limit Φ_{∞} equips the pair*

$$\left(C_{\text{CE}}^*(T^*[-1]\mathfrak{h}), \widehat{U}_{\hbar}(\mathfrak{h}) \right)$$

with a complete homotopy Swiss-cheese algebra. Its closed color is the full derived centre of the universal boundary algebra.

Proof. Every fixed polynomial weight receives finitely many brace insertions, so the finite-level Swiss-cheese operations commute with the jet projections. Their separated inverse limit gives the complete action. Morita theory identifies Hochschild cochains of the boundary algebra with its derived centre. $\square$

Finite Chan–Paton ranks

56.1 RANK ONE

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

56.2 RANK TWO AND CAYLEY–HAMILTON REDUCTION

For a 2×2 matrix X ,

$$X^2 - (\text{Tr } X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X -degree at most one after adjoining $\text{Tr } X$ and $\det X$. The same statement holds for Y . On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 56.1 (First finite-rank trace relation). *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \quad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20}p_{02} = -(x_1y_2 - x_2y_1)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

56.3 STABLE PASSAGE

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

BOOK **XI**

Global holomorphic symplectic descent

Fedosov descent

57.1 THE WEYL ALGEBROID

Let (S, ω) be a holomorphic symplectic surface. The bundle of formal Weyl algebras has fibre the completed Weyl algebra of a formal symplectic plane. A holomorphic symplectic connection and a Fedosov one-form define a flat connection

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -]$$

whose flat sections form a deformation-quantization algebroid $\mathcal{W}_{S, \hbar}$.

Theorem 57.1 (Globalized formality on a holomorphic symplectic surface). *Fix a stable formality morphism and a compatible homotopy- E_2 refinement. There is a sheafwise homotopy- E_2 quasi-isomorphism*

$$\Phi_S : (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -]) \xrightarrow{\sim} C_{\mathrm{HH}}^\bullet(\mathcal{W}_{S, \hbar}, \mathcal{W}_{S, \hbar}),$$

where $\pi = \omega^{-1}$. A change of holomorphic symplectic connection or Fedosov one-form gives a gauge-equivalent resolution and a homotopic quasi-isomorphism. Holomorphic symplectomorphisms act equivariantly on the resulting homotopy class.

Proof. Fedosov resolutions replace both sheaves by acyclic sheaves of formal fibrewise polyvectors and polydifferential operators. The fixed covariant formality morphism acts fibrewise and intertwines the Fedosov differentials. A change of connection or Fedosov one-form is a gauge transformation in the pronilpotent resolution and therefore gives a homotopy. $\square$

Corollary 57.2 (Global mixed-HT centre). *The deformation-quantization category $\mathrm{Perf}(\mathcal{W}_{S, \hbar})$ has Morita centre*

$$Z_{\mathrm{Mor}}(\mathrm{Perf}(\mathcal{W}_{S, \hbar})) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[[\hbar]], [\hbar\pi, -]).$$

For a $K3$ surface this gives a global open-closed equivalence for the Hamiltonian BF sector.

Proof. The Morita centre of perfect modules over an algebroid is the derived global section complex of its Hochschild cochains. Apply the globalized formality equivalence and then derived global sections. A $K3$ surface is holomorphic symplectic, so the stated Hamiltonian sector is the resulting Poisson-BV complex. $\square$

Factorization and descent

58.1 THE ALGEBRAIC MIXED-HT REALIZATION

The sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -])$$

is a sheaf of complete homotopy- E_2 algebras. Dunn additivity gives its locally constant factorization realization in two oriented topological directions. The holomorphic sheaf variable retains the dependence on S .

Theorem 58.1 (Algebraic factorization realization). *The homotopy- E_2 sheaf $\mathcal{P}_S$ determines a complete mixed holomorphic-topological factorization algebra*

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \quad \text{on} \quad \mathbb{R}_{\mathrm{top}}^2 \times S.$$

Its universal boundary category is $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$, and the bulk-boundary morphism is a local equivalence

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \xrightarrow{\sim} Z_{\mathrm{Mor}}^{\mathrm{der}}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})).$$

On every formal Darboux chart the map is the completed Kontsevich–Duflo equivalence.

Proof. The factorization envelope of a complete homotopy- E_2 algebra is locally constant on oriented two-disks. Sheafwise application to $\mathcal{P}_S$ gives the first assertion. The Fedosov algebroid is the boundary quantization of the same linear Poisson data. Globalized formality identifies its Hochschild cochains with $\mathcal{P}_S$, and Morita invariance identifies Hochschild cochains with the centre of the perfect-module category. The construction is local on S and therefore respects factorization descent. $\square$

58.2 BV REALIZATION

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \left\langle B, (\mathrm{d} + \bar{\partial})A + \frac{1}{2}[A, A] \right\rangle.$$

Its classical observables map to $\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}}$ by the Chevalley–Poisson identification. A compatible gauge fixing, extension of graph weights to compactified configurations, and solution of the quantum master equation produce a perturbative factorization realization of the same algebraic bulk-boundary map. The finite jet zero modes carry the exact quantum measure constructed above.

BOOK **XII**

Matrix branes and trace geometry

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \quad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$\mathcal{A}_N = C_{\text{CE}}^*(\mathfrak{gl}_N, \text{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])).$$

Proposition 59.1 (Representation of the universal brane). *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \text{Der}(\mathcal{A}_N), \quad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U}_\hbar(\mathfrak{h}) \longrightarrow \text{End}(\mathcal{A}_N)[[\hbar]], \quad f \longmapsto \hbar \delta_f.$$

Thus the N -brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f , cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial_Y^{\text{cyc}} f(X, Y), \quad \delta_f(Y) = \partial_X^{\text{cyc}} f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient $\mathcal{A}_N$. The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f, g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar \delta_f$ satisfies

$$[\hbar \delta_f, \hbar \delta_g] = \hbar^2 \delta_{\{f, g\}} = \hbar (\hbar \delta_{\{f, g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N -dimensional Chan–Paton representation. $\square$

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y \rangle, \quad F_{\text{cyc}} = F/[F, F].$$

For a cyclic word f , define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\partial_f(x) = -\partial_y^{\text{cyc}} f, \quad \partial_f(y) = \partial_x^{\text{cyc}} f.$$

Theorem 60.1 (Necklace Lie algebra). *The bracket*

$$\{f, g\}_{\text{neck}} = [\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g]_{\text{cyc}}$$

makes $F_{\text{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \partial_f$ is a Lie morphism. It preserves the moment map:

$$\partial_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\text{cyc}} f] + [y, \partial_y^{\text{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\partial_f, \partial_g] = \partial_{\{f, g\}_{\text{neck}}}$. $\square$

Theorem 60.2 (Trace moment map). *For*

$$J_N(f) = \text{Tr } f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N(\{f, g\}_{\text{neck}}),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\text{Tr}(dX \wedge dY)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. $\square$

Stable traces

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i) . Define

$$p_{ab}^{(N)} = \sum_{i=1}^N x_i^a y_i^b.$$

Theorem 61.1 (Stable multisymmetric Poisson algebra). *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[[p_{ab} \mid a, b \geq 0, a + b > 0]].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc) p_{a+c-1, b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab} . The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. $\square$

Remark 61.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \quad v = y_1 - y_2, \quad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 62.1 (The $\mathcal{A}_1$ Kirillov–Kostant cone). *With $\{u, v\} = 2$,*

$$\{H, E\} = 4E, \quad \{H, F\} = -4F, \quad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ Lie–Poisson bracket on the nilpotent cone.

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

BOOK XIII

Wheel corrections and exact quantum
structure

The Duflo wheel

For $\mathfrak{sl}_2$ take generators

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \quad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 63.1 (The zero-point shift). The Duflo map gives

$$\text{Duf}(q) = \Omega + \frac{1}{2}.$$

On the irreducible highest-weight- n module,

$$\left(\Omega + \frac{1}{2}\right)\Big|_{V_n} = \frac{(n+1)^2}{2} \text{id}.$$

Proof. For $\mathfrak{sl}_2$, the second wheel in $j^{1/2}(\partial)$ is the constant shift by $\rho^2 = 1/2$ in the stated normalization. The Casimir eigenvalue is $n(n+2)/2$. $\square$

The shift is the algebraic one-loop term in the open–closed map. Higher wheels assemble into $j^{1/2}$.

The full local theorem

Let

$$\mathcal{A}_{\text{bulk}}^{\text{alg}} = \text{Fact}_{\mathbb{E}_2} \left(C_{\text{CE,cont}}^* (T^*[-1]\mathfrak{h})[[\hbar]] \right)$$

be the factorization envelope in the two topological directions, with the formal holomorphic coefficient retained at the Darboux point.

Theorem 64.1 (Universal mixed holomorphic–topological Deligne theorem). *Let*

$$C_{\partial} = \text{Perf}(\widehat{U}_{\hbar}(\mathfrak{h})).$$

The formality morphism induces a quasi-isomorphism of complete homotopy- $\mathbb{E}_2$ factorization algebras

$$\mathcal{A}_{\text{bulk}}^{\text{alg}} \xrightarrow{\sim} Z_{\text{Mor}}^{\text{der}}(C_{\partial}).$$

At the fixed formal Darboux insertion point this map is Φ_{∞} . Its restriction to every finite matrix brane is the Rees-scaled necklace representation followed by the Hochschild action.

Proof. The continuous formality theorem gives the complete homotopy- $\mathbb{E}_2$ equivalence at the Darboux point. Factorization envelopes preserve equivalences between complete $\mathbb{E}_2$ algebras. Morita invariance identifies the centre of the perfect-module category with continuous Hochschild cochains. The matrix representation $f \mapsto \hbar \delta_f$ is a module map, so restriction is functorial. $\square$

Corollary 64.2 (Global K_3 Hamiltonian sector). *Let S be a K_3 surface with holomorphic symplectic form. The Fedosov category $\text{Perf}(\mathcal{W}_{S,\hbar})$ has derived centre equivalent to the global algebraic Hamiltonian bulk $\mathcal{A}_{S,\text{bulk}}^{\text{alg}}$. The equivalence carries the Duflo wheel, cyclic chains, and the selected homotopy Swiss-cheese action.*

Proof. Apply globalized formality and the algebraic factorization realization. Fedosov descent transports the fixed graph operations, including wheel and brace terms, while Morita invariance transports cyclic chains and the centre action. $\square$

The harmonic square

The four complexes

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{h}), \quad C_{\text{HH}}^*(\widehat{U}\mathfrak{h}, \widehat{U}\mathfrak{h}), \quad O(\text{Rep}_\infty B)^{GL_\infty}, \quad H_*^{\text{CE}}(\mathfrak{gl}_\infty B)$$

are connected by explicit maps. The first two are equivalent by formality. The stable trace map reaches the third. Loday–Quillen–Tsygan identifies the primitives of the fourth with shifted cyclic homology. Their common input is the cyclic Hamiltonian class, and each construction retains a different amount of structure.

Theorem 65.1 (Four realizations of cyclic primitives). *For a unital algebra B and a Hamiltonian Lie map $\mathfrak{h} \rightarrow HC_0(B)$, the following maps are functorial:*

$$\begin{aligned} C_{\text{CE}}^*(T^*[-1]\mathfrak{h}) &\xrightarrow{\Phi_\infty} C_{\text{HH}}^*(\widehat{U}\mathfrak{h}, \widehat{U}\mathfrak{h}), \\ \mathfrak{h} &\longrightarrow HC_0(B) \longrightarrow O(\text{Rep}_N B)^{GL_N}, \\ \widehat{\text{Sym}} HC_0(B) &\longrightarrow O(\text{Rep}_\infty B)^{GL_\infty}, \\ \text{Sym}(HC_{*-1}(B)) &\xrightarrow{\sim} H_*^{\text{CE}}(\mathfrak{gl}_\infty B). \end{aligned}$$

The first map preserves the homotopy Gerstenhaber structure, the second preserves the necklace Poisson bracket, the third preserves multiplication, and the fourth preserves the Hopf structure.

Proof. The first arrow is the inverse limit of the finite-jet Kontsevich–Duflo formality maps. The graph formulas preserve braces and the Schouten bracket up to their canonical coherent homotopies, and compatibility with jet quotients passes this structure to the complete limit.

A cyclic class $[b] \in HC_0(B) = B/[B, B]$ defines the invariant function

$$\text{Tr}_N([b])(\rho) = \text{Tr}(\rho(b))$$

on the representation scheme. Cyclicity makes this function independent of the representative of $[b]$. The double-Poisson or Hamiltonian hypothesis on $\mathfrak{h} \rightarrow HC_0(B)$ converts the necklace bracket into the Poisson bracket of trace functions, which proves the second assertion.

Products of single traces define compatible multisymmetric functions as N grows. The universal property of the completed symmetric algebra therefore gives the third map. Its multiplicativity is built into the passage from a cyclic primitive to its multi-trace exponential assembly.

The stable Lie algebra $\mathfrak{gl}_\infty B$ has block-sum coproduct. The Loday–Quillen–Tsygan theorem identifies its primitive homology with $HC_{*-1}(B)$ and identifies the full stable homology with the free graded-commutative

Hopf algebra on those primitives. This gives the fourth isomorphism and its Hopf compatibility. Naturality of formality, trace evaluation, stabilization, and the Loday–Quillen–Tsygan map proves functoriality of the four constructions. $\square$

Synthesis

The finite jets supply exact BV theories. Their boundary quantizations supply universal branes. Linear Poisson formality identifies bulk and centre. Universal graph formulas pass to the formal limit. Fedosov descent globalizes the equivalence. Matrix substitution produces finite Chan–Paton branes, necklace Hamiltonians, and stable traces. The Duflo wheels measure the quantum difference between symmetric and enveloping coordinates.

The complete chain is

$$\begin{array}{ccc}
 C_{\text{CE}}^*(T^*[-1]\mathfrak{h}) & \xrightarrow[\sim]{\Phi_\infty} & C_{\text{HH}}^*(\widehat{U}_\hbar\mathfrak{h}, \widehat{U}_\hbar\mathfrak{h}) \\
 \text{representation} \downarrow & & \downarrow \text{module action} \\
 \mathcal{O}(\text{Rep}_N)^{GL_N} & \xrightarrow{\text{central action}} & C_{\text{HH}}^*(\mathcal{A}_N, \mathcal{A}_N).
 \end{array}$$

The upper arrow is the universal open–closed equivalence. The lower row is its finite-brane image. This order retains the centre before taking traces and retains the field theory before taking its character.

BOOK **XIV**

**Technical appendices to the preceding
book**

Signs in the shifted cotangent complex

Let e_a have degree zero and let $e^a[-1]$ have degree -1 . The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C_{\text{CE}}^*(T^*[-1]\mathbf{I}) = \text{Sym}(\mathbf{I}^\vee[-1] \oplus \mathbf{I}).$$

The degree-zero polynomial factor is $\text{Sym } \mathbf{I} = \mathcal{O}(\mathbf{I}^*)$. The differential on a polynomial p is

$$(d_{\text{CE}} p)(x) = \text{ad}_x p,$$

which is the linear Poisson differential.

A direct Jacobi check for Hamiltonian jets

For monomials

$$f = z_1^a z_2^b, \quad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a, b), (c, d)) \det((a + c - 1, b + d - 1), (e, f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\text{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

Bibliographic note

The finite and global formality maps used above are the linear and covariant forms of Kontsevich formality and its Fedosov globalization [14, 27, 53]. The Hochschild model of enveloping algebras is the Cartan–Eilenberg comparison. The all-arity centre action is the Deligne–Tamarkin brace action [54, 61]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book entrance: The Igusa Square Root

The Jacobi seed, Borcherds denominator, root superalgebra, local-wrapped currents, quantum group, Hall model, and Fredholm realization.

BOOK **XV**

The automorphic determinant

The protected Jacobi input

70.1 THE K_3 ELLIPTIC GENUS

Let S be a K_3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K_3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^4 \frac{\mathfrak{I}_a(\tau, z)^2}{\mathfrak{I}_a(\tau, 0)^2} \in J_{0,1}^w$$

[30]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 70.1 (Exact low Fourier coefficients). *The first four q -rows are*

$$\begin{aligned} \phi_{0,1} = & r^{-1} + 10 + r \\ & + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2) \\ & + q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) \\ & + q^3(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \\ & \quad - 2752r + 808r^2 - 64r^3) + O(q^4). \end{aligned}$$

In particular,

$$f(0, 0) = 10, \quad \frac{f(0, 0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x . The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 100. $\square$

70.2 THE VIRTUAL HALF-INDEX

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\text{sdim } U_{n,\ell} = f(n, \ell).$$

This choice represents the coefficient in $K_o(\text{sVect}_{\text{fd}})$. The determinant construction depends on the superdimension.

Lemma 70.2 (Fock character and Berezinian). *Let $V = \bigoplus_{\gamma} V_{\gamma}$ be a locally finite graded super vector space. Then*

$$\text{sch}(\text{Sym}(V_{\bar{0}}) \otimes \bigwedge (V_{\bar{1}})) = \prod_{\gamma} (1 - x^{\gamma})^{-\text{sdim } V_{\gamma}},$$

and

$$\text{sdet}_V(1 - x) = \prod_{\gamma} (1 - x^{\gamma})^{\text{sdim } V_{\gamma}}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^{\gamma})^{-1}$. An odd one-dimensional summand contributes $1 - x^{\gamma}$. Multiplication over a homogeneous basis proves both identities. $\square$

The Borcherds product

71.1 THE EFFECTIVE CHAMBER

Define

$$\Gamma_{\text{eff}} = \{(n, \ell, m) \in \mathbb{Z}^3 : m > 0, n \geq 0\} \\ \cup \{(n, \ell, 0) : n > 0\} \cup \{(0, \ell, 0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n, \ell, m)$ put

$$V_\gamma = U_{nm, \ell}, \quad \text{sdim } V_\gamma = f(nm, \ell).$$

Definition 71.1 (Monic virtual determinant). The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \text{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^\ell s^m)^{f(nm, \ell)}.$$

Theorem 71.2 (Borcherds–Gritsenko–Nikulin product). *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5 . The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0, 0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [8, 9, 41, 43]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. $\square$

71.2 THE THETA CONSTANT NORMALIZATION

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 71.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	scalar	monomial
(0, 0)	4	1	1
(1, 0)	2	2	$q^{1/8}$
(0, 1)	2	2	$s^{1/8}$
(1, 1)	2	2	$q^{1/8} r^{1/4} s^{1/8}$.

Multiplication gives

$$1^4 2^2 2^2 2^2 q^{1/2} r^{1/2} s^{1/2} = 64 q^{1/2} r^{1/2} s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0, 0)$ the zero lattice vector contributes one. For each nonzero a , the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. $\square$

Proposition 71.4 (Boundary factor of the Borchers product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0, \ell \in \mathbb{Z}} (1 - q^n r^\ell)^{f(0, \ell)} \prod_{\ell < 0} (1 - r^\ell)^{f(0, \ell)} = (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1}) (1 - q^n)^{10} (1 - q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2} r^{1/2} (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1}) (1 - q^n)^{10} (1 - q^n r) = \eta(\tau)^9 \mathfrak{g}_{\text{II}}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\mathfrak{g}_{\text{II}}(\tau, z) = -q^{1/8} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r) (1 - q^n r^{-1}) (1 - q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0, -1) = 1$, $f(0, 0) = 10$, and $f(0, 1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ give the second, including its leading sign and monomial. $\square$

Corollary 71.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5^2 = 2^{-12} \Delta_5^{\theta^2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. $\square$

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = o\} / \mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 72.1 (Diagonal divisor). *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$ -orbit of H_1 , with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [17, 47, 48]. $\square$

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

72.1 AUTOMORPHIC SECTION, ORIENTATION LINE, AND PFAFFIAN OPERATOR

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\begin{aligned} &\text{automorphic section} \longrightarrow \text{determinant-line identification} \\ &\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.} \end{aligned}$$

Proposition 72.2 (Logical separation of the four levels). *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. $\square$

The title names the monic automorphic section D_5 , characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

BOOK XVI

The denominator algebra

The Lorentzian root lattice

Let

$$L = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \quad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus 2\mathbb{Z}f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\text{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \quad \alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2(2nm' + 2n'm - \ell\ell'),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

73.1 REAL SIMPLE ROOTS AND WEYL VECTOR

Put

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3.$$

Proposition 73.1 (Type-II Cartan matrix). *The Gram matrix of the real simple roots is*

$$((\delta_i, \delta_j))_{i,j=1}^3 = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. $\square$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \quad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$.

The Gritsenko–Nikulin Lie superalgebra

The Borchers product determines a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ with real simple roots $\delta_1, \delta_2, \delta_3$ and an imaginary correction system determined by $\phi_{0,1}$.

Theorem 74.1 (Root supermultiplicities and denominator). *On the active positive support,*

$$\text{smult } \mathfrak{g}_{\Delta_5, \alpha(n, \ell, m)} = f(nm, \ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6} \Delta_5^\theta(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borchers product has exponent $f(nm, \ell)$. The resulting product is the doubled-variable monic Igusa form [41–43]. $\square$

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_s,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_s}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H} \mathfrak{n}_{\Delta_s,+} = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta_s, \alpha}, \quad \mathfrak{n}_{\Delta_s,+}^{(H)} = \mathfrak{n}_{\Delta_s,+} / F^{>H} \mathfrak{n}_{\Delta_s,+}.$$

Height additivity makes $F^{>H} \mathfrak{n}_{\Delta_s,+}$ an ideal. Properness and local finiteness make $\mathfrak{n}_{\Delta_s,+}^{(H)}$ finite-dimensional and nilpotent.

Theorem 75.1 (Height-complete Bar–Chevalley comparison). *For every H , antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\mathrm{Bar}(U(\mathfrak{n}_{\Delta_s,+}^{(H)})) \simeq C_*(\mathfrak{n}_{\Delta_s,+}^{(H)}).$$

The quotient maps preserve these comparisons and induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^{\wedge}(U(\mathfrak{n}_{\Delta_s,+})) := \varprojlim_H \mathrm{Bar}(U(\mathfrak{n}_{\Delta_s,+}^{(H)})) \simeq C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_s,+}) := \varprojlim_H C_*(\mathfrak{n}_{\Delta_s,+}^{(H)}).$$

Its root-graded Euler supercharacter is

$$\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_s,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\mathrm{sdim} \mathfrak{g}_{\Delta_s, \alpha}}.$$

Proof. For each H , the antisymmetrization map sends $\mathrm{Sym}(\mathfrak{n}_{\Delta_s,+}^{(H)}[1])$ to the normalized bar of $U(\mathfrak{n}_{\Delta_s,+}^{(H)})$. The PBW filtrations have the same associated-graded Koszul resolution of $\mathrm{Sym}(\mathfrak{n}_{\Delta_s,+}^{(H)})$, so antisymmetrization is a filtered quasi-isomorphism. Naturality gives a morphism of inverse systems. Every fixed root degree is visible in all sufficiently large height quotients and is unchanged thereafter; hence the inverse-limit map is a quasi-isomorphism coefficientwise. An even root vector becomes odd after the Chevalley shift and contributes $1 - e^{-\alpha}$; an odd root vector becomes even and contributes its reciprocal. Multiplication over the finite-dimensional root spaces gives the product. $\square$

Let $\mathbb{T}_{L_I}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \quad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

Corollary 75.2 (Bar denominator identity). *On the formal root torus,*

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\text{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [41, 43]. Hence

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. $\square$

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

Finite-height currents

Retain the finite-height quotients $\mathfrak{n}_{\Delta_s,+}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}(\mathfrak{n}_{\Delta_s,+}^{(H)}) = \mathbb{k}[\partial] \otimes \mathfrak{n}_{\Delta_s,+}^{(H)}, \quad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\left(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+}^{(H)})\right).$$

Theorem 76.1 (Height-complete current envelope). *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_s} = \varprojlim_H V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\mathrm{sch} \widehat{V}_{\Delta_s} = \prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta_s, \alpha}}.$$

Moreover,

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(V_H) \simeq \mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}\left(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+}^{(H)})\right),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. $\square$

76.1 CLASSICAL LOCAL THEORY AND EXACT QUANTUM ZERO MODES

For every H , the finite-dimensional nilpotent Lie superalgebra $\mathfrak{n}_{\Delta_s,+}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_s,+}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 76.2 (Finite-height BF realization). *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the $\mathfrak{n}_{\Delta,+}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U(\mathfrak{n}_{\Delta,+}^{(H)}), \quad \mathbf{I} \otimes_{U(\mathfrak{n}_{\Delta,+}^{(H)})} \mathbf{I} \simeq C_*(\mathfrak{n}_{\Delta,+}^{(H)}), \quad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta,+}), \quad C_{*,h}^\wedge(\mathfrak{n}_{\Delta,+}), \quad \widehat{V}_{\Delta}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U(\mathfrak{n}_{\Delta,+}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*(\mathfrak{n}_{\Delta,+}^{(H)})$. Holomorphic descendants generate $U^{\text{ch}}(\text{Cur}_E \mathfrak{n}_{\Delta,+}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. $\square$

BOOK XVII

The compact scalar character

Reduced curve counting on $K_3 \times E$

Let S be a nonsingular projective K_3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 77.1 (Primitive Igusa cusp-form theorem). *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [64]; their normalization is the monic Borcherds product χ_{10} . The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. $\square$

The theorem is a scalar trace identity and fixes every primitive reduced invariant. The Borcherds root algebra supplies the fixed algebraic target whose PBW character is the same product.

Corollary 77.2 (Exact character comparison). *Let $j : \mathbb{H}_2 \rightarrow \mathbb{H}_2$ be the half-coordinate map $j(Z) = Z/2$, and let*

$$\text{Fock}(\mathfrak{n}_{\Delta_5,+}) = \text{Sym}(\mathfrak{n}_{\Delta_5,+_{\bar{0}}}) \otimes \bigwedge (\mathfrak{n}_{\Delta_5,+_{\bar{1}}}).$$

Then

$$\mathcal{Z}_{S \times E}^{\text{red}} = -j^* \left(e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) \right) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_5, \alpha}}.$$

The denominator theorem gives $\text{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\text{sch Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) = \text{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by j sends $\text{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10} . The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. $\square$

The equality is an identity of completed protected characters. The universal root algebra supplies the primitive bracket, and reduced curve counting supplies the geometric trace.

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \tilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\text{phys}} = \Lambda \oplus \Lambda, \quad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\text{ind}}.$$

Proposition 78.1 (Polarization identity). *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = (Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B . □

78.1 THE POLARIZED CENTRAL EXTENSION

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\text{ind}} \longrightarrow \tilde{\Gamma}_{\Pi} \longrightarrow \Gamma_{\text{phys}} \longrightarrow 0, \quad \tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\text{phys}}} A_c$ be a charge-graded associative algebra, and let $\mathbb{k}[\Gamma_{\text{ind}}]$ have monomials u^λ . Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \mathbb{k}[\Gamma_{\text{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \tilde{\Gamma}_\Pi$.

Theorem 78.2 (Graph-multiplicative polarization). *The product $\star$ is associative and $\tilde{\Gamma}_\Pi$ -graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \quad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c, d) + B(c + d, e) - B(d, e) - B(c, d + e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. $\square$

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Point-local and wrapped sectors

Let $p_E : S \times E \rightarrow E$ be projection.

Lemma 79.1 (Positive elliptic degree is wrapped). *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E . Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C , is all of E . $\square$

Projection-finite objects define point-supported sectors on $\text{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 79.2 (The local–wrapped factorization operad). Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$l, \quad w_m \quad (m \geq 1)$$

by the following operation objects.

(i) Operations with local output are

$$\text{Hyb}_E(l^{\otimes k}; l) = \text{Ch}_E(k).$$

(ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\text{Hyb}_E(l^{\otimes k}, w_{m_1}, \dots, w_{m_r}; w_m) = \text{Ch}_E(k) \otimes \text{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 79.3 (Hybrid local–wrapped algebra). A hybrid local–wrapped algebra over E is an algebra over Hyb_E . It consists of a factorization algebra $\mathbb{F}_{\text{loc}}$, a height- or charge-complete family $\mathcal{W} = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1, \dots, m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1 + \cdots + m_r},$$

and factorization-compatible actions of the local operation objects $\text{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E -algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

BOOK XVIII

The Borcherds algebra and its quantum
group

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m . Set

$$\mathfrak{n}_{\Delta_s, +_{\text{loc}}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) = 0}} \mathfrak{g}_{\Delta_s, \alpha}, \quad \mathfrak{n}_{\Delta_s, +_{\text{wr}}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) > 0}} \mathfrak{g}_{\Delta_s, \alpha}.$$

Proposition 80.1 (Semidirect local–wrapped decomposition). *The space $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_s, +} = \mathfrak{n}_{\Delta_s, +_{\text{loc}}} \ltimes \mathfrak{n}_{\Delta_s, +_{\text{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. $\square$

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_s} = \widehat{U}_h(\mathfrak{n}_{\Delta_s, +}), \quad \mathbb{V}_{\Delta_s, E} := \widehat{V}_{\Delta_s} = \varprojlim_H U^{\text{ch}}(\text{Cur}_E(\mathfrak{n}_{\Delta_s, +}^{(H)})).$$

The adjoint action of $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ on $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 80.2 (Current-generated local–wrapped operad). The colored operad $\text{Hyb}_{E, \Delta}^{\text{cur}}$ is generated by the chiral current operations of $\text{Cur}_E(\mathfrak{n}_{\Delta_s, +_{\text{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 80.3 (Local–wrapped Borchers decomposition). *Let $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_s, +} = \mathfrak{n}_{\Delta_s, +_{\text{loc}}} \ltimes \mathfrak{n}_{\Delta_s, +_{\text{wr}}},$$

where $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_s, +}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_s,+}) \simeq \text{Tot } C_*(\mathfrak{n}_{\Delta_s,+\text{loc}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_s,+\text{wr}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_s,+}_h} = \varprojlim_H \mathfrak{n}_{\Delta_s,+}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_s(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. $\square$

Remark 80.4. The theorem constructs a Borchers root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borchers Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee, \text{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted $\mathfrak{g}_{\Delta, \text{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 81.1 (Height-admissible Lie superbialgebra). A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 81.2 (Height admissibility). *The paired Gritsenko–Nikulin Borchers Lie superbialgebra $\mathfrak{g}_{\Delta, \text{pair}}$ is height-admissible.*

Proof. The Borchers root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. $\square$

Theorem 81.3 (Quantized Borchers superalgebra). *Choose the symmetrizable Borchers–Cartan superdatum and multiplicity index set that present $\mathfrak{g}_{\Delta}$. Hong’s construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta, \text{pair}})$$

over $\mathbb{Q}(q)$, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition

$$U_q(\mathfrak{g}_{\Delta, \text{pair}}) \cong U_q^-(\mathfrak{g}_{\Delta, \text{pair}}) \otimes U_q^0(\mathfrak{g}_{\Delta, \text{pair}}) \otimes U_q^+(\mathfrak{g}_{\Delta, \text{pair}}).$$

Its PBW root spaces have the same graded ranks as the classical Borchers superalgebra. The canonical pairing of the two Borel halves yields a universal R -matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta, \text{pair}}) \widehat{\otimes} U_q^-(\mathfrak{g}_{\Delta, \text{pair}}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$.

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borchers–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borchers superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R -matrix are established by Hong [46]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. $\square$

Remark 81.4. The quantum group is attached to the Borchers generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

A holomorphic–topological BF realization

For every height bound H , let $\mathfrak{n}_{\Delta,+}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{\circ,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta,+}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 82.1 (Classical local and exact zero-mode realization). *For every H :*

- (i) S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;
- (ii) harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;
- (iii) after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U(\mathfrak{n}_{\Delta,+}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes_{U(\mathfrak{n}_{\Delta,+}^{(H)})}^{\mathbb{L}} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta,+}^{(H)})$, and the holomorphic boundary current envelope is V_H .

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta,+})$, interval complex $C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta,+})$, and current envelope $\widehat{V}_{\Delta_s}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. $\square$

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha >_0 \\ h(\alpha) \leq H}} \mathfrak{g}_{\Delta_s \alpha}$$

and define

$$\mathcal{D}_H|_{\mathfrak{g}_{\Delta_s \alpha}} = (1 - e^{-\alpha}) \text{id}.$$

Theorem 83.1 (Universal determinant section). *The finite-height Berezinian is*

$$\text{Ber}(\mathcal{D}_H) = \prod_{\substack{\alpha >_0 \\ h(\alpha) \leq H}} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_s \alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho} \text{Ber}(\mathcal{D}_\infty) = D_5(2Z).$$

After pullback by $j(Z) = Z/2$,

$$j^*(e^{-\rho} \text{Ber}(\mathcal{D}_\infty)) = D_5(Z), \quad D_5(Z)^2 = \chi_{\text{io}}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. $\square$

83.1 TRACE-CLASS REALIZATION ON THE ROOT HILBERT SUPERSPACE

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus_{\alpha >_0} \mathfrak{g}_{\Delta_s \alpha}}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\text{abs}}$ be the subchamber on which

$$\sum_{\alpha >_0} \dim \mathfrak{g}_{\Delta_s \alpha} \left| e^{-\alpha(Z)} \right| < \infty.$$

For $Z \in \mathfrak{C}_{\text{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta_5, \alpha}} = e^{-\alpha(Z)} \text{id}, \quad D_Z = \mathbf{1} - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z, \bar{\epsilon}} = \begin{pmatrix} \circ & D_{Z, \bar{\epsilon}} \\ -D_{Z, \bar{\epsilon}}^t & \circ \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = \circ$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\text{sPf}_F(\mathbb{A}_Z) = \frac{\text{Pf}_F(\mathbb{A}_{Z, \bar{\delta}})}{\text{Pf}_F(\mathbb{A}_{Z, \bar{\mathbf{i}}})}.$$

Theorem 83.2 (Polarized Fredholm realization). *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_5, \alpha}} = e^{-\alpha(Z)} \text{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = \mathbf{1} - K_Z$,*

$$\text{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{\delta}})}{\det_F(D_Z|_{\bar{\mathbf{i}}})} = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{\text{sdim } \mathfrak{g}_{\Delta_5, \alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_{\Delta} \oplus \mathcal{H}_{\Delta}^$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z, \bar{\delta}}$ and $\mathbb{A}_{Z, \bar{\mathbf{i}}}$ have an oriented Fredholm super-Pfaffian*

$$\text{sPf}_F(\mathbb{A}_Z) = \frac{\text{Pf}_F(\mathbb{A}_{Z, \bar{\delta}})}{\text{Pf}_F(\mathbb{A}_{Z, \bar{\mathbf{i}}})} = \text{sdet}_F(D_Z).$$

Hence

$$e^{-\rho} \text{sdet}_F(D_Z) = D_5(2Z).$$

Proof. The preceding convergence theorem gives the trace-class criterion for the diagonal operator. The Fredholm determinant of $\mathbf{1} - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z . Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. $\square$

Remark 83.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This theorem constructs the square-root operator on the universal Borchers root Hilbert superspace. A geometric realization is a determinant-line intertwiner from a protected geometric Hilbert complex to this universal operator family.

What the automorphic character determines

Theorem 84.1 (Denominator data and reconstruction data). *The Borchers product determines:*

- (i) *the signed root multiplicity $\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ in every active root degree;*
- (ii) *the Weyl vector and denominator character;*
- (iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borchers–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. $\square$

Recognition Theorem 84.2 (Complete primitive recognition). *Let $\mathfrak{p}$ be a positive graded Lie superalgebra and let $\Psi : \mathfrak{n}_{\Delta, +} \rightarrow \mathfrak{p}$ be a graded Lie map. The map is an isomorphism if either:*

- (i) *the parity-refined Hilbert series of $U(\mathfrak{n}_{\Delta, +})$ and $U(\mathfrak{p})$ agree; or*
- (ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. $\square$

BOOK **XIX**

**The universal Borchers Hall and
three-Calabi–Yau models**

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borchers superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{i \in I_\Delta : h(i) \leq H\}.$$

The restriction of the invariant form gives a finite symmetric Borchers–Cartan matrix

$$\begin{aligned} A_H &= (a_{ij})_{i,j \in I_H}, \\ a_{ii} &= 2 && \text{for real } i, \\ a_{ii} &\leq 0 && \text{for imaginary } i, \\ a_{ij} &\leq 0 && \text{for } i \neq j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 85.1 (The Igusa Borchers quiver). Order I_H . For $i < j$ place $-a_{ij}$ arrows from i to j . At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 85.2 (Euler form). *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H :*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \neq j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij} . At i the value is $2 - 2g_i = a_{ii}$. $\square$

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3,$$

with Gram matrix

$$A_{\text{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 86.1 (Real-root Hall seed). *The symmetrized Euler form of Q_{re} is A_{re} . The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re} .*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2 . The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. $\square$

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

86.1 THE FIRST GINZBURG DIFFERENTIAL

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \rightarrow j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \rightarrow i$ of degree -1 and loops t_i of degree -2 . The canonical three-Calabi–Yau completion has

$$da_{ij}^{(r)} = 0, \quad d(a_{ij}^{(r)})^* = 0,$$

and

$$dt_i = \sum_{a:s(a)=i} aa^* - \sum_{a:t(a)=i} a^*a.$$

The identity $d^2 = 0$ follows immediately. Formula (86.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 86.2 (Vertex one). At vertex 1,

$$dt_1 = \sum_{r=1}^2 \left(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \right).$$

At vertex 3 the same four terms occur with reverse order and negative sign. Summing the three equations gives zero, as required by the trace of the moment map.

Proof. Vertex 1 is the source of the four displayed arrows and the target of none, so the Ginzburg moment-map formula contains exactly the four positive terms. Vertex 3 is their terminal counterpart. Every arrow–reverse-arrow pair occurs once with each sign in the sum over vertices. $\square$

Hall realization of the positive algebra

Let $\mathcal{A}_H = \text{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N \rangle} \sum_{[L]} \frac{|\{N' \subset L : N' \cong N, L/N' \cong M\}|}{|\text{Aut } M| |\text{Aut } N|} [L], \quad v^2 = q.$$

Theorem 87.1 (Finite-window Hall theorem). *The composition subalgebra generated by the simple representations of $Q_{\Delta,H}$ is the positive quantum generalized Kac–Moody algebra attached to the underlying even Borcherds–Cartan matrix A_H . Semisimple perverse sheaves on the representation varieties give its canonical basis. The semi-derived Hall algebra of two-periodic complexes realizes the corresponding full ordinary quantum generalized Kac–Moody algebra after Cartan reduction.*

Proof. The positive statement is the quiver-with-loops construction of Kang–Schiffmann. Their perverse-sheaf category gives the canonical basis. Lu’s semi-derived Hall theorem realizes the full quantum generalized Kac–Moody algebra from the same quiver with loops; the Cartan reduction identifies the two torus copies [3, 50, 59]. $\square$

87.1 PARITY AND THE QUANTIZED SUPERDATUM

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. Extend it additively to the root lattice. Hong’s construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta,}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta,}^{(H)})$.

Theorem 87.2 (Parallel Hall and supergroup realizations). *The quiver Hall algebra and Hong’s quantum supergroup are two functorial realizations of the same finite Borcherds–Cartan window:*

- (i) *the quiver Hall algebra realizes the underlying ordinary generalized Kac–Moody presentation and its canonical basis;*
- (ii) *Hong’s algebra realizes the parity-refined super presentation and its universal R -matrix;*
- (iii) *their root lattices, Cartan forms, height filtrations, and Weyl denominators agree after applying the parity supertrace.*

Proof. The first statement is the finite-window Hall theorem. The second is Hong’s quantized Borcherds-superalgebra theorem. Both presentations use A_H and the same simple-root lattice. The parity map changes ordinary dimensions into superdimensions in the PBW character. The denominator theorem identifies this signed PBW product with the Gritsenko–Nikulin product. $\square$

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its completed three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Equivalently, it is the Ginzburg dg algebra of the doubled quiver with a degree -1 reverse arrow for every arrow and a degree -2 loop at every vertex, with differential determined by the canonical potential.

Theorem 88.1 (Borcherds–Ginzburg tower). *For every H :*

- (i) $\Gamma_{\Delta,H}$ is a complete three-Calabi–Yau dg algebra;
- (ii) the degree-zero inclusion $kQ_{\Delta,H} \rightarrow \Gamma_{\Delta,H}$ is functorial in full-subquiver inclusions;
- (iii) the Hall composition algebra of $kQ_{\Delta,H}$ is the underlying ordinary positive quantum generalized Kac–Moody algebra;
- (iv) the inclusions $H \leq H'$ form a compatible tower of dg categories and Hall algebras.

Proof. The Calabi–Yau assertion is Keller’s theorem on deformed Calabi–Yau completions, in the Ginzburg model [35, 51]. Full-subquiver inclusions extend to reverse arrows, loops, and differentials. The Hall statement is the finite-window Hall theorem. $\square$

Define the proper-height objects

$$\Gamma_{\Delta} = \varinjlim_H \Gamma_{\Delta,H}, \quad \widehat{U}_q(\mathfrak{g}_{\Delta}) = \varprojlim_H U_q(\mathfrak{g}_{\Delta}^{(H)}).$$

The first is locally three-Calabi–Yau: every finite collection of objects lies in a finite three-Calabi–Yau window. The second is Hong’s root-height-complete quantum symmetry.

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the canonical-basis construction. Their equivariant hypercohomology groups give finite Hall state spaces in every dimension vector.

Definition 89.1 (Igusa Hall category). For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_H \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta,\alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta,+}^{(H)})$.

Theorem 89.2 (Hall and protected-state comparison). *The Hall Grothendieck algebra is the finite-window ordinary positive quantum generalized Kac–Moody algebra. The protected one-particle superspace has signed root multiplicities*

$$\mathrm{sdim} \mathcal{H}_{\mathrm{ip},H,\alpha} = \mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\mathrm{sch} \mathrm{Fock}(\mathfrak{n}_{\Delta,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}},$$

and its Weyl-shifted inverse is $D_{\mathfrak{s}}(2Z)$. The Hall canonical basis and the protected superbasis share the simple-root indexing and the Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The Hall assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. $\square$

Coordinate-framed orientation on finite critical charts

Fix H and a dimension vector $\mathbf{d}$. The representation space of the graded doubled quiver underlying $\Gamma_{\Delta,H}$ is affine. Order its vertex, arrow, reverse-arrow, and degree-2 loop coordinates. The corresponding volume line is trivial. The derived moment-map equations form a symmetric three-Calabi–Yau deformation complex whose determinant is the square of this ordered coordinate line.

Theorem 90.1 (Multiplicative coordinate-framed orientation). *A choice of order for the vertex, arrow, reverse-arrow, and loop coordinates trivializes the ambient canonical line and supplies an orientation of every finite-dimensional critical representation chart of $\Gamma_{\Delta,H}$. Under direct sum and the Hall extension correspondence, the orientation line is the tensor product of the two diagonal orientation lines. The construction is associative for three summands and compatible with full-subquiver inclusions.*

Proof. On the extension chart, every arrow matrix is block upper triangular. The tangent complex splits into two diagonal deformation complexes and two off-diagonal complexes exchanged by three-Calabi–Yau duality. The paired off-diagonal part carries the standard determinant square root induced by the three-Calabi–Yau duality pairing. The ordered diagonal coordinate lines therefore multiply. A three-block decomposition gives the same tensor order in both parenthesizations. Full-subquiver inclusion appends an ordered set of coordinates. $\square$

Corollary 90.2 (Oriented critical chart tower). *The finite critical charts carry compatible coordinate-framed vanishing-cycle complexes. Their Grothendieck groups refine the quiver Hall tower by the three-Calabi–Yau critical coefficients.*

Proof. The chosen coordinate framings are compatible with direct sum, extension correspondences, and full-subquiver inclusions in the fixed root order. Vanishing cycles are functorial for these oriented critical charts, so their convolution complexes and Grothendieck groups form the stated tower. $\square$

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \bigsqcup_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta, +} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta, +}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta, +}[m], \mathfrak{n}_{\Delta, +}[n]] \subset \mathfrak{n}_{\Delta, +}[m + n].$$

Construction 91.1 (Hybrid factorization algebra). Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}_{\Delta, E}^{\mathrm{hyb}} = C_*^{\mathrm{ch}}(\mathrm{Cur}_E(\mathfrak{n}_{\Delta, +})).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree- m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 91.2 (Local–wrapped factorization). $\mathcal{F}_{\Delta, E}^{\mathrm{hyb}}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta, +}[0]$. Every degree- m color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta, +})$, and constant modes recover $\mathfrak{n}_{\Delta, +}$.

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. $\square$

This colored Ran space is the exact carrier of the positive elliptic-degree direction. The wrapped label survives when the point support is empty and interacts with local insertions through the same Hall bracket.

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c, d) = \Pi(c + d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}}$$

with multiplication

$$(c, \lambda)(d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

Theorem 92.1 (Graph homomorphism). *The map*

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys} -graded algebra A , the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\tilde{\Gamma}_{\Pi}$ -grading with cocycle $u^{-B(c, d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c) j_{\Pi}(d) = (c + d, \Pi(c) + \Pi(d) + B(c, d)) = j_{\Pi}(c + d).$$

The algebra identity follows immediately. $\square$

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K_3 \times E$ theory enters through a charge functor to $\tilde{\Gamma}_{\Pi}$.

The universal protected field theory

For each H take cotangent BF theory with gauge Lie superalgebra $\mathfrak{n}_{\Delta,+}^{(H)}$ on $\mathbb{R}_{\text{top}}^2 \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \widehat{\otimes} \Omega^{\circ,\bullet}(E) \widehat{\otimes} \left(\mathfrak{n}_{\Delta,+}^{(H)}[1] \oplus (\mathfrak{n}_{\Delta,+}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (d + \bar{\delta})A + \frac{1}{2}[A, A] \rangle.$$

Theorem 93.1 (Classical local theory and exact quantum harmonic sector). *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}_{\text{top}}^2 \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U(\mathfrak{n}_{\Delta,+}^{(H)})$, the two-sided bar complex $C_*(\mathfrak{n}_{\Delta,+}^{(H)})$, and the current envelope on E . The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. $\square$

Absolute convergence of the virtual root operator

For each positive root α , let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ when this integer is nonnegative and purely odd of dimension $-\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus_{\alpha > 0} V_\alpha}.$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta, \alpha}$.

Let $h(\alpha)$ be the proper height. Jacobi-form coefficient estimates give constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = \left| \text{sdim } \mathfrak{g}_{\Delta, \alpha} \right| \leq C_0 \exp(C_1 h(\alpha))$$

for every positive root. The number of Gram triples of height H is polynomial in H .

Theorem 94.1 (Nonempty trace-class chamber). *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha > 0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty.$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re \alpha(Z_0) \geq c(Z_0)h(\alpha)$. Combine this with (94). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp((C_1 - tc(Z_0))H).$$

The series converges when $tc(Z_0) > C_1$. □

Corollary 94.2 (Trace-class root Hamiltonian). *On the conical tail, the diagonal operator*

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borchers product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

The root Dirac operator

Let

$$\mathcal{H}_\Delta = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta, \alpha}}$$

with Hilbert norm chosen root space by root space. In the absolute-convergence chamber define

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}, \quad D_Z = 1 - K_Z.$$

Theorem 95.1 (Oriented Fredholm square root). *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\text{sdet}_F(D_Z) = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^t & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\text{sPf}_F(\mathbb{A}_Z) = \text{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = D_\zeta(2Z).$$

Proof. Absolute convergence of the Borchers product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. $\square$

95.1 AUTOMORPHIC ORIENTATION

The product defines a section of the automorphic line $\mathcal{L}^\zeta \otimes \nu_\zeta$. On the absolute-convergence chamber, the oriented Fredholm Pfaffian is a trivialization of this line. Analytic continuation uses the automorphy factor of D_ζ and therefore glues the chamberwise Fredholm models with transition character ν_ζ .

Corollary 95.2 (Orientation character). *The automorphic orientation line of the root Dirac family is the sign local system ν_ζ . On every real simple reflection it has value -1 .*

Proof. The Fredholm product equals D_ζ on the convergence chamber, and D_ζ transforms with automorphy factor ν_ζ . This transformation law is the gluing rule for the automorphic orientation line. A real simple root occurs with multiplicity one, so reflection reverses its oriented normal line. $\square$

The universal Igusa protected effective theory

Definition 96.1 (Universal Igusa completion). The protected completion selected by the K_3 half-index is the quintuple

$$\mathfrak{R}_\Delta = \left(\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\text{ip}}, \mathcal{F}_{\Delta, E}^{\text{hyb}}, \mathbb{A}_Z \right).$$

Its components are the locally three-Calabi–Yau completion, the ordinary quiver Hall category, the Borchers one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the virtual root Dirac family.

Theorem 96.2 (Igusa protected completion). *The quintuple $\mathfrak{R}_\Delta$ satisfies:*

- (i) *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta, \zeta}$;*
- (ii) *its ordinary Hall presentation and Hong’s parity-refined quantum supergroup share the Borchers–Cartan grading and signed PBW denominator;*
- (iii) *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta, \zeta}$;*
- (iv) *its raw multiparticle supercharacter is P^{-1} , where $e^{-\rho} P = \text{Den}_\Delta$; equivalently $e^\rho \text{sch Fock}(\mathfrak{n}_{\Delta, +}) = \text{Den}_\Delta^{-1}$;*
- (v) *the half-coordinate pullback of the doubled Weyl-normalized character is $-\chi_{10}^{-1}$ in the Oberdieck–Pixton normalization;*
- (vi) *its oriented first-order determinant is Den_Δ and its orientation monodromy is ν_ζ .*

Proof. The first three statements are the Borchers root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = D_\zeta^2$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. $\square$

This theorem constructs the protected effective theory selected by the Igusa denominator. A geometric type-IIA or topological-string model determines a realization functor into this universal completion by its primitive states and extension amplitudes. Equivalence of protected observables is equivalence of the oriented Hall and factorization objects carried by that functor.

Universal property

A presentation-preserving protected realization consists of an oriented root-graded theory $\mathfrak{T}$, a map from each Igusa simple-root space to its primitive states, and chosen comparison maps from the Hall, Ginzburg, current, quantum-double, and determinant functors applied to that simple-root map.

Theorem 97.1 (Universal presentation map). *Every presentation-preserving protected realization carries a unique Lie-superalgebra morphism*

$$\mathfrak{n}_{\Delta,+} \longrightarrow \text{Prim}(\mathfrak{T})$$

extending the simple-root map. Functoriality produces compatible morphisms from the Hall algebra, the Ginzburg completion, the local-wrapped current theory, the completed quantum group, and the virtual root determinant of $\mathfrak{R}_{\Delta}$ to their selected targets in $\mathfrak{T}$. These morphisms are equivalences when the primitive map and the selected functorial comparison maps are equivalences.

Proof. The Borchers generators and relations give the unique primitive Lie morphism. Hall convolution, Calabi–Yau completion, chiral Chevalley chains, quantized doubling, and the diagonal Fredholm construction are functorial in the root presentation. PBW promotes a primitive equivalence to the completed enveloping and Fock objects. The remaining equivalences are exactly the selected comparison maps in the definition of a presentation-preserving realization. $\square$

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [17]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [37–39, 41–43]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borchers–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_o^{(2)}(2)$	3	2
∇_2	$\Gamma_o^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_o^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 98.1 (Eight-row algebraic realization). *Each diagonal-divisor row has a proper-height virtual root superspace, a super-Fock determinant, a colored abelian current theory, and a trace-class Fredholm denominator operator whose character is the corresponding automorphic product. For every row equipped with its reflective denominator presentation, the finite-window Borchers quiver, ordinary Hall category, three-Calabi–Yau completion, and parity-refined quantum supergroup refine this product model and recover the same denominator character.*

Proof. Local finiteness of the product exponents gives the graded superspace and its proper-height completion. The Fock and Fredholm constructions depend only on these signed multiplicities and reproduce the product coefficientwise. A reflective denominator presentation supplies a symmetrizable Borchers superdatum; the quiver, Hall, Ginzburg, current, and Hong constructions then apply to that datum and preserve its PBW denominator. $\square$

Algebraic and physical closure

Theorem 99.1 (Universal Igusa realization theorem). *The following objects are canonically connected by constructed morphisms:*

$$\begin{array}{llll}
 \phi_{o,I} \xrightarrow{\text{Borch}} D_5, & \phi_{o,I} \xrightarrow{\text{root datum}} \mathfrak{g}_{\Delta_5}, & A_{\Delta} \xrightarrow{\text{quiver}} Q_{\Delta,H} \xrightarrow{\text{Hall}} \mathcal{B}_{\Delta}, \\
 \mathcal{B}_{\Delta} \xrightarrow{\Pi_3} \Gamma_{\Delta}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Cur}_E} \mathcal{F}_{\Delta,E}^{\text{hyb}}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}), \\
 \mathfrak{g}_{\Delta_5} \xrightarrow{\text{root spectrum}} \mathbb{A}_Z, & e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = \text{Den}_{\Delta}(Z) = D_5(2Z), \\
 \mathcal{Z}_{K_3 \times E}^{\text{red}} = -(J^* \text{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}.
 \end{array}$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borchers root bracket, a local-wrapped collision law, a quantum supergroup, an automorphic orientation line, a first-order Fredholm determinant, and the reduced scalar square.

Proof. Each arrow is one of the preceding constructions. Compatibility follows from their common root grading and PBW filtration. The final scalar identity is the primitive reduced curve-counting theorem. $\square$

BOOK **XX**

**Technical appendices to the preceding
book**

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\begin{aligned}\vartheta_2(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1}, \\ \vartheta_3(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n}, \\ \vartheta_4(\tau, z) &= \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.\end{aligned}$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r) .

The root reflection matrices

In the basis (f_2, f_3, f_{-2}) , the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \quad S_i^\top G S_i = G.$$

Concordance with the earlier Igusa draft

The reconstitution retains the half-index determinant, theta normalization, Borchers denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum. It reorganizes their dependencies as follows:

- (i) the automorphic square root is completed entirely within the Borchers product theorem;
- (ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;
- (iii) the bar–Chevalley comparison gives the denominator a homological source;
- (iv) the current and BF constructions give an exact first-order algebraic model;
- (v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;
- (vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;
- (vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Book entrance: Calabi–Yau Quantum Groups

Calabi–Yau categories, geometric BV theories, Hall correspondences, quantum doubles, K_3 symmetries, and compact-background comparison.

BOOK **XXI**

**Categorical and field-theoretic
foundations**

Calabi–Yau categories

103.1 SMOOTHNESS, PROPERNESS, AND THE CALABI–YAU CLASS

Let C be a small stable $\mathbb{k}$ -linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\mathbb{k}$. A d -Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{CY} \in HC_d^-(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d .

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \mathrm{RHom}_C(E, E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 103.1 (The two universal categorical outputs). *Let C be a smooth proper d -Calabi–Yau category.*

- (i) *The Hochschild cochains $CC^*(C)$ carry a framed E_2 -algebra structure.*
- (ii) *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2-d)$ -shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [11]. The shifted form at E is

$$\mathrm{RHom}(E, E)[1]^{\otimes 2} \xrightarrow{\circ} \mathrm{RHom}(E, E)[2] \xrightarrow{\mathrm{Tr}} \mathbb{k}[2-d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [10, 66]. $\square$

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2 . These structures meet through deformation theory while retaining their separate operadic degrees.

Local BV theories and factorization observables

104.1 CYCLIC ELLIPTIC LOCAL LIE ALGEBRAS

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\text{Obs}^{\text{cl}}(U) = C_{\text{CE}}^*(\mathcal{L}_\epsilon(U)).$$

Disjoint union of supports gives the prefactorization product.

Theorem 104.1 (Factorization observables). *An elliptic classical BV theory produces a factorization algebra Obs^{cl} . A renormalized quantization satisfying the quantum master equation produces a factorization algebra Obs^{q} over $\mathbb{k}[[\hbar]]$.*

Proof. The Chevalley cochains of a direct sum are the completed tensor product of the cochains, giving the disjoint-support multiplication. Elliptic descent supplies the Weiss cosheaf property. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [19, 20]. $\square$

104.2 BOUNDARY AND DEFECT RESTRICTION

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \text{RHom}(b, b).$$

Theorem 104.2 (Ordered boundary algebra). *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \text{Alg}_{\text{E}_1}(\text{Fact}_D(C), \otimes^!).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b -modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C . The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. $\square$

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 105.1 (Curve-level faces). A transverse face is

$$\mathcal{A}^\perp \in \text{Alg}_{\mathbf{E}_1}(\text{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$\mathcal{A}_{\text{Ass}}^{\text{ch}} \in \text{Alg}_{\mathbf{E}_1}(\mathcal{D}_C, \star^{\text{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\text{ch}} \in \text{Alg}_{\mathbf{g}_{\text{Lie}}}(\mathcal{D}_C, \star^{\text{ch}}).$$

Their bars are $\text{Bar}^\perp$, $\text{Bar}_{\text{Ass}}^{\text{ch}}$, and $\text{Bar}_{\text{Lie}}^{\text{ch}}$.

Theorem 105.2 (Source separation). *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$ satisfies*

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(U^{\text{ch}}(\mathfrak{g})) \simeq \text{Bar}_{\text{Lie}}^{\text{ch}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. $\square$

This theorem replaces every dimension-based dispatch of an $\mathbf{E}_n$ level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X , and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega_X^{\circ,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\text{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 106.1 (Classical hCS factorization algebra). *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X .*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2} [A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. $\square$

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [21].

Pushforward to a curve

Let $p : X \rightarrow C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X , define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 107.1 (Factorization pushforward). *The assignment $p_*\mathbb{F}$ is a factorization algebra on C . Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C , Riemann–Hilbert algebraization gives a factorization D -module on C .

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D -module setting. $\square$

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

The derived centre and the physical bulk

For an ordered boundary algebra $\mathcal{A}$, define

$$Z_{\text{ord}}(\mathcal{A}) = \text{RHom}_{\mathcal{A} \otimes \mathcal{A}^{\text{op}}}(\mathcal{A}, \mathcal{A}).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(\mathcal{A}).$$

Definition 108.1 (Open–closed completeness). A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase “bulk equals centre” into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

BOOK XXII

Hall and BPS algebras

Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}_{\alpha, \beta}^{\text{ext}}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha, \beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha + \beta}.$$

Theorem 109.1 (Hall associativity). *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^*(a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3 . $\square$

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeyer [49]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [52].

Theorem 110.1 (Threefold Hall algebra from strong orientation). *Let $\mathcal{C}$ be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [52]. $\square$

Definition 110.2 (Reduced $K_3 \times E$ Hall datum). A reduced $K_3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

The geometric E -equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\leq 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X , and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\geq 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ . Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E -equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E, \mathrm{mot}} = \widehat{\bigoplus_{\gamma \in \Gamma_X} K_o^E(\mathrm{St}/\mathcal{M}_\gamma)},$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem III.1 (Geometric reduced Hall algebra). *The extension correspondence in $\mathrm{Coh}_{\leq 1}(X)$ defines an associative E -equivariant motivic Hall product on $\mathcal{H}_X^{E, \mathrm{mot}}$. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E, \mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X][\epsilon]/(\epsilon^2)}$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E -orbits.

Proof. The stack of short exact sequences is E -equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E -equivariant motivic Hall theorem of Oberdieck–Shen [65]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ . $\square$

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\text{loc}} = \widehat{\bigoplus_{m(\gamma)=0} K_o^E(\text{St}/\mathcal{M}_\gamma)}, \quad \mathcal{H}_X^{\text{wr}} = \widehat{\bigoplus_{m(\gamma)>0} K_o^E(\text{St}/\mathcal{M}_\gamma)}.$$

A pure one-dimensional support of degree zero projects to a finite subset of E . A support of positive degree projects onto E . The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 112.1 (Geometric local–wrapped decomposition). *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\text{loc}}$ is a subalgebra, $\mathcal{H}_X^{\text{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E, \text{mot}} \cong \mathcal{H}_X^{\text{loc}} \ltimes \mathcal{H}_X^{\text{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E . $\square$

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borchers root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeyer orientation therefore produces

$$\mathcal{H}_X^{E, \text{coh}} = \widehat{\bigoplus_{\gamma \in \Gamma_X} H_*^{\text{BM}, E}(\mathcal{M}_\gamma, \Phi_\gamma)},$$

with its cohomological Hall product.

Theorem 113.1 (Strong-oriented cohomological lift). *For every strong orientation enhancement on the E -equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E, \text{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [52]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. $\square$

For the dyonic charge lattice

$$\Gamma_{\text{phys}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}),$$

write

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

The polarized central extension of the companion volume makes the graph $c \mapsto (c, \Pi(c))$ additive. Applying this graph to the reduced integration target defines the Gram-polarized character

$$I_{X, \Pi}^E : \mathcal{H}_X^{E, \text{mot}} \longrightarrow \mathbb{Q}[\widehat{\Gamma_{\text{phys}}}] [\epsilon] / (\epsilon^2).$$

Theorem 113.2 (Exact geometric character map). *The map $I_{X,\Pi}^E$ is multiplicative. Its logarithm is primitive for the completed Hall coproduct whenever that coproduct is defined in the selected localization. In every reduced Donaldson–Thomas sector covered by the equivariant integration theorem, its coefficients are the reduced invariants of $K_3 \times E$. The primitive reduced Gromov–Witten theory supplies the independent automorphic character $-\chi_{10}^{-1}$ in classes primitive in the K_3 factor.*

Proof. Multiplicativity of I_X^E is the reduced integration theorem. The cocycle identity for the polarization cross-term makes the graph of Π multiplicative. The logarithm of a group-like series is primitive. The enumerative identifications are the reduced Hall calculations of Oberdieck–Shen and the primitive Igusa theorem of Oberdieck–Pixton [64, 65]. $\square$

The geometric Hall source and the universal Igusa avatar now occupy explicit objects. A primitive root framing is a morphism from the simple BPS sheaves of $\mathcal{H}_X^{E,\text{coh}}$ to the simple-root spaces of $\mathfrak{g}_{\Delta}$, that preserves the Euler pairing, parity, and extension brackets. The universal presentation theorem then extends this framing through Hall, current, quantum-double, and determinant constructions.

BPS primitives

A connected graded bialgebra has primitive Lie algebra

$$\mathrm{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem 11.4.1 (Two-Calabi–Yau BPS algebra). *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on $K3$ and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [23]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. $\square$

This result supplies a global $K3$ generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \quad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 115.1 (Integral protected Lie algebra). The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X \boxtimes \Delta} = \mathfrak{b}_X \widehat{\Pi}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 115.2 (Hall–Borcherds amalgamation). *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X \boxtimes \Delta} \longrightarrow \mathfrak{b}_X, \quad r_\Delta : \mathfrak{b}_{X \boxtimes \Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X \boxtimes \Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X , while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ .

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \rightarrow \mathfrak{h}_\Pi \rightarrow \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. $\square$

Definition 115.3 (Comparison ideal). A primitive comparison $\varphi : \mathfrak{n}_{\Delta,+} \rightarrow \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta,+} \rangle} \subset \mathfrak{b}_{X \boxtimes \Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X \simeq \Delta, \varphi} = \mathfrak{b}_{X \boxtimes \Delta} / \mathcal{I}_\varphi.$$

Theorem 115.4 (Universal comparison quotient). *A Cartan-preserving primitive comparison is equivalent to a quotient of the integral amalgam that identifies the Borchers generators with geometric BPS primitives. If φ is an isomorphism, the quotient is canonically isomorphic to both Borels. In every case, the kernel $\mathcal{I}_\varphi$ measures the complete discrepancy between the geometric and automorphic primitive theories. PBW carries the quotient to enveloping algebras, Chevalley chains, current envelopes, and Fock characters.*

Proof. A quotient identifying $i_\Delta(x)$ with an element of $\mathfrak{p}_X^+$ defines φ , and the Lie relations in the quotient make it a Lie morphism. Conversely, a Lie morphism makes the generators of $\mathcal{I}_\varphi$ compatible with every bracket, so the quotient exists. An isomorphism φ gives mutually inverse maps through the two retractions. Functoriality of enveloping algebras, Chevalley chains, currents, and symmetric-superalgebra completion gives the final statement. $\square$

This amalgam is the maximal theory that retains both sources before a duality identification is imposed. The geometric Hall source and the automorphic Borchers source are its two canonical retracts. A microscopic duality is a diagonal quotient rather than an equality inferred from a scalar character.

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H° be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a nondegenerate Hopf pairing.

Definition 116.1 (Completed Hall double). The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \widehat{\otimes} H^\circ \widehat{\otimes} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R -matrix whenever the canonical tensor converges.

Theorem 116.2 (Double construction). *Let H^+ and H^- be paired complete graded bialgebras, let H° be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \widehat{\otimes} H^\circ \widehat{\otimes} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R -matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [29, 32]. $\square$

Theorem 116.3 (Quantum Hall–Borcherds amalgam). *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H° , with continuous Hopf retractions onto H° . Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \widehat{*}_{H^\circ} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H° and has canonical Hopf retractions to both factors. A mixed universal R -matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R -matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H° , so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H° . A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. $\square$

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Ordered chiral bialgebras

Let $\mathcal{A}$ be an ordered chiral algebra on a curve C . An ordered chiral bialgebra consists of a coproduct

$$\Delta : \mathcal{A} \longrightarrow \mathcal{A} \widehat{\otimes} \mathcal{A}$$

that is a morphism of ordered chiral algebras, together with counit and the chosen completion. An antipode upgrades it to a Hopf object. A spectral coproduct may take values in Laurent series in a difference parameter.

Theorem 117.1 (Representation fusion). *A completed coproduct defines a tensor product of continuous $\mathcal{A}$ -modules by restriction of scalars. A universal R -matrix satisfying quasitriangularity defines a braid operator on that module category.*

Proof. For modules M, N , pull back the $\mathcal{A} \widehat{\otimes} \mathcal{A}$ action on $M \widehat{\otimes} N$ along Δ . Coassociativity supplies the associator. The relation $R\Delta(a) = \Delta^{\text{op}}(a)R$ makes the flip composed with R an intertwiner, and the Yang–Baxter equation gives the braid relation. $\square$

BOOK XXIII

Exact model cases

The A_2 Hall algebra

For the quiver $1 \rightarrow 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2, 1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [40, 70].

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z , with potential

$$W = \text{Tr } x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\text{Tr } W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 119.1 (Affine-Yangian positive half). *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of $\mathfrak{gl}_1$. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [22, 24, 71, 72]. $\square$

The vacuum representation produces a $W_{1+\infty}$ -type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

Surface CoHAs and ADE Yangians

Let $S_\Gamma \rightarrow \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 120.1 (Curve-supported surface CoHA). *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [26]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. $\square$

This theorem is a local K_3 model whenever an ADE configuration occurs on a K_3 surface. The global K_3 surface adds compact curve classes and monodromy beyond the local root lattice.

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [12]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

BOOK XXIV

K₃ symmetries

The Mukai lattice

For a K_3 surface S , the Mukai lattice is

$$\tilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with pairing

$$((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's.$$

It is even unimodular of signature $(4, 20)$ and rank 24.

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 123.1 (K_3 Heisenberg action). *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m \delta_{m+n,0} \int_S \alpha \cup \beta \text{ id}.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^(S^{[0]})$.*

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha \beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [63]. $\square$

This action is global, compact, and abelian at the Lie level. It is an exact K_3 symmetry and supplies the rank-twenty-four Fock sector of any broader K_3 algebra.

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 12.4.1 (Orthogonal LLV symmetries). *For a K_3 surface S , the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\text{LLV}}(S) \cong \mathfrak{so}(\widetilde{H}(S), (-, -)_{\text{Muk}}) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K_3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\text{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K_3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S , the extension has signature $(4, 20)$. For $K_3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [58, 73]. $\square$

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Local ADE Yangians

An ADE configuration of (-2) -curves on S gives a root sublattice of $H^2(S, \mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K_3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

The K_3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4, 20)$	$H^*(S)$; separately $\mathfrak{so}(4, 21)$ on $K_3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K_3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

The K_3 correspondence category

Let S be a projective K_3 surface. Let $\mathcal{M}_S$ contain the smooth projective moduli spaces $\mathcal{M}_\sigma(v)$ of stable objects for primitive Mukai vectors and generic stability conditions, the Hilbert schemes $S^{[n]}$, and the equivariant curve-supported moduli at every ADE enhancement. Close this collection under finite products.

Definition 127.1 (K_3 correspondence category). The category $\text{Corr}_{K_3}(S)$ has the spaces of $\mathcal{M}_S$ as objects and

$$\text{Hom}(M, N) = H_*^{\text{BM}}(M \times N)$$

for proper objects, with the corresponding localized equivariant Borel–Moore theory on ADE charts. Composition is convolution:

$$\beta \circ \alpha = (p_{13})_* (p_{12}^* \alpha \cap p_{23}^* \beta).$$

The tensor product is Cartesian product of moduli spaces and external product of correspondences.

Theorem 127.2 (Associativity and monoidality). *Transverse convolution makes $\text{Corr}_{K_3}(S)$ a graded monoidal category. The cohomology functor*

$$\omega_S(M) = H^*(M)$$

and its localized equivariant form on ADE charts is strong monoidal and carries convolution to composition of linear operators.

Proof. The two parenthesizations of three correspondences are the two push–pull descriptions on $M_1 \times M_2 \times M_3 \times M_4$. Proper base change and the projection formula identify them. Cartesian products and external products satisfy the interchange law. The standard action of a kernel on cohomology is compatible with both operations. $\square$

Nakajima incidence varieties are morphisms in this category. Cup product and Lefschetz adjoints are diagonal kernels. ADE Hecke correspondences are equivariant morphisms. Thus the three K_3 symmetry families occupy one correspondence category before passage to any enveloping algebra.

The common K_3 action algebra

Define the completed state space

$$\mathcal{H}_S = \widehat{\bigoplus_{M \in \mathcal{M}_S} \omega_S(M)},$$

completed by Mukai norm, instanton number, and cohomological degree. Every correspondence acts continuously on $\mathcal{H}_S$.

Definition 128.1 (K_3 action algebra). The K_3 action algebra is the image

$$\mathcal{A}_{K_3}(S) = \text{im} \left(\bigoplus_{M, N} \text{Hom}_{\text{Corr}_{K_3}(S)}(M, N) \longrightarrow \text{End}_{\text{cont}}(\mathcal{H}_S) \right),$$

closed in the operator topology.

Theorem 128.2 (Exact K_3 symmetry envelope). *The algebra $\mathcal{A}_{K_3}(S)$ is separated and complete. It contains the Nakajima Heisenberg, every LLV Lefschetz triple, monodromy transport, and every local ADE Yangian action. The cross-relations are the intersection identities of their kernels. This description is faithful by construction.*

Proof. Each named operator is induced by a correspondence in $\text{Corr}_{K_3}(S)$. Convolution gives their products. The image quotient removes precisely the correspondences acting trivially on all selected cohomologies, and closure gives completeness. $\square$

The three symmetries retain their intrinsic forms inside one algebra. Their interaction is calculated by intersections of the corresponding cycles. On a Noether–Lefschetz locus, an LLV operator and an ADE Hecke operator meet in the same correspondence space; the resulting convolution is the geometric cross-relation.

Tannaka reconstruction from K_3 correspondences

Let $\mathcal{C}_S^{\text{geom}}$ be the rigid Karoubi envelope of the dualizable correspondence objects, modulo the tensor ideal of kernels acting trivially on cohomology, with Tate and parity shifts adjoined. For a finite Mukai-norm truncation, let $\mathcal{C}_S$ be the exact rigid tensor envelope of its image under cohomology: it is closed inside $\text{sVect}_{\text{fd}}^{\text{gr}}$ under finite sums, tensor products, duals, summands, kernels, and cokernels. The induced functor

$$\omega_S : \mathcal{C}_S \longrightarrow \text{sVect}_{\text{fd}}^{\text{gr}}$$

is faithful, exact, and strong monoidal.

Definition 129.1 (K_3 correspondence quantum group). The coordinate bialgebra is the coend

$$O(\mathbb{G}_{K_3}) = \int^{M \in \mathcal{C}_S} \omega_S(M)^* \otimes \omega_S(M).$$

Its continuous dual is denoted $U(\mathbb{G}_{K_3})$.

Theorem 129.2 (K_3 Tannaka theorem). *On every finite Mukai-norm truncation, $\mathcal{C}_S$ is equivalent to the rigid exact tensor subcategory of finite comodules over $O(\mathbb{G}_{K_3})$ generated by the universal coactions on the fibres $\omega_S(M)$. Every geometric K_3 correspondence induces a comodule morphism. The continuous dual acts naturally on every cohomology fibre and represents tensor-compatible natural endomorphisms of the fibre functor. A braided transport on $\mathcal{C}_S$ produces a coquasitriangular structure on the coend.*

Proof. Exact rigid coend reconstruction applies to the faithful exact strong monoidal functor ω_S . The coend carries the universal coaction on each fibre. The exact tensor envelope used to define $\mathcal{C}_S$ is precisely the subcategory generated from these comodules by the operations preserved by ω_S , which gives the stated equivalence. The quotient map from geometric correspondences makes each correspondence a comodule morphism. Dualizing the universal coaction represents tensor-compatible natural endomorphisms. A braiding supplies the bilinear form whose hexagon identities are the coquasitriangular axioms. $\square$

The action algebra $\mathcal{A}_{K_3}(S)$ and the Tannaka algebra $U(\mathbb{G}_{K_3})$ are complementary. The first is generated by geometric kernels. The second consists of tensor symmetries compatible with every kernel. Their commutants record two complementary aspects of the K_3 correspondence geometry: geometric operators and tensor-compatible natural symmetries.

BOOK XXV

The exact $K_3 \times E$ protected completion

The protected input

The K_3 elliptic genus supplies the Jacobi form

$$Z_{K_3} = 2\phi_{0,1}.$$

Its Borchers half determines the weight-five product D_5 , the Gritsenko–Nikulin Borchers superalgebra $\mathfrak{g}_{\Delta,5}$, and its simple-root superdatum. The companion volume constructs the Borchers quiver $Q_{\Delta,H}$ in every proper-height window, its coordinate-framed Hall category, its formal three-Calabi–Yau Ginzburg completion, its colored elliptic Ran factorization algebra, its quantum double, and its virtual root Dirac operator.

Definition 130.1 (Protected avatar). The protected avatar of $K_3 \times E$ is

$$\mathfrak{P}(K_3 \times E) = (\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{\text{ip}}, \mathcal{F}_{\Delta,E}^{\text{hyb}}, \widehat{U}_q(\mathfrak{g}_{\Delta,5}), \mathbb{A}_Z).$$

Here $\mathcal{B}_{\Delta}$ is the ordinary quiver Hall category and $\mathcal{H}_{\text{ip}}$ is the parity-refined Borchers root superspace.

Theorem 130.2 (Existence of the universal protected avatar). *The protected avatar exists and has the following structures:*

- (i) Γ_{Δ} is a locally finite three-Calabi–Yau dg category;
- (ii) $\mathcal{B}_{\Delta}$ is the quiver Hall category of the underlying even Borchers–Cartan datum, while $\mathcal{H}_{\text{ip}}$ carries the parity-refined root spaces of $\mathfrak{g}_{\Delta,5}$;
- (iii) $\mathcal{F}_{\Delta,E}^{\text{hyb}}$ is a local–wrapped factorization algebra with primitive current algebra $\text{Cur}_E(\mathfrak{g}_{\Delta,5})$;
- (iv) $\widehat{U}_q(\mathfrak{g}_{\Delta,5})$ is a root-height-complete quasitriangular Hopf superalgebra;
- (v) $\mathbb{A}_Z$ is an oriented virtual Fredholm family with Pfaffian $D_5(2Z)$ after the Weyl shift.

Proof. The assertions are respectively the Calabi–Yau completion theorem, the quiver Hall theorem, the Borchers root-space construction, the colored-Ran Chevalley theorem, Hong’s quantization theorem, and the root-space Fredholm theorem of the companion volume. $\square$

Definition 130.3 (Integral protected avatar). Let $\mathfrak{p}_X^+$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E -equivariant Hall source. The integral protected avatar is

$$\mathfrak{P}_{\text{int}}(X, \Delta) = (\mathcal{H}_X^{E,\text{mot}}, \mathfrak{P}(K_3 \times E), \mathfrak{b}_{X \boxtimes \Delta}, r_X, r_{\Delta}, I_{X,\Pi}^E).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}_X^+$ and hence the amalgamated primitive component.

Theorem 130.4 (Integral source–target completion). *The integral protected avatar retains the geometric extension theory, the universal Igusa root theory, and their common polarized charge lattice in one functorial system. The retraction r_X recovers the geometric BPS Borel, while r_Δ recovers the Borchers Borel and its denominator. The character morphism factors through the common Cartan grading. Every primitive comparison $\varphi : \mathfrak{n}_{\Delta,+} \rightarrow \mathfrak{p}_X^+$ produces a diagonal quotient of the integral avatar, and every diagonal quotient arises in this way.*

Proof. The geometric Hall algebra and its character were constructed before the universal root avatar. The polarized graph map identifies their Cartan gradings. Hall–Borchers amalgamation gives the common primitive object and its two retractions. The universal comparison-quotient theorem identifies primitive comparisons with diagonal quotients. Each step is natural under morphisms preserving the charge lattice, Hall correspondences, and root presentation. $\square$

The noncommutative Calabi–Yau field theory

Every finite-height Ginzburg algebra $\Gamma_{\Delta,H}$ has a cyclic three-Calabi–Yau pairing. Its open string field is a degree-one element a in the completed endomorphism algebra. The action is

$$S_H(a) = \frac{1}{2} \langle a, m_1 a \rangle + \sum_{n \geq 2} \frac{1}{n+1} \langle a, m_n(a^{\otimes n}) \rangle.$$

Theorem 131.1 (Master equation of the universal Igusa open string). *The cyclic A_∞ identities imply*

$$\{S_H, S_H\} = 0.$$

The fixed root ordering and coordinate framing orient each finite Borchers–Ginzburg representation chart and define its vanishing-cycle Hall theory. The proper-height limit gives a complete noncommutative Calabi–Yau field theory whose ordinary boundary Hall algebra is the quiver Hall presentation, whose protected primitive superspace is $\mathfrak{g}_{\Delta,+}$, and whose quantum symmetry is $\widehat{U}_q(\mathfrak{g}_{\Delta_s})$.

Proof. The cyclic A_∞ relations are the coefficients of the classical master equation. The three-Calabi–Yau pairing gives the odd symplectic form. The coordinate-framed orientation theorem of the companion volume defines the finite-chart Hall product. Proper height makes every fixed coefficient finite, so the inverse limit preserves the master equation and the Hall operations. $\square$

The mixed holomorphic–topological boundary theory

Apply cotangent BF theory to the finite positive Borchers Lie superalgebras $\mathfrak{g}_{\Delta,+}^{(H)}$ on $\mathbb{R}_{\text{top}}^2 \times E$. The boundary polarization has algebra $U(\mathfrak{g}_{\Delta,+}^{(H)})$ and derived centre

$$C_{\text{HH}}^{\bullet}\left(U(\mathfrak{g}_{\Delta,+}^{(H)}), U(\mathfrak{g}_{\Delta,+}^{(H)})\right).$$

The shifted-cotangent bulk is

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{g}_{\Delta,+}^{(H)}).$$

Theorem 132.1 (Finite-height mixed-HT open–closed equivalence). *In every finite height window, linear Poisson formality gives a homotopy- E_2 equivalence*

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{g}_{\Delta,+}^{(H)})[\![\hbar]\!] \xrightarrow{\sim} C_{\text{HH}}^{\bullet}\left(U_{\hbar}(\mathfrak{g}_{\Delta,+}^{(H)}), U_{\hbar}(\mathfrak{g}_{\Delta,+}^{(H)})\right).$$

The maps are compatible with height truncation. Their inverse limit identifies the completed algebraic bulk with the completed Hochschild centre, root degree by root degree. Applying the current functor on E and retaining the elliptic-degree colour produces compatible local–wrapped factorization objects; the open–closed map is defined on each current coefficient by the same universal formality graphs.

Proof. Each finite algebra is finite-dimensional and nilpotent. Kontsevich formality for its linear Poisson structure gives the first equivalence, and the brace refinement supplies the homotopy- E_2 structure. Universal graph formulas commute with quotient maps. Proper height makes the inverse limit exact in each root degree. The current functor is applied after this algebraic equivalence; root-degree addition preserves the local–wrapped colour. $\square$

The exact scalar and first-order observables

Set

$$\text{Den}_\Delta(Z) = D_5(2Z), \quad j(Z) = Z/2, \quad j^* \text{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\text{sch Fock}(\mathfrak{g}_{\Delta,+}) = \prod_{\alpha>0} (1 - e^{-\alpha})^{-\text{sdim } \mathfrak{g}_{\Delta,\alpha}}.$$

The oriented root operator gives

$$e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}_{K_3 \times E}^{\text{red}} = -(j^* \text{Den}_\Delta)^{-2} = -\chi_{\text{to}}^{-1}.$$

Theorem 133.1 (Protected square-root theorem). *Within $\mathfrak{P}(K_3 \times E)$, the denominator section Den_Δ is simultaneously:*

- (i) *the Weyl denominator of $\mathfrak{g}_{\Delta,+}$;*
- (ii) *the Euler character of the bar–Chevalley complex;*
- (iii) *the oriented Fredholm Pfaffian of $\mathbb{A}_Z$;*
- (iv) *the PBW determinant of the parity-refined root superspace $\mathcal{H}_{\text{ip}}$;*
- (v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borchers Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $j^* \text{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. $\square$

Physical meaning

The protected avatar is the complete universal algebraic model selected by the Igusa root datum. The Ginzburg category supplies its noncommutative three-Calabi–Yau sector. The quiver Hall category supplies its ordinary extension geometry. The Borchers root superspace supplies its protected one-particle states. The colored Ran object supplies its local and wrapped current algebra on the elliptic curve. Hong’s quantum Borchers algebra supplies its line-operator symmetry. The root Dirac family supplies its first-order determinant, and the Fock construction supplies the scalar square.

A compact string construction with specified simple-root states and extension amplitudes maps to the protected avatar through the presentation universal property. The comparison is determined on primitives; PBW propagates it to multiparticle states, the Hall product, the current algebra, the quantum double, and the determinant.

Universal property of the protected avatar

A presentation-preserving protected theory consists of a three-Calabi–Yau category, an oriented Hall boundary, a colored elliptic factorization algebra, a quasitriangular quantum symmetry, an oriented determinant, and comparison functors from the Borchers generators-and-relations construction to these five objects. The chosen simple-root states satisfy the Gritsenko–Nikulin Cartan, parity, and Serre relations.

Theorem 135.1 (Presentation universal property). *Every presentation-preserving protected theory carries a unique primitive Lie-superalgebra morphism*

$$\mathfrak{n}_{\Delta,+} \longrightarrow \text{Prim}(\mathfrak{T})$$

extending the chosen simple-root states. The Hall, Ginzburg, current, quantum-group, and determinant functors extend this primitive morphism to the five selected comparison maps. A primitive isomorphism together with equivalences of the selected functorial realizations gives an equivalence of protected data.

Proof. The Borchers generators and relations give the unique primitive morphism. Each remaining construction is functorial in a presentation map: quiver Hall convolution in the quiver map, Calabi–Yau completion in the dg-algebra map, chiral Chevalley theory in the Lie map, Hong quantization in the Cartan superdatum, and the diagonal Fredholm operator in the root-graded map. PBW promotes the primitive isomorphism to the completed enveloping and Fock objects. $\square$

The Borchers-to-Calabi–Yau completion

Theorem 136.1 (Exact root-datum construction). *The sequence*

$$\begin{aligned} \{\text{finite Borchers simple-root superdatum}\} &\longmapsto Q_{\Delta,H} \longmapsto \Gamma_{\Delta,H}, \\ \Gamma_{\Delta,H} &\longmapsto \mathcal{B}_{\Delta,H} \longmapsto U_q(\mathfrak{g}_{\Delta}^{(H)}). \end{aligned}$$

is functorial under full inclusions of finite root windows. Proper-height passage gives a locally three-Calabi–Yau dg category, its Hall presentation, and a complete quantized Borchers superalgebra. The simple-root datum extracted from $\phi_{o,I}$ gives the universal Igusa protected avatar.

Proof. A full inclusion of root data adds vertices and arrows to the Borchers quiver. Keller completion and the quiver Hall functors preserve exact full subcategories. The generator-and-relation map to Hong’s quantized Borchers superalgebra is compatible with root truncation. Proper height makes each coefficient eventually constant. The Gritsenko–Nikulin theorem identifies the resulting root datum with the Igusa denominator algebra. $\square$

BOOK XXVI

Quantum groups, exchange, and
representation theory

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^- , carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+$. Its universal R -matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\mathbb{k}$ -linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \rightarrow \mathbf{Vect}_{\text{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M)$$

exists and that the canonical comparison functor to finite-dimensional H -comodules is comonadic. Tensor product makes H a bialgebra, and rigidity supplies its antipode. A braiding gives a coquasitriangular form; a meromorphic braiding gives its spectral version.

Theorem 138.1 (Rigid Tannaka reconstruction). *Under the preceding finite and comonadic hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathbf{Comod}_{\text{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H .

Proof. Matrix coefficients define the universal H -coaction on every $\omega(M)$. Tensor product and duality give the bialgebra and antipode structures. The Barr–Beck comonadic theorem identifies $\mathcal{L}$ with coalgebras for the induced comonad, and the coend identifies that comonad with $H \otimes -$. The hexagon identities transport to the coquasitriangular identities. $\square$

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra alone provides fusion; the plane of exchange supplies braiding.

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R -matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

The centre and the double

The Hochschild centre $Z_{\text{ord}}(\mathcal{A})$ is the universal E_2 algebra acting centrally on an ordered algebra. The Drinfeld centre $Z(\text{Mod}_{\mathcal{A}})$ is the braided category of modules equipped with half-braidings. Under dualizability hypotheses these are related by the passage from algebraic Hochschild cochains to categorical endomorphisms of the identity. Derived loop-space descriptions of centres provide the geometric form of this relation [7].

The centre and the double answer complementary questions. The centre classifies central bulk actions. The double reconstructs a full quantum group from paired positive and negative halves. A field theory can supply a comparison by realizing both as line and bulk operator constructions.

BOOK **XXVII**

***A compact six-dimensional protected
source***

Holomorphic cotangent BF theory on $K_3 \times E$

Let S be a K_3 surface with holomorphic symplectic form σ_S , let E be an elliptic curve with holomorphic form dz , and put

$$X = S \times E, \quad \Omega_X = \sigma_S \wedge dz.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta,+}^{(H)}$ of the positive Igusa Borchers algebra, take fields

$$A \in \Omega^{\circ,\bullet}(X, \mathfrak{n}_{\Delta,+}^{(H)})[1], \quad B \in \Omega^{\circ,\bullet}(X, (\mathfrak{n}_{\Delta,+}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A, B) = \int_X \Omega_X \wedge \langle B, \bar{\partial} A + \frac{1}{2}[A, A] \rangle.$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta,+}^{(H)}$ -valued Dolbeault forms.

Theorem 141.1 (Classical compact-background source). *The action (141) satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X .*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta,+}^{(H)}$ together with $\bar{\partial}^2 = 0$ and the derivation property of $\bar{\partial}$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. $\square$

Tree exactness and the quantum master equation

Every interaction vertex in (141) has one B -leg and two A -legs. The propagator pairs an A -leg with a B -leg.

Lemma 142.1 (Loop bound). *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B -half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. $\square$

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta,+}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Theorem 142.2 (Perturbative exactness). *Every one-loop Lie weight of the finite-height theory vanishes. The renormalized quantum master equation holds with zero counterterm, and the perturbative effective action is represented by tree graphs. The statement is compatible with the height maps $H \rightarrow H + 1$.*

Proof. The loop bound leaves tree graphs and one-loop graphs. Their Lie factor is a cyclic trace of a product of adjoint or coadjoint operators. Positive height makes each such product strictly triangular, so its trace is zero. The anomaly and the one-loop effective vertex therefore vanish. Tree graphs satisfy the classical master equation by the usual boundary-stratum argument. Every map preserves root height. $\square$

Corollary 142.3 (Height-complete quantum source). *The coefficientwise inverse limit*

$$\text{Obs}_X^\Delta = \varprojlim_H \text{Obs}_X^{(H)}$$

is a perturbative quantum factorization algebra on $K_3 \times E$. Its primitive gauge algebra is the complete positive Igusa Borchers algebra.

Proof. Every fixed root height stabilizes in the finite tower. The factorization maps and quantum differential are compatible with the truncations, so the separated inverse limit preserves the quantum master equation and the primitive Lie algebra. $\square$

Harmonic reduction along K_3

The Dolbeault algebra of a K_3 surface has the strict harmonic model

$$H^{\circ,\bullet}(S, \mathcal{O}_S) = \mathbb{C}\mathbf{1} \oplus \mathbb{C}\bar{\sigma}_S[-2], \quad \bar{\sigma}_S^2 = 0.$$

The trace pairs $\mathbf{1}$ with $\bar{\sigma}_S$.

Theorem 143.1 (K_3 harmonic reduction). *Factorization pushforward along $p_E : S \times E \rightarrow E$, followed by the K_3 harmonic deformation retract, gives a quasi-isomorphism*

$$p_{E,*} \text{Obs}_X^{(H)} \simeq \text{Obs}_E \left(T^*[-1] (\text{Cur}_E(\mathfrak{n}_{\Delta,+}^{(H)})) \otimes H^{\circ,\bullet}(S, \mathcal{O}_S) \right).$$

The primitive degree-zero current sector is $\text{Cur}_E(\mathfrak{n}_{\Delta,+}^{(H)})$. The degree-two K_3 class is its cotangent partner.

Proof. Kähler Hodge theory gives a deformation retract of the Dolbeault complex onto harmonic forms. The harmonic representatives $\mathbf{1}$ and $\bar{\sigma}_S$ are closed under multiplication, so the retract is a cyclic dg-algebra quasi-isomorphism. Tensoring the local cyclic Lie model with this retract and applying homological transfer gives the displayed cotangent current theory. The product $\bar{\sigma}_S^2$ vanishes by complex dimension, so higher transferred products vanish. $\square$

Corollary 143.2 (Exact curve shadow). *The K_3 harmonic reduction of the compact six-dimensional source is the cotangent current factorization algebra of the Igusa positive root algebra on E . On a formal disc, its chiral enveloping mode algebra is $\widehat{U}_h(\mathfrak{n}_{\Delta,+})$, and the associative chiral bar of this envelope is the completed Chevalley complex. A topological collar supplies the corresponding transverse interval algebra by the boundary polarization of Volume II.*

Proof. The harmonic reduction theorem identifies the finite-height curve theories. Chiral PBW identifies the associative chiral bar of the current envelope with chiral Chevalley chains. Proper height passes this statement to the complete limit. Boundary polarization of the cotangent theory gives the transverse enveloping algebra on a collar. $\square$

This construction gives a compact effective field-theoretic source whose gauge algebra is selected by the Igusa root datum. The geometric E -equivariant Hall algebra constructed above gives an independent compact D-brane source. A primitive root framing compares its extension algebra with the effective source.

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\text{ip}} = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta, \alpha}}$$

with its Borchers parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\text{Fock}} = \widehat{\text{Sym}}(\mathcal{H}_{\text{ip}, \bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\text{ip}, \bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-HZ} act by $e^{-\alpha(Z)}$ on the root space of degree α .

Theorem 14.1 (Protected Hamiltonian trace). *The supertrace is*

$$\text{str}_{\mathcal{H}_{\text{Fock}}}(e^{-HZ}) = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{-\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\text{Den}_{\Delta}(Z) = D_5(2Z)$. After the half-coordinate pullback and doubling, the disconnected protected scalar is $-(j^ \text{Den}_{\Delta})^{-2} = -\chi_{\text{io}}^{-1}$ in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. $\square$

The compact-source comparison

Let $\mathfrak{P}_\Delta(X)$ denote the compact cotangent theory above, its K_3 harmonic shadow, its Hall boundary category from the Borchers quiver, and its root Dirac determinant.

Theorem 145.1 (Compact protected source theorem). *The system $\mathfrak{P}_\Delta(K_3 \times E)$ contains, in one functorial construction:*

- (i) *a quantum factorization algebra on the compact Calabi–Yau threefold;*
- (ii) *a curve-level local–wrapped current algebra on E ;*
- (iii) *an ordinary quiver Hall boundary, a parity-refined root superspace, and a three-Calabi–Yau Ginzburg completion;*
- (iv) *Hong’s height-complete quantized Borchers superalgebra;*
- (v) *the Fredholm Pfaffian Den_Δ and the scalar trace $-(j^* \text{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$.*

Every arrow among these objects preserves root degree, parity, product, and completion.

Proof. The compact BV theory and its quantum exactness were proved above. Harmonic reduction gives the curve theory. The Borchers quiver and Ginzburg completion give the Hall boundary. Hong’s construction gives the quantum group. The root Hamiltonian gives the determinant. Each construction is functorial in finite root windows, and proper height gives the common limit. $\square$

The theorem realizes the complete universal root-datum protected model over the $K_3 \times E$ background. The geometric D-brane theory enters through a monoidal functor from its oriented extension category to the Hall boundary of $\mathfrak{P}_\Delta$. Such a functor is determined by primitive branes and their extension amplitudes.

BOOK XXVIII

Physics of the comparison maps

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathbf{I})) \simeq C_*(\mathbf{I}).$$

For $\mathbf{I} = \mathfrak{n}_{\Delta,+}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Six-dimensional hCS on $K_3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{\circ,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS} . Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S .

A protected reduction to a chiral algebra consists of a deformation retract of the K_3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S \times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \quad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R -matrix. Toric three-folds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant $K3$ fibers to the selected generic fiber.

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

BOOK **XXIX**

**Automorphic characters and scalar
shadows**

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi^{C_*}(\mathfrak{n}_+) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

The universal Borcherds weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\text{wt}(\text{Borch}(\varphi)) = \frac{c(o, o)}{2}.$$

For the K_3 half-index, $c(o, o) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1$$

with weights

$$5, 2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [17]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Scalarization hierarchy

The constructions descend through four levels:

factorization or Hall object $\longrightarrow$ primitive Lie algebra
 $\longrightarrow$ PBW character $\longrightarrow$ scalar partition function.

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

BOOK **XXX**

The unified theorem system

The foundational theorem

Definition 154.1 (Complete Calabi–Yau quantum-group datum). A complete datum is a governing tuple $\mathfrak{D}$ together with:

- (i) PTVV-representable moduli for the smooth proper Calabi–Yau category;
- (ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b ;
- (iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;
- (iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;
- (v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;
- (vi) compatible open–closed and Hall–boundary morphisms;
- (vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 154.2 (Calabi–Yau quantum-group assembly). *Let a complete datum consist of:*

- (i) *a Calabi–Yau category and a representable moduli-of-objects problem;*
- (ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*
- (iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*
- (iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*
- (v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

- (i) *framed E_2 Hochschild cochains and the shifted-symplectic moduli structure;*
- (ii) *the three curve-level algebraic faces and their three bars;*
- (iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*
- (iv) *the completed quantum double and braided representation category;*

- (v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*
- (vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [10, 11, 66]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [52]. The paired bialgebra data give the double and universal R -matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. $\square$

The local quiver, equivariant surface, K_3 correspondence, and Igusa Borchers chapters construct objects in their stated categories. A geometric $K_3 \times E$ Hall theory determines a morphism from its primitive and determinant-line data to the universal Igusa objects.

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y^+(\widehat{\mathfrak{gl}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
K_3	$U(\mathfrak{g}_{\text{BPS}}^+)$	Heisenberg and LLV actions	K_3 Hall characters
$K_3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_s})$ and the height-complete current/BF model	$-\chi_{10}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K_3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row combines exact automorphic, Lie, bar, current, BF, Fredholm, and enumerative constructions; geometric Hall data enter through the primitive comparison morphism.

Closure theorem for the established and universal layers

Theorem 156.1 (Foundational closure). *The following layers are complete mathematical constructions:*

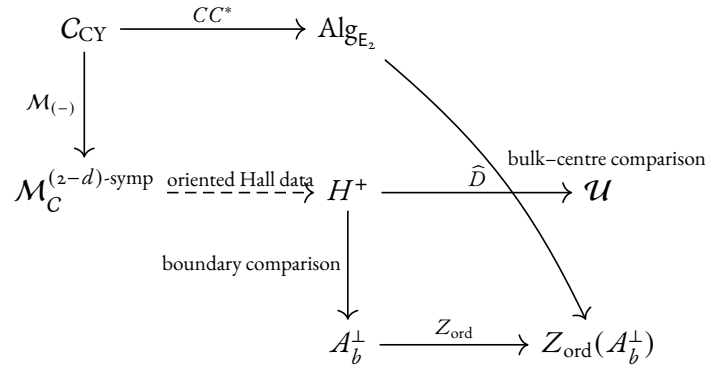
- (i) *framed Hochschild and shifted-symplectic categorical data;*
- (ii) *perturbative factorization observables from a quantized cyclic elliptic BV theory;*
- (iii) *the three curve-level algebraic faces and their bars;*
- (iv) *Hall products in the established quiver, surface, 2-Calabi–Yau, and strong-oriented 3-Calabi–Yau settings;*
- (v) *the K_3 Heisenberg, LLV, and local ADE Yangian symmetry atlas;*
- (vi) *the Igusa Borchers root algebra, its quantized Borchers superalgebra, proper-height bar/current system, classical local BF theory, exact finite zero-mode BV theory, and analytic determinant;*
- (vii) *the primitive reduced $K_3 \times E$ scalar character.*

A compact $K_3 \times E$ quantum-group realization is exactly an instance of the compact comparison theorem with a constructed reduced Hall datum.

Proof. Items (1)–(7) are the theorems cited and proved in the corresponding chapters. The last sentence is the definition of a compact realization together with the preceding recognition theorem. $\square$

Common diagram

The relationships are summarized by



Here $A_b^\perp \in \text{Alg}_{E_1}(\text{Fact}_D(C), \otimes^!)$. Every arrow is a construction or a supplied comparison morphism. The diagram separates universal categorical outputs, geometric field-theory outputs, Hall operations, and quantum-group assembly.

Concordance with the original Calabi–Yau volume

The reconstitution preserves the original vision of a bridge among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products. It realizes that vision through the following replacements:

- (i) framed E_2 Hochschild cochains and $(2 - d)$ -shifted symplectic moduli replace a dimension-dependent Hochschild E_n assertion;
- (ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;
- (iii) pushforward along an actual map $p : X \rightarrow C$ replaces formal cycle-specialization notation;
- (iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;
- (v) coproduct, Cartan, Hopf pairing, and topology define the double;
- (vi) K_3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;
- (vii) the graph of the $K_3 \times E$ Gram map becomes multiplicative in the polarized central extension;
- (viii) point-local and wrapped sectors form a hybrid factorization problem;
- (ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;
- (x) anomaly identities are transported through explicit maps between deformation complexes.

These constructions retain the scale of the original program while placing every operation in its natural category.

BOOK XXXI

Technical appendices to the preceding
book

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^!$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\text{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull-push product
Bialgebra	product, coproduct, and counit	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R -matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

The three K_3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\text{Corr}_{K_3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Exact arithmetic checks

The companion verification calculation establishes:

- (i) the low Fourier rows of $\phi_{o,1}$ through q^3 by exact theta-series division;
- (ii) the vector-valued cocycle identity for the Gram polarization;
- (iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;
- (iv) the A_2 Hall flag counts at ranks zero and one;
- (v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Book entrance: Universal Chiral BV and Einstein Completion

The complete mixed boundary datum, its centre, deformation algebra, BV envelope, gravitational completion, and anomaly geometry.

BOOK XXXII

The full mixed boundary datum

Three products and one interface

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-Mod}(\text{Ran } X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\text{ch}}$ for chiral convolution.

Definition 162.1 (Complete mixed boundary datum). A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, \mathcal{A}^{\text{ch}}, \mathcal{A}^\perp, K, \iota, \lambda)$$

with the following components.

- (i) L is a complete chiral Lie algebra.
- (ii) $\mathcal{A}^{\text{ch}}$ is a complete associative algebra for $\star^{\text{ch}}$.
- (iii) $\mathcal{A}^\perp$ is a complete E_1 algebra in $(\text{Fact}_D(X), \otimes^!)$.
- (iv) K is an $(\mathcal{A}^{\text{ch}}, \mathcal{A}^\perp)$ interface bimodule.
- (v) $\iota : L \rightarrow (\mathcal{A}^{\text{ch}})_{\text{Lie}}$ is a chiral Lie morphism.
- (vi) λ is the aligned mixed distributive law relating the two module actions on K .

Every structure map is continuous for the chosen weight filtrations.

The three native bars are

$$\text{Bar}_{\text{Lie}}^{\text{ch}}(L), \quad \text{Bar}_{\text{Ass}}^{\text{ch}}(\mathcal{A}^{\text{ch}}), \quad \text{Bar}^\perp(\mathcal{A}^\perp).$$

The interface has a fourth bar. Its degree- r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 162.2 (Interface bar). Let $\overline{\mathcal{A}}^{\text{ch}}$ and $\overline{\mathcal{A}}^\perp$ be the augmentation ideals. Set

$$\text{Bar}^{\text{mix}}(\mathbb{A}; K) = \bigoplus_{p, q \geq 0} (s\overline{\mathcal{A}}^{\text{ch}})^{\otimes p} \otimes sK \otimes (s\overline{\mathcal{A}}^\perp)^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K , the right action on K , and the mixed terms supplied by λ .

Theorem 162.3 (Four-colour differential). *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. $\square$

Compatible modules and the mixed centre

A compatible module is a triple $(M^{\text{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K . Let $\text{Mod}_{\mathbb{A}}^{\text{mix}}$ be their stable category.

Definition 163.1 (Mixed derived centre). The mixed derived centre is

$$Z_{\text{mix}}(\mathbb{A}) = \text{End}\left(\text{id}_{\text{Mod}_{\mathbb{A}}^{\text{mix}}}\right).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 163.2 (Universal interior action). $Z_{\text{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every coherent interior action on the mixed boundary category factors uniquely through this centre. Morita-equivalent complete mixed data have equivalent centres.

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. An interior operator acts naturally on every compatible module and therefore defines an endomorphism of the identity. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. $\square$

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\text{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V , modulo the chiral relations already present in V . Weight completion gives $\widehat{\text{Coll}}(V)$.

Theorem 164.1 (Universal collar). *For every complete transversely ordered chiral algebra B and every chiral morphism $V \rightarrow B$, there is a unique continuous E_1 morphism*

$$\widehat{\text{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. $\square$

The full envelope of V is obtained by taking $L = V_{\text{Lie}}$, $A^{\text{ch}} = U^{\text{ch}}(L)$, $A^\perp = \widehat{\text{Coll}}(V)$, the regular interface, and the canonical aligned distributive law.

BOOK XXXIII

The deformation algebra and its BV envelope

The centre-extended deformation algebra

Fix a cofibrant coloured operad $\mathcal{P}_{\text{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\text{mix}}^!$ be its bar cooperad. The convolution complex

$$\text{Def}_{\mathbb{A}}^{\text{full}} = \text{Hom}_{\mathbb{S}}(\mathcal{P}_{\text{mix}}^!, \text{End}_{\mathbb{A}})$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 165.1 (Complete deformation L_{∞} algebra). *The convolution complex has a canonical complete L_{∞} structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_{∞} identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\text{mix}}$ to $\text{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. $\square$

165.1 FIELD EQUATIONS AND BIANCHI IDENTITIES

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_{\alpha}x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 165.2 (Universal Bianchi identity). *For every α ,*

$$d_{\alpha}\mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_{\alpha}^2 = 0$.

Proof. Both identities are the exponential form of the L_{∞} Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha}x$. $\square$

Shifted cotangent transgression

Let $\mathbf{I}$ be a complete elliptic L_∞ algebra on a three-manifold M . Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathbf{I}[1] = \mathbf{I}[1] \oplus \mathbf{I}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 166.1 (Universal BV envelope). *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \quad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. $\square$

Definition 166.2 (Universal chiral BV envelope). For a complete mixed datum $\mathbb{A}$, let $\mathbf{I}_\mathbb{A}^{\text{full}}$ be the analytic localization of $\text{Def}_\mathbb{A}^{\text{full}}$ on $X^{\text{an}} \times \mathbb{R}$. Define

$$\mathbf{BEnv}_3^{\text{full}}(\mathbb{A}) = T^*[-1] \text{MC}(\mathbf{I}_\mathbb{A}^{\text{full}}).$$

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ : the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T\mathfrak{X})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 167.1 (Oriented BV complex). *The operator Δ_μ has order two and satisfies*

$$\begin{aligned} \Delta_\mu^2 &= 0, & [d_{\mathfrak{X}}, \Delta_\mu] &= 0, \\ \Delta_\mu(fg) &= \Delta_\mu(f)g + (-1)^{|f|}f\Delta_\mu(g) + (-1)^{|f|}\{f, g\}. \end{aligned}$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[[\hbar]], d_{\mathfrak{X}} + \hbar\Delta_\mu \right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. $\square$

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}} \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 167.2 (Intrinsic quantum complex of a mixed datum). For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\text{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\text{MC}(\mathbf{I}_{\mathbb{A}}^{\text{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 167.3 (Quantum master equation). *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_{\mu}S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_{\mu})e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

Perturbative factorization realization

Choose elliptic gauge-fixing data for $\mathcal{I}_{\mathbb{A}}^{\text{full}}$. The heat kernel gives a propagator on $\mathcal{M} \times \mathcal{M} \setminus \Delta$. Fulton–MacPherson compactification resolves simultaneous collisions of vertices.

Theorem 168.1 (Graphwise factorization). *Renormalized graph weights that extend to the compactified configuration spaces and satisfy the quantum master equation define a factorization algebra of quantum observables. The boundary restriction of this factorization algebra realizes the original mixed datum $\mathbb{A}$.*

Proof. Disjoint unions of configurations multiply graph weights. Stokes’ theorem on each compactified configuration space expresses the differential of a weight as the sum over boundary contractions. The quantum master equation cancels the codimension-one contractions. The resulting cochains satisfy Weiss descent. Restriction to a collar recovers the generating operations used to define the deformation algebra. $\square$

BOOK XXXIV

Boundary duality, exchange, and reduction

The Koszul–Morita square

Let P be a dualizable boundary state for $\mathcal{A}^\perp$. Its boundary dual is

$$\mathcal{A}_P^\dagger = \mathrm{RHom}_{\mathcal{A}^\perp}(P, P).$$

The two-sided bar sends the diagonal bimodule to a diagonal bicomodule.

Theorem 169.1 (Mixed Koszul–Morita transport). *On the positive complete locus, the universal twisting cochains identify complete mixed modules with compatible comodules over the four bar objects. Derived tensor products of interfaces become derived cotensor products. The mixed centre and mixed Hochschild chains are preserved.*

Proof. Apply one-colour bar–cobar duality to each associative colour. The interface bar is a bimodule bar and its filtration is finite in each total weight. The aligned distributive law identifies the two orders of transformation. Eilenberg–Moore comparison sends tensor to cotensor. Endomorphisms of the diagonal and its trace are invariant under the induced Morita equivalence. $\square$

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathcal{L}$ with a faithful fibre functor ω has coend

$$O(\mathbb{G}) = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M).$$

Theorem 170.1 (Quantum symmetry from exchange). *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R -matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathcal{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. $\square$

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 171.1 (BRST–bar bicomplex). *The total differential*

$$D = Q + b_{\text{ch}} + b_{\perp} + b_{\text{mix}}$$

satisfies $D^2 = 0$. A bounded-below complete filtration gives a convergent spectral sequence

$$E_1 = H_Q(\text{Bar}^{\text{mix}}(\mathbb{A}; K)) \implies H_D(\text{Bar}^{\text{mix}}(\mathbb{A}; K)).$$

When Q -cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_1 differential with the bar differential of the reduced operations. $\square$

BOOK **XXXV**

Conformal covariance and Einstein completion

Projective connections and the Virasoro tangent complex

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c . At a projective connection p , its tangent complex is locally

$$\Omega^{\circ, \bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{\circ, \bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon \partial p + 2(\partial \epsilon) p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 172.1 (Projective covariance). *A conformal map $\text{Vir}_c \rightarrow \mathcal{A}$ defines a formal family of mixed structures over $\mathcal{P}_c(X)$. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. $\square$

Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^c J_c, \quad [J_a, P_b] = \epsilon_{ab}{}^c P_c, \quad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^c J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 173.1 (Cartan–Einstein equivalence). *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \quad g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [1, 74]. $\square$

Four equivalent linear complexes

Fix a constant-curvature solution (e, ω) . There are four deformation complexes.

- (i) The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.
- (ii) The Einstein–Cartan complex on $(\delta e, \delta \omega)$ with linearized torsion and curvature.
- (iii) The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.
- (iv) The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 174.1 (Chain equivalence of gravitational presentations). *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\delta \omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. $\square$

Brown–Henneaux and the projective boundary

Write the AdS connection as

$$A^\pm = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 175.1 (Brown–Henneaux central charge). *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [13]. $\square$

Einstein completion of a conformal pair

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 176.1 (Einstein BV completion). The Einstein completion is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1] \mathrm{MC}(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L, A_R}).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L, A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 176.2 (Einstein completion theorem). *The completion has a canonical BV action. Its pure gravitational sector is three-dimensional Einstein gravity. Its matter sector is the relative deformation theory of the conformal pair. The coupling is the moment map for the projective and diffeomorphism actions. The Bianchi identity of the full system is the L_∞ identity of the semidirect product.*

Proof. The homotopy fibre product matches the projective gauge fields and retains the relative matter directions. The gravitational algebra acts on those directions by conformal transport. The semidirect product is therefore an L_∞ algebra. Shifted cotangent transgression gives the BV action and its master equation. Restriction to the gravitational summand gives the Cartan action. The mixed term is the Hamiltonian pairing of the gauge field with the matter moment map. $\square$

Higher-spin completion

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^N V_{2s-2}, \quad \sum_{s=2}^N (2s-1) = N^2 - 1.$$

Theorem 177.1 (Principal higher-spin spectrum). *The principal embedding gives one massless gauge field of each integer spin $2, 3, \dots, N$. Classical Drinfeld–Sokolov reduction of the boundary affine Poisson algebra gives the principal $\mathcal{W}_N$ algebra. A pair of principal $\mathcal{W}_N$ boundary data therefore has an $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ Einstein–Cartan completion with the same spin spectrum.*

Proof. The irreducible $\mathfrak{sl}_2$ module V_{2s-2} corresponds to a frame-like field of spin s . The dimension identity exhausts the adjoint. Principal Drinfeld–Sokolov reduction produces one generating current in each conformal weight $2, \dots, N$. The boundary currents and bulk gauge fields therefore match representation by representation [15, 44]. $\square$

BOOK XXXVI

Exact gravitational complexes and quantum tests

The four complexes in coordinates

178.1 FLAT ADJOINT COMPLEX

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M, \mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

178.2 EINSTEIN–CARTAN COMPLEX

Write a gauge parameter as (ξ^a, λ^a) and a fluctuation as $(u^a, v^a) = (\delta e^a, \delta \omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a{}_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2} \epsilon^a{}_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a{}_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2} \epsilon^a{}_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 178.1 (Chain identity). *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ . Replace the background curvature by $-\ell^{-2} e \wedge e$ and use $D_\omega e = 0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. $\square$

178.3 METRIC COMPLEX

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab} e^a{}_{(\mu} u^b{}_{\nu)}.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2} \left(-\nabla^2 h_{\mu\nu} - \nabla_\mu \nabla_\nu h + \nabla_\mu \nabla^\rho h_{\rho\nu} + \nabla_\nu \nabla^\rho h_{\rho\mu} \right. \\ \left. - g_{\mu\nu} (\nabla^\rho \nabla^\sigma h_{\rho\sigma} - \nabla^2 h) \right) - \Lambda (h_{\mu\nu} - g_{\mu\nu} h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 178.2 (Explicit metric reduction). *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v . The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. $\square$

178.4 CALABI COMPLEX

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

Boundary BFV theory and the Virasoro moment map

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_{\partial} = \frac{k}{4\pi} \int_{\partial M} \langle \partial A \wedge \partial A \rangle.$$

A chiral polarization selects $A_{\bar{z}}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 179.1 (Boundary current algebra). *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k . Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k + h^{\vee})} : J^a J_a : (z) + \text{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_{∂} is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. $\square$

Matter stress tensor and the Einstein moment map

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\text{matt}} : \mathfrak{m} \longrightarrow \Omega^{\circ, \bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 180.1 (Conservation from the master equation). *The coupled BV action*

$$S_{\text{tot}} = S_{\text{grav}} + S_{\text{matt}} + \langle h, \mu_{\text{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^\mu T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. $\square$

One-loop gravity and analytic torsion

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 181.1 (One-loop determinant). *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{1-loop}(\mathcal{A})| = \tau_{RS}(M, \text{ad } \mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [69, 74]. $\square$

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

BTZ, Cardy, and the area law

For a unitary modular boundary theory with left and right central charges c_L, c_R , the asymptotic entropy is

$$S_{\text{Cardy}} = 2\pi \sqrt{\frac{c_L}{6} \left(L_o - \frac{c_L}{24} \right)} + 2\pi \sqrt{\frac{c_R}{6} \left(\bar{L}_o - \frac{c_R}{24} \right)}.$$

For the BTZ black hole,

$$L_o - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \quad \bar{L}_o - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \quad c = \frac{3\ell}{2G}.$$

Theorem 182.1 (Cardy–Bekenstein equality). *Substitution gives*

$$S_{\text{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy's theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [5, 16]. $\square$

Higher-spin holonomy and boundary charges

In $\mathfrak{sl}_N$ Chern–Simons gravity, a smooth Euclidean saddle is selected by the conjugacy class of the thermal holonomy. The invariant polynomials

$$\mathrm{Tr}(\mathrm{Hol}^s), \quad 2 \leq s \leq N,$$

fix the spin- s charges. Drinfeld–Sokolov reduction identifies the same charges with zero modes of the W_N currents.

Theorem 183.1 (Bulk–boundary charge equality). *For a principal $\mathfrak{sl}_N$ connection in Drinfeld–Sokolov gauge, the invariant polynomials of the spatial holonomy equal the classical W_N zero-mode Hamiltonians. Their Poisson brackets are the classical W_N brackets.*

Proof. Drinfeld–Sokolov gauge writes the connection as the principal raising operator plus lowest-weight components. The coefficients of the characteristic polynomial are the invariant polynomials and are polynomial functions of those components. Hamiltonian reduction gives the W_N Poisson structure on the same functions. $\square$

BOOK XXXVII

**Anomalies, Hall sources, and automorphic
completion**

Four native anomaly classes

The construction carries four curvature classes.

- (i) A local BV anomaly lies in the degree-one local deformation complex.
- (ii) A projective anomaly is the curvature of the determinant line over parameter space.
- (iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.
- (iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 184.1 (Curvature-killing extension). *Let κ be a closed central degree-two class in the deformation complex of a theory. Adjoining a degree-one generator u with $du = \kappa$ produces a central extension in which the lifted curvature vanishes. Tensoring with the inverse invertible field theory gives the geometric realization of the same cancellation.*

Proof. The extended differential squares to zero because $d\kappa = 0$ and κ is central. The new Maurer–Cartan equation contains the compensating field u . An invertible theory with curvature $-\kappa$ adds the opposite class under tensor product. $\square$

Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+ . A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Theorem 185.1 (Source-to-BV functor). *A morphism from the Hall boundary datum to a complete mixed boundary datum induces morphisms*

$$H^+ \longrightarrow A^{\text{ch}}, \quad Z_{\text{Hall}} \longrightarrow Z_{\text{mix}}(\mathbb{A}), \quad \mathbf{BEnv}_{\text{Hall}} \longrightarrow \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}).$$

Compatibility with coproduct, pairing, and braiding propagates to the quantum line category. Compatibility with cyclic traces propagates to the quantum and modular amplitudes.

Proof. The Hall map preserves associative multiplication. Relative operadic induction extends it to the mixed colours and interface. Natural transformation of module actions gives the centre map. Functoriality of convolution deformation algebras and shifted cotangent gives the BV map. The remaining structures are natural under monoidal and cyclic functors. $\square$

For $X = K_3 \times E$, Volume IV constructs the geometric BPS Borel $\mathfrak{b}_X$, the Igusa Borchers Borel $\mathfrak{b}_\Delta$, and their completed Lie amalgam. Applying the enveloping functor gives the cocommutative Hopf algebra

$$H_{X \boxtimes \Delta} = \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{b}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q -deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

Theorem 185.2 (Amalgamated source envelope). *Relative mixed induction sends $H_{X \boxtimes \Delta}$ to a complete mixed boundary datum $\mathbb{A}_{X \boxtimes \Delta}$. Its universal BV envelope has two canonical retractions*

$$\mathbf{BEnv}_3^{\text{full}}(\mathbb{A}_{X \boxtimes \Delta}) \longrightarrow \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}_X), \quad \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}_{X \boxtimes \Delta}) \longrightarrow \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}_\Delta).$$

A primitive comparison quotient induces the diagonal BV theory in which geometric BPS primitives and Igusa roots act as one field. The kernel of the quotient is the complete mixed interaction ideal.

Proof. Relative induction preserves coproducts and the common Cartan action, so the amalgamated Hopf algebra produces the amalgamated mixed datum. The Hopf retractions induce maps of mixed data. Functoriality of the centre, convolution deformation algebra, and shifted cotangent transgression gives the two BV retractions. A primitive comparison quotient is a quotient of mixed data, and functoriality carries its defining ideal to the corresponding ideal in the deformation and BV complexes. $\square$

The Igusa protected insertion

Let $\mathfrak{R}_\Delta$ be the Borchers–Ginzburg protected category of Volume III. Its ordinary Hall boundary is the quiver Hall category, its parity-refined line symmetry is the quantized Borchers superalgebra, its local–wrapped current object lies on the elliptic curve, and its first-order determinant is the oriented virtual-root Pfaffian.

Theorem 186.1 (Igusa insertion into the universal envelope). *The protected category $\mathfrak{R}_\Delta$ defines a complete mixed boundary datum on the elliptic curve. Put $\text{Den}_\Delta(Z) = D_5(2Z)$ and $j(Z) = Z/2$. Its universal BV envelope has primitive algebra $\mathfrak{g}_\Delta$, line symmetry $U_q(\mathfrak{g}_\Delta)$, and Weyl-shifted one-loop determinant Den_Δ . The doubled half-coordinate trace is $-(j^* \text{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ in the reduced enumerative normalization.*

Proof. The colored Ran current algebra supplies the chiral and transverse colours. The Hall category supplies ordered particle fusion and the interface to the three-Calabi–Yau completion. The quantum double supplies exchange. The root Dirac family supplies the determinant. The identities follow from the Borchers denominator and the primitive reduced $K_3 \times E$ theorem. $\square$

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\text{diag}, \text{ch}, \perp, \text{int}, \text{ex}, \text{cyc}, \text{mod}, \text{BV}, \text{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\text{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S .

Theorem 187.1 (Carrier descent). *When S and T form a relation-excise pair, the square*

$$\begin{array}{ccc} \text{Str}_{S \cup T}(\mathbb{A}) & \longrightarrow & \text{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \text{Str}_T(\mathbb{A}) & \longrightarrow & \text{Str}_{S \cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. $\square$

Definition 187.2 (Relation-complete carrier cover). Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i \in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Čech homotopy colimit

$$\text{hocolim}_{\emptyset \neq J \subseteq I} \mathcal{I}_{\bigcap_{j \in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 187.3 (Power-set coherence). *The assignment*

$$S \longmapsto \text{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i \in I}$ of U , the restriction map is an equivalence

$$\text{Str}_U(\mathbb{A}) \xrightarrow{\sim} \text{holim}_{\emptyset \neq J \subseteq I} \text{Str}_{\bigcap_{j \in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. $\square$

Definition 187.4 (Exact-support deformation complex). For $S \subseteq \mathfrak{C}$, let $\text{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S , and set

$$\text{Def}_{=S} = \text{cofib} \left(\text{hocolim}_{T \subsetneq S} \text{Def}_{\leq T} \longrightarrow \text{Def}_{\leq S} \right).$$

Thus $\text{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S .

Theorem 187.5 (Carrier-support spectral sequence). *Let $\text{Def}_{\mathbb{A}}^{\text{full}}$ be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S|=p}} H^{p+q}(\text{Def}_{=S}) \implies H^{p+q}(\text{Def}_{\mathbb{A}}^{\text{full}}).$$

A class on the S -summand is a deformation or curvature class whose first complete geometric carrier is S . The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its p th associated-graded quotient is the direct sum of the exact-support complexes $\text{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. $\square$

Sixteen structural laws

The theory is governed by the following laws.

- (i) A geometric carrier determines the type of its operation.
- (ii) The diagonal stores the complete principal-part filtration.
- (iii) The interval stores ordered fusion.
- (iv) The plane stores exchange.
- (v) The circle stores trace.
- (vi) The stable curve stores modular sewing.
- (vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.
- (viii) Chiral PBW compares the last two on enveloping objects.
- (ix) The interface bar records the mixed distributive law.
- (x) The derived centre is the universal interior action.
- (xi) The deformation algebra is the tangent algebra of the centre-extended datum.
- (xii) Maurer–Cartan curvature is the field equation.
- (xiii) The L_∞ identity is the Bianchi identity.
- (xiv) Shifted cotangent is the universal classical BV envelope.
- (xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.
- (xvi) Einstein completion is projective matching followed by gravitational cotangent transgression.

Final synthesis

The construction proceeds by a single chain of functors:

$$\begin{aligned} \mathbb{A} &\xrightarrow{Z_{\text{mix}}} Z_{\text{mix}}(\mathbb{A}) \xrightarrow{\text{Def}} \mathfrak{l}_{\mathbb{A}}^{\text{full}} \xrightarrow{T^*[-1]} \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}), \\ \mathbf{BEnv}_3^{\text{full}}(\mathbb{A}) &\xrightarrow{\text{projective matching}} \mathbf{EinBV}_3(\mathcal{A}_L, \mathcal{A}_R). \end{aligned}$$

Hall and Calabi–Yau sources enter on the left. Quantum groups enter through planar exchange and the line category. Automorphic products enter as characters of root-graded protected objects. Each arrow preserves the structure carried by its source and adds the structure specified by its geometric target.

BOOK **XXXVIII**

**Technical appendices to the preceding
book**

The Chern–Simons normalization

With invariant pairing $\langle J_a, P_b \rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\text{CS}}[\mathcal{A}] = \frac{k}{4\pi} \int_M \langle \mathcal{A} \wedge d\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \rangle.$$

For $\mathcal{A}^\pm = \omega \pm \ell^{-1}e$ one finds

$$S_{\text{CS}}[\mathcal{A}^+] - S_{\text{CS}}[\mathcal{A}^-] = \frac{k}{2\pi\ell} \int_M \epsilon_{abc} e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2} e^b \wedge e^c \right) + \int_{\partial M} (\cdots).$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \dots, N$.

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \quad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\text{ch}} \oplus E_{\perp}) = \det(E_{\text{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

BOOK XXXIX

Detailed treatment of Volume I

Five carriers, three noncommutativities, one type discipline

193.1 THE LOCAL EXPERIMENT

Let X be a smooth complex algebraic curve. Place observables in a product chart $U \times I \subset X^{\text{an}} \times \mathbb{R}$, with U a small disc and I an oriented interval. Two independent short-distance motions are visible before any algebra is chosen.

In the first motion, the points in U collide while their interval positions remain fixed. The result is not merely a residue. It is a distribution supported on a partial diagonal, with all normal jets and their coordinate-change law. In the second motion, two subintervals become adjacent while the points of U remain separated. The result is an ordered associative composition. Thickening I to a plane permits one interval to pass around another and creates braid transport. Closing the interval creates a circle trace. Allowing the complex curve to degenerate creates clutching and sewing.

Thus the five carriers are

diagonal, interval, plane, circle, stable curve.

Their native algebraic structures are, respectively, a chiral distribution, an E_1 -product, braid transport, Hochschild/cyclic trace, and a modular action. They are related by geometry, but none is a synonym for another.

Theorem 193.1 (Stokes-to-master principle). *Let C be an oriented manifold with corners, let (E, d_E) be a cochain complex, and assign to every oriented codimension-one face F an odd operator ρ_F on E . Assume the assignment restricts functorially to corners and*

$$d_E \rho_F + \rho_F d_E = 0.$$

For $D = d_E + \sum_F \rho_F$, the component of D^2 supported on a codimension-two stratum S is the oriented sum of the composites $\rho_{F_2} \rho_{F_1}$ over flags $S \subset F_2 \subset C$. Hence $D^2 = 0$ exactly when every corner identity holds.

Proof. Expand D^2 . The internal square vanishes and the mixed terms cancel by the chain-map hypothesis. The remaining terms are ordered pairs of incident faces. Grouping by the common codimension-two stratum produces the cellular coefficient of $\partial^2 C$, and the two induced corner orientations are opposite. This is the entire sign mechanism. The algebraic content enters only through the operators decorating the faces. $\square$

Associativity, chiral Jacobi or chiral associativity, the Yang–Baxter equation, the BV master equation, and modular sewing all have this form, but they live on different compactified parameter spaces. A universal boundary argument explains the shape of the equations; it does not identify their coefficient maps.

193.2 THE RAN COEFFICIENT AND ITS TWO PRODUCTS

Let $\text{Fin}_{\neq \emptyset}^{\text{surj}}$ denote nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a partial diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \quad (x_j)_{j \in J} \longmapsto (x_{\alpha(i)})_{i \in I}.$$

A right D -module on $\text{Ran } X$ is represented by a coherent family $F_I \in \text{DMod}(X^I)$ with extraordinary pullback identifications along these diagonals. The unital completion adjoins the empty set.

For a partition $\pi = \{B_1, \dots, B_r\}$ of I , let $U_\pi \subset X^I$ be the locus where points in distinct blocks are disjoint, and write $j_\pi : U_\pi \hookrightarrow X^I$.

Definition 193.2 (Factorisable coefficient). A factorisable right D -module is a Ran family F equipped with coherent equivalences

$$j_\pi^! F_I \simeq j_\pi^! \left(\boxtimes_{B \in \pi} F_B \right)$$

for every partition, compatible with refinement, permutation, and diagonal transition maps. The chosen stable presentable enhancement is denoted $\text{FactD}(X)$.

There are two monoidal products.

The pointwise coefficient product is

$$(F \otimes! G)_I = F_I \otimes^! G_I.$$

It does not add support cardinalities. It is the product in which endomorphisms of a fixed boundary condition compose.

Chiral convolution is the Day convolution for disjoint union:

$$(F \star_{\text{ch}} G)_I = \bigoplus_{I=I_1 \sqcup I_2} (j_{I_1, I_2})_* j_{I_1, I_2}^! (F_{I_1} \boxtimes G_{I_2}).$$

It adds support cardinalities and is the ambient product for the Ran-space chiral Chevalley theory.

Proposition 193.3 (Cardinality pro-nilpotence). *Let $\mathcal{R}_{\geq r}$ be Ran objects whose components vanish below cardinality r . Then*

$$\mathcal{R}_{\geq r} \star_{\text{ch}} \mathcal{R}_{\geq s} \subseteq \mathcal{R}_{\geq r+s}.$$

Consequently, reduced chiral convolution is nilpotent modulo cardinality $> N$, and its inverse limit is pro-nilpotent.

Proof. Every summand is indexed by a disjoint union of the supporting finite sets, so cardinalities add. An $(N+1)$ -fold product of reduced objects has support cardinality at least $N+1$ and vanishes in the N -th quotient. $\square$

The same statement is false for $\otimes!$, whose cardinality degree is zero. This elementary distinction is the first reason that the two bar constructions must not be conflated.

193.3 THREE FORMS OF NONCOMMUTATIVITY

Definition 193.4 (Transversely ordered chiral algebra). A transversely ordered chiral algebra is an augmented algebra

$$\mathcal{A} \in \text{Alg}_{\mathbb{E}_1}^{\text{aug}}(\text{FactD}(X), \otimes_{\dagger}).$$

The factorisable coefficient carries collision on X . The internal $\mathbb{E}_1$ -multiplication $m_{\perp} : \mathcal{A} \otimes_{\dagger} \mathcal{A} \rightarrow \mathcal{A}$ carries order along a transverse line.

Definition 193.5 (Associative chiral algebra). An associative chiral algebra is an augmented $\mathbb{E}_1$ -algebra for chiral convolution,

$$\mathcal{C} \in \text{Alg}_{\mathbb{E}_1}^{\text{aug}}(\mathcal{R}_X, \star_{\text{ch}}),$$

or, in a local geometric presentation, an algebra over the non-symmetric associative chiral operad Ass_X^{ch} . Its multiplication composes ordered chiral insertions and may be nonlocal in the vertex-algebraic sense.

Definition 193.6 (Braided or exchange enhancement). An exchange enhancement is an $\mathbb{E}_2$ -lift, a flat meromorphic transport on labelled configuration spaces, or equivalent braided categorical data. It supplies paths that exchange two ordered factors and is not part of a bare $\mathbb{E}_1$ -algebra.

These are three separate noncommutativities:

- (i) noncommutativity of an ordered word in the transverse direction;
- (ii) noncommutativity or nonlocality of chiral fields under formal-disc composition;
- (iii) nontrivial braid exchange after a second transverse direction or meromorphic connection is supplied.

A theory may carry any one, any compatible pair, or all three.

Definition 193.7 (Doubly associative chiral algebra). A doubly associative chiral algebra is a Ran/factorisation object $\mathcal{A}$ equipped with

$$m_{\perp} : \mathcal{A} \otimes_{\dagger} \mathcal{A} \rightarrow \mathcal{A}, \quad m_X : \mathcal{A} \star_{\text{ch}} \mathcal{A} \rightarrow \mathcal{A},$$

coherent $\mathbb{E}_1$ -structures for each multiplication, and a mixed locality datum on every common disjoint decomposition. In a cofibrant model, this is an algebra over the two-coloured operadic tensor product

$$W\text{Ass}_{\perp} \otimes W\text{Ass}_X^{\text{ch}}$$

with the mixed cells retained rather than collapsed.

The phrase “the mixed locality datum” is deliberate. The ambient products do not admit a global interchange isomorphism in general. The exact common-decomposition retract is constructed in [chapter 194](#).

No-go. *There is no functor, natural on all factorisable coefficients, that reconstructs m_X from $m_{\perp}$, or $m_{\perp}$ from m_X . Indeed, fix a coefficient F . Distinct associative algebras B give distinct transverse products on $B \otimes F$ with the same chiral coefficient. Conversely, distinct nonlocal chiral products can be placed on a fixed underlying coefficient while the transverse product remains the same. A comparison requires extra geometric data, not a change of notation.*

193.4 BOUNDARY AND DEFECT ORIGIN

Let $\mathcal{C}$ be a $\text{FactD}(X)$ -linear factorisation category and $b \in \mathcal{C}$ a factorisation-compatible boundary condition. The internal endomorphism object

$$A_b = \text{RHom}_{\mathcal{C}}(b, b)$$

composes in an order and varies factorisably in X .

Theorem 193.8 (Factorisation Morita chart). *Assume that b restricts to a compact dualisable generator on a factorising basis, that these local generator equivalences commute with restriction, and that both sides satisfy Weiss descent. Then A_b is a transversely ordered chiral algebra and*

$$\text{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \xrightarrow{\sim} \text{Mod}_{A_b}(\text{FactD}(X))$$

is an equivalence of factorisation categories.

Proof. Composition gives the internal E_1 -structure. Factorisation of b on disjoint discs makes composition a morphism of coefficient systems. The ordinary compact-generator Morita theorem applies on each basis element. Compatibility on overlaps and Weiss descent glue the local equivalences. $\square$

The algebra A_b is a chart, not the Morita-invariant totality. Changing the generator changes the algebra while preserving the factorisation category. This distinction becomes essential when comparing a boundary algebra with a physical bulk.

193.5 THE COMPLETE DIAGONAL COEFFICIENT

Let $D = \Delta(X) \subset X^2$. The meromorphic extension $\mathcal{O}_{X^2}(*D)$ has pole filtration

$$P_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}((m+1)D), \quad m \geq 0,$$

and

$$\text{gr}_m^P \mathcal{O}_{X^2}(*D) \simeq \mathcal{O}_D((m+1)D) \simeq N_{D/X^2}^{\otimes(m+1)}.$$

Because $N_{\Delta/X^2} \simeq T_X$, the coefficient of $(z-w)^{-m-1}$ is the local splitting of the $(m+1)$ -st normal symbol. Coordinate changes act upper triangularly on these splittings. The filtered distribution is invariant; individual modes are local coordinates on it.

Proposition 193.9 (Complete-coefficient principle). *A chiral operation is the entire diagonal extension and structure morphism. Residues, jets, delta derivatives, propagator coefficients, and formal modes are functorial projections or splittings of this object. No finite list of residues reconstructs the entire operation without a stated bounded-pole hypothesis.*

No-go. *There is no coordinate-free linear map from logarithmic one-forms to arbitrary finite principal parts that is inverse to inclusion. A logarithmic residue detects the simple normal coefficient. Higher principal parts transform through higher jets of a coordinate change and therefore require the full filtered normal object. Any formula that extracts all modes from $d \log(z-w)$ has silently chosen additional coordinates or a connection.*

Aligned decompositions and the exact locality law

194.1 LINEAR SPECIES

Let $\mathcal{V}$ be an additive stable symmetric monoidal category with finite direct sums. A linear species is a functor $F : \text{Fin}^{\sim} \rightarrow \mathcal{V}$. It carries the Hadamard product

$$(F \boxtimes_H G)(I) = F(I) \otimes G(I)$$

and the Cauchy product

$$(F \boxtimes_C G)(I) = \bigoplus_{I=I_1 \sqcup I_2} F(I_1) \otimes G(I_2),$$

where the sum is over ordered decompositions.

The component of

$$(F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2)$$

is indexed by a pair (d, e) of ordered decompositions of I , while the component of

$$(F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2)$$

is indexed by one decomposition d . The latter is exactly the diagonal sector (d, d) of the former.

Theorem 194.1 (Aligned-decomposition splitting). *There are canonical natural transformations*

$$\begin{aligned} \iota : (F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2) &\longrightarrow (F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2), \\ \pi : (F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2) &\longrightarrow (F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2), \end{aligned}$$

with

$$\pi \iota = \text{id}, \quad e := \iota \pi, \quad e^2 = e.$$

The image of e is the common-decomposition summand. These maps are compatible with associativity, units, and the symmetry of $\boxtimes_H$.

Proof. At arity I , include the summand indexed by d into the summand indexed by (d, d) ; this is ι . Project the double sum onto those pairs with equal decompositions; this is π . The identities are immediate componentwise. For three Cauchy factors, both coherence composites include, respectively project onto, triples with one common ordered decomposition. Hence every coherence diagram reduces to equality of the same direct-summand map. $\square$

No-go. *Unless all independent-decomposition summands vanish, ι is not surjective and π is not injective. Therefore Hadamard and Cauchy products are not related by a universal interchange isomorphism. Any theorem using a strong duoidal law must either restrict to an aligned subcategory or retain the idempotent e .*

Definition 194.2 (Aligned window). A full replete subcategory of species is aligned if it is closed under both products and if the projector e acts as the identity on every iterated mixed product of its objects. On such a window, ι and π are inverse and the two products form a strong duoidal pair.

Finite support in a fixed decomposition type gives an elementary aligned window. The full Ran theory is not one.

194.2 SHEAF-LEVEL LOCALITY

Fix an ordered decomposition $d = (I_1, I_2)$ and the corresponding disjointness inclusion $j_d : U_d \hookrightarrow X^I$. A factorisation structure gives

$$\delta_d : j_d^! A_I \xrightarrow{\sim} j_d^! (A_{I_1} \boxtimes A_{I_2}).$$

An ambient pointwise multiplication $m : A \otimes! A \rightarrow A$ belongs to $\text{FactD}(X)$ exactly when the square

$$\begin{array}{ccc} j_d^! (A_I \otimes! A_I) & \xrightarrow{j_d^! m} & j_d^! A_I \\ \delta_d \otimes! \delta_d \downarrow & & \downarrow \delta_d \\ j_d^! ((A_{I_1} \otimes! A_{I_1}) \boxtimes (A_{I_2} \otimes! A_{I_2})) & \xrightarrow{m \boxtimes m} & j_d^! (A_{I_1} \boxtimes A_{I_2}) \end{array}$$

commutes for every d .

Theorem 194.3 (Exact locality criterion). *A pointwise E_1 -multiplication on the underlying Ran object defines a transversely ordered chiral algebra precisely when all of the displayed disjoint-locus squares and their higher E_1 coherences commute. Under finite-set indexing, these equations lie in the image of the aligned projector of theorem 194.1.*

Proof. A morphism in the category of factorisation objects is by definition a Ran morphism intertwining all factorisation equivalences. Applying this definition to every operation of the E_1 -algebra gives the stated squares. The same decomposition d occurs before and after multiplication, which is exactly the common-decomposition sector. $\square$

This criterion is stronger and more honest than saying that $\mathcal{A}$ is a “bimonoid” in a globally duoidal category. The bialgebra equation is a family of locality squares, not an unsupported isomorphism of mixed products.

194.3 THE TWO COPRODUCTS ON THE TRANSVERSE BAR

Once $m_\perp$ is a morphism of factorisation objects, every face and degeneracy in the simplicial bar is a morphism of factorisation objects. Geometric realization therefore retains the chiral factorisation coproduct. Independently, the normalized word complex has the deconcatenation coproduct

$$\Delta_{\text{dec}}[a_1 | \cdots | a_r] = \sum_{p=0}^r [a_1 | \cdots | a_p] \otimes [a_{p+1} | \cdots | a_r].$$

The first coproduct separates support on the curve; the second cuts the transverse word. They commute on the aligned locality sector because they operate on independent indices. They should not be called the same coproduct.

194.4 ALIGNED PROJECTION AND MIXED LOCALITY

The aligned projector has four consequences.

- (i) It prevents an independent pair of decompositions from being mistaken for one laboratory decomposition.
- (ii) It explains why locality squares are exact even though the ambient products are not strongly duoidal.
- (iii) It makes the bar inherit factorisation without pretending that all mixed species summands participate.
- (iv) It identifies the precise extra hypothesis needed to iterate transverse and chiral associative bars.

The idempotent records algebraically that the same decomposition is used in both directions.

The fourth item is a theorem rather than an observation.

Proposition 194.4 (Where a mixed distributive law can live). *The sector on which a mixed distributive law between $\otimes_!$ and $\star_{\text{ch}}$ is geometrically defined is the common-decomposition sector, that is the image of the idempotent e of theorem 194.1. Off that sector the two products admit no interchange, only the retraction of theorem 194.1, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. By theorem 194.1 the canonical comparison in the additive model is ι , with retraction π , and $\pi\iota = \text{id}$ while $\iota\pi = e \neq \text{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e acts as the identity, which is $\text{im } e$ by definition 194.2. $\square$

Remark 194.5 (The law is located, not constructed). Proposition 194.4 does not exhibit a λ . No example in the present corpus does. Zamolodchikov–Faddeev algebras of chapter 208 are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction; producing λ for one of them, and verifying that it lands in $\text{im } e$, is the outstanding construction on which the doubly noncommutative object depends for content.

Rectangular recognition and its exact hypotheses

195.1 PRODUCT-DISK FACTORISATION

Let Disk_X be a factorising basis of sufficiently small analytic discs in X^{an} , and let $\text{Disk}_I^{\text{ord}}$ be the little oriented intervals operad. The product basis consists of rectangles $U \times I$. A multimorphism is a family of pairwise disjoint rectangular embeddings, ordered in the interval coordinate.

Definition 195.1 (Rectangular holomorphic–topological factorisation algebra). A rectangular factorisation algebra $\mathcal{O}$ on $X^{\text{an}} \times \mathbb{R}$ satisfies:

- (i) Weiss descent on the product-disk basis;
- (ii) local constancy under isotopy in the interval variable;
- (iii) a holomorphic algebraisation in the X -variable by a factorisable right D -module;
- (iv) compatibility of that algebraisation with the finite colimits and tensor products used to form the structure maps.

Recognition Theorem 195.2 (Rectangular recognition). *Under the preceding algebraisation and descent hypotheses, restriction to a standard interval induces an equivalence*

$$\text{Fact}_{\text{hol,lc}}^{\text{rect}}(X \times \mathbb{R}) \simeq \text{Alg}_{E_1}(\text{FactD}(X), \otimes_!).$$

The internal multiplication is fusion of ordered subintervals; every operation is a morphism of chiral coefficient systems.

Proof. A product-disk algebra is an algebra over the Boardman–Vogt tensor product $\text{Disk}_X \otimes \text{Disk}_I^{\text{ord}}$. The operadic exponential law identifies its algebras with E_1 -algebras internal to Disk_X -algebras. Local constancy identifies all sufficiently small intervals through a contractible space of choices, and ordered embeddings produce the E_1 -operations. Conversely, an internal E_1 -algebra evaluates a family of rectangles by external factorisation in X , followed by ordered multiplication. The tensor-product relation is exactly the locality square. Weiss descent and the algebraisation hypothesis extend the equivalence from the basis to the asserted categories. $\square$

Warning 195.3. The theorem is not an equivalence between arbitrary analytic factorisation algebras and algebraic D -modules. Irregular singularities, infinite products, non-algebraic monodromy, and failure of continuity under completion are genuine obstructions. These are hypotheses, not notation.

195.2 BLOCK-RAN RECOGNITION

An ordered block profile is a finite ordered list $\mathbf{I} = (I_1, \dots, I_r)$ of finite chiral label sets. Its carrier is $X^{I_1} \times \dots \times X^{I_r}$. Chiral refinements occur inside a block; transverse fusion joins adjacent blocks. Let $\mathbf{BRan}_X$ be the Grothendieck construction of the simplicial associative indexing category acting on the Ran diagram.

Theorem 195.4 (Block-Ran recognition). *Transversely ordered chiral algebras are equivalent to coCartesian sections over $\mathbf{BRan}_X$ that are Segal in the block direction and factorising inside each chiral block. Marking a left endpoint, right endpoint, or both endpoints gives left modules, right modules, and bimodules. Gluing marked intervals computes relative tensor products.*

Proof. The simplicial nerve of an internal E_1 -algebra has r -simplices $A^{\otimes r}$ and satisfies the Segal condition. The active map $[2] \rightarrow [1]$ recovers multiplication, $[0] \rightarrow [1]$ recovers the unit, and the simplicial identities recover all coherence. Factorisation inside each block says these maps live in $\mathbf{FactD}(X)$. Straightening and unstraightening provide inverse constructions. The marked statements are the corresponding coloured operads. $\square$

195.3 OPPOSITES AND THREE REVERSALS

Reflection of the interval, $t \mapsto -t$, gives $A^{\text{op}, \perp}$ and reverses only the transverse multiplication. Reversal of an exchange path acts on a braid or R -matrix. Verdier duality acts on the D -module coefficient. These operations can agree in a rigid unitary model, but in general they have different domains and cannot be merged into a single “opposite” operation.

No-go. *The ordered chamber $\text{Conf}_n^<(\mathbb{R})$ is contractible and contains no path exchanging two labelled points. Hence an E_1 -product determines associativity but no braid generator and no R -matrix. Exchange requires a second transverse direction, a meromorphic connection, or braided categorical input.*

Associative chiral algebras and nonlocal vertex fields

196.1 WHY THE SECOND E_1 IS NECESSARY

The transverse theory keeps a conventional chiral coefficient and adds a line-order product. This is the correct language for boundary operators in a holomorphic–topological theory. It does not, by itself, construct an associative chiral multiplication whose formal fields are nonlocal. The original motivation for an E_1 -chiral Koszul duality therefore requires an additional operad.

Let $\text{Ass}_X^{\text{ch}}(I)$ denote the complex of ordered chiral operations with inputs labelled by I . Geometrically, an operation is defined on the complement of collision diagonals, carries an ordering of the input flags, and extends as an allowed distribution across a specified diagonal. Operadic substitution is composition along nested partial diagonals, with the order of flags retained. Forgetting the order and antisymmetrising maps to the usual chiral Lie operad; imposing braided locality gives quantum-vertex-type quotients.

Definition 196.1 (Associative chiral multiplication). An associative chiral multiplication on a coefficient C is an algebra structure over Ass_X^{ch} . In arity two it is a distributional map

$$\mu_X : j_* j^*(C \boxtimes C) \longrightarrow \Delta_! C$$

with ordered inputs. In arity three, the two nested collision composites are related by the specified associative homotopy rather than by antisymmetrised Jacobi alone.

The ordinary Beilinson–Drinfeld chiral algebra is Lie-like: its binary operation is skew after the appropriate transposition and satisfies Jacobi. An associative chiral algebra contains more data. Taking the commutator of ordered chiral multiplication gives a chiral Lie operation when the transposition and singular extensions are defined.

196.2 FORMAL-DISC REALIZATION

Let V be a vector space with vacuum $\mathbf{1}$ and a field map

$$Y(-, z) : V \longrightarrow \text{End}(V)((z)), \quad a \longmapsto Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}.$$

Definition 196.2 (Nonlocal vertex algebra). A nonlocal vertex algebra consists of $(V, Y, \mathbf{1})$ such that

- (i) $Y(\mathbf{1}, z) = \text{id}_V$;

- (ii) $Y(a, z)\mathbf{1} \in V[[z]]$ and its constant term is a ;
- (iii) for every $a, b, c \in V$, there exists $N \geq 0$ with

$$(z_0 + z_2)^N Y(a, z_0 + z_2)Y(b, z_2)c = (z_0 + z_2)^N Y(Y(a, z_0)b, z_2)c.$$

No commutativity or locality axiom is imposed.

The weak associativity identity is the formal-disc image of an associative chiral three-corolla after denominators have been cleared. It is the field-theoretic avatar of an E_1 -chiral structure, not of a transverse E_1 -word.

Definition 196.3 (Quantum vertex enhancement). A quantum vertex enhancement consists of a non-local vertex algebra together with an invertible rational quantum Yang–Baxter operator

$$S(z) : V \otimes V \longrightarrow V \otimes V \otimes \mathbb{k}((z))$$

satisfying:

$$\begin{aligned} S_{21}(-z)S(z) &= \text{id}, \\ S_{12}(z_1)S_{13}(z_1 + z_2)S_{23}(z_2) &= S_{23}(z_2)S_{13}(z_1 + z_2)S_{12}(z_1), \\ [D \otimes \text{Id}, S(z)] &= -\frac{d}{dz}S(z), \end{aligned}$$

plus the braided locality and hexagon compatibility relating S to Y . Here $D(a) = \partial_z Y(a, z)\mathbf{1}|_{z=0}$.

Different authors package nondegeneracy and braided locality in slightly different but equivalent ways on suitable topologically free categories. The invariant content is fixed: the exchange operator is extra structure and its three-body compatibility is Yang–Baxter.

Recognition Theorem 196.4 (Formal-disc recognition). *Suppose an associative chiral algebra on the formal disc is translation equivariant, unital, complete for the pole/weight filtration, and its binary operations admit Laurent expansions with finite negative part on every vector. Then its state space carries a nonlocal vertex algebra structure. Conversely, a complete nonlocal vertex algebra satisfying the corresponding covariance and continuity conditions reconstructs an associative chiral operation on the formal disc. A compatible S -operator reconstructs the braided/quantum chiral enhancement.*

Proof sketch with the exact construction. Expand the binary chiral distribution in a formal coordinate to define the modes $a_{(n)}b$ and hence $Y(a, z)b$. The unit and translation-equivariance supply the vacuum and creation axioms. The two boundary strata of the compactified ordered three-point collision give the two iterates; chiral associativity says their expansions agree after multiplication by a sufficient power of the common denominator, which is weak associativity. Conversely, weak associativity gives compatible expansions on nested formal configuration domains. Completeness glues them into the required distributional maps. The Yang–Baxter, unitarity, and shift equations are exactly the three-point, two-point reversal, and translation compatibilities for braided exchange. $\square$

No-go. *An object of $\text{Alg}_{E_1}(\text{FactD}(X), \otimes_1)$ contains no field map $Y(a, z)b$ whose ordered chiral iterates satisfy weak associativity, and it contains no operator $S(z)$. These may be supplied by the coefficient or by extra exchange data, but cannot be reconstructed from transverse multiplication alone.*

196.3 THE GENUINELY DOUBLY NONCOMMUTATIVE OBJECT

The local formal version of definition 193.7 consists of:

- (i) a nonlocal vertex operation $Y_X(a, z)b$, possibly with $\mathcal{S}(z)$;
- (ii) a transverse associative product $a \cdot_{\perp} b$;
- (iii) mixed identities saying transverse multiplication is a morphism of nonlocal vertex algebras, or equivalently

$$Y_X(a \cdot_{\perp} b, z)(c \cdot_{\perp} d) = \sum (Y_X(a, z)c) \cdot_{\perp} (Y_X(b, z)d)$$

in the aligned decomposition sector, with the appropriate braiding when factors are exchanged;

- (iv) all higher homotopies resolving the failure of strict equality in a derived model.

This is at least doubly noncommutative: the field product need not be local, and the transverse word need not commute. If $\mathcal{S} \neq \mathbf{1}$, exchange supplies a third noncommutative layer.

196.4 ELEMENTARY FREE CONSTRUCTION

Let $\mathcal{M}$ be a complete translation module. The free nonlocal vertex algebra $T_{\text{nv}}(\mathcal{M})$ is generated by fields corresponding to $\mathcal{M}$, with only vacuum, translation, and weak associativity imposed. Let B be an augmented associative algebra. The tensor product

$$\mathcal{A} = B \otimes T_{\text{nv}}(\mathcal{M})$$

has transverse multiplication on B and nonlocal chiral multiplication on the second factor. Both are explicit, and the mixed law is strict because the operations act on separate tensor factors. Quotienting by an exchange relation generated by an R -matrix produces the Zamolodchikov–Faddeev models of chapter 208.

Homotopy-coherent generators and relations

197.1 WHY ORDINARY QUOTIENTS ARE INSUFFICIENT

A presentation $T(V)/(R)$ remembers generators and strict equations in an ordinary category. In a derived coefficient category, a relation can hold by a chosen null-homotopy, two null-homotopies can differ by a second homotopy, and these choices affect bar and deformation complexes. The correct quotient is therefore a homotopy pushout of free algebras.

Fix a combinatorial monoidal model category $\mathcal{M}$ presenting the desired stable symmetric monoidal ∞ -category. Assume the relevant coloured operads are admissible, well pointed, and Σ -cofibrant after replacement. Let

$$\mathcal{P} = W\mathrm{Ass}_{\perp} \otimes W\mathrm{Ass}_X^{\mathrm{ch}}$$

be a cofibrant two-direction resolution.

Definition 197.1 (Derived presentation). Let V be a collection of generators, R a collection of relation cells, and $\partial_0, \partial_1 : \mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)$ the two sides of each relation. The derived algebra presented by (V, R) is the homotopy coequalizer

$$\mathcal{A} \simeq \mathrm{hcoeq}(\mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)),$$

equivalently the corresponding homotopy pushout after adjoining relation disks.

A cofibrant replacement of this homotopy coequalizer is the generators-and-relations object needed for derived mapping spaces. A map $\mathcal{A} \rightarrow B$ is a choice of images of generators, null-homotopies of relations, and all higher compatibility data encoded by $\mathcal{P}$.

197.2 TWO-COLOURED LEVELLED FORESTS

A cell of $\mathcal{P}$ is represented by a levelled forest whose internal edges are coloured $\perp$ or X . A transverse edge resolves associative composition along the ordered line. A chiral edge resolves nested collision. For a forest T , set

$$\mathrm{or}(T) = \det \mathbb{k}E_{\perp}(T) \otimes \det \mathbb{k}E_X(T).$$

The cellular complex is

$$\mathrm{Word}_{\mathcal{P}}(I) = \bigoplus_T \mathrm{Dec}(T) \otimes \mathrm{or}(T)$$

with

$$d = d_{\text{dec}} + d_{\perp} + d_X.$$

Theorem 197.2 (Mixed boundary identity). *With the product orientation,*

$$d_{\perp}^2 = d_X^2 = 0, \quad d_{\perp}d_X + d_Xd_{\perp} = 0,$$

and d_{dec} supercommutes with both edge differentials. Hence $d^2 = 0$.

Proof. A codimension-two cell has two contracted edges. If the colours agree, the two orders give the opposite boundary orientations of the same cellular operad. If the colours differ, both orders give the same decorated forest but exchange the two one-dimensional determinant factors, producing a minus sign. Naturality of the decoration differential gives the remaining anticommutators. $\square$

When vertices are decorated by operations, $d_{\perp}^2 = 0$ is the Stasheff system, $d_X^2 = 0$ is associative chiral coherence, and the mixed anticommutator is the entire hierarchy of locality/interchange homotopies.

197.3 RECTIFICATION AND THE ∞ -CATEGORY

Theorem 197.3 (Cofibrant comparison). *Assume:*

- (i) $W\text{Ass}_{\perp}, W\text{Ass}_X^{\text{ch}}$, and their tensor product are admissible;
- (ii) the tensor product is cofibrant;
- (iii) the augmentations to $\text{Ass}_{\perp}$ and Ass_X^{ch} tensor to a weak equivalence;
- (iv) change of operad along that weak equivalence is a Quillen equivalence.

Then strict algebras over the target operad and homotopy-coherent algebras over $\mathcal{P}$ present equivalent ∞ -categories. Derived presentations computed in either model represent the same mapping functor.

Proof. The hypotheses make $\mathcal{P} \rightarrow \text{Ass}_{\perp} \otimes \text{Ass}_X^{\text{ch}}$ a cofibrant resolution and identify the localized algebra categories by change of operad. Homotopy pushouts between cofibrant diagrams compute colimits in the localized ∞ -category. $\square$

This theorem is the proper relation between model-category tooling and ∞ -category semantics. A model is used to calculate; the localized mapping space is the invariant answer.

197.4 LOW ARITY

At arity two there are two primitive binary operations $m_{\perp}$ and m_X . At arity three, the codimension-one faces produce

$$\begin{aligned} m_{\perp}(m_{\perp}(a, b), c) - m_{\perp}(a, m_{\perp}(b, c)), \\ m_X(m_X(a, b), c) - m_X(a, m_X(b, c)), \end{aligned}$$

and the mixed square compares

$$m_X(m_{\perp}(a, b), m_{\perp}(c, d)) \quad \text{with} \quad m_{\perp}(m_X(a, c), m_X(b, d))$$

on a common decomposition. In a braided model, the middle factors are transported by the exchange operator before the second expression is evaluated. At arity four the pentagons, chiral nested-collision cells, and mixed prisms appear automatically. No independent “coherence conjecture” is needed: it is the cellular boundary of the cofibrant operad.

197.5 HOMOTOPY TRANSFER

Given a deformation retract

$$(H, d_H) \underset{p}{\overset{i}{\rightleftarrows}} (V, d_V), \quad i p - \text{id}_V = d_V h + h d_V,$$

evaluate a decorated forest by placing original operations at vertices and h on internal edges. The mixed boundary identity proves the Maurer–Cartan equation for the transferred two-direction structure. The same formula is the algebraic skeleton of perturbative Feynman rules: interaction vertices decorate vertices, propagators decorate edges, and boundary restrictions of compactified graph configurations implement edge contraction.

No-go. *The nerve of the incidence poset of collision strata can be contractible even when the geometric screen has nonzero cohomology. For three points on the affine complex line, the projectivized screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, whose first cohomology has rank two, whereas the incidence fibre with terminal corolla is contractible. Therefore a poset resolution cannot replace screen chains unless a comparison theorem proves that the omitted rows are acyclic.*

The transverse interval bar

198.1 THE SIMPLICIAL OBJECT

Let

$$A \in \text{Alg}_{\mathbf{E}_1}^{\text{aug}}(\text{FactD}(X), \otimes_l), \quad \varepsilon : A \rightarrow \mathbf{1}, \quad \bar{A} = \text{fib}(\varepsilon).$$

The two-sided simplicial bar is

$$B_r^\perp(A) = \mathbf{1} \otimes_l A^{\otimes_l r} \otimes_l \mathbf{1}, \quad r \geq 0.$$

Interior faces multiply adjacent factors, outer faces use the augmentation, and degeneracies insert the unit. Its geometric realization is

$$\text{Bar}_X^\perp(A) = |B_\bullet^\perp(A)| \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1}.$$

Every simplicial map belongs to $\text{FactD}(X)$ by the exact locality criterion; hence the entire bar remains a chiral coefficient system.

198.2 NORMALIZED WORDS AND SIGNS

Write $s\bar{A}$ for cohomological suspension. Normalization gives the tensor coalgebra

$$T^c(s\bar{A}) = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes_l r}.$$

For homogeneous $a_i \in \bar{A}$, abbreviate $[a_1 | \cdots | a_r] = sa_1 \otimes \cdots \otimes sa_r$. We fix

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

The multiplication coderivation is therefore

$$b_m[a_1 | \cdots | a_r] = \sum_{i=1}^{r-1} (-1)^{\sum_{j < i} (|a_j| + 1) + |a_i|} [a_1 | \cdots | a_i a_{i+1} | \cdots | a_r].$$

The internal differential is the tensor differential on suspended letters. For an A_∞ -algebra, add the coderivations induced by every m_k .

Proposition 198.1 (Bar identity). *The total bar differential $d_B = d_{\text{int}} + \sum_{k \geq 2} b_{m_k}$ squares to zero exactly when the A_∞ -relations hold. In the strict ungraded case, the square on a three-letter word is the suspended associator*

$$b_m^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Proof. A coderivation of the cofree tensor coalgebra is determined by its projection to cogenerators. The projection of d_B^2 in arity n is precisely the signed sum of all substitutions $m_r(\mathbf{i}^{\otimes i} \otimes m_s \otimes \mathbf{i}^{\otimes j})$ with $r + s = n + 1$, which is the n -th Stasheff identity. $\square$

The deconcatenation coproduct

$$\Delta_{\text{dec}}[a_1 | \cdots | a_r] = \sum_{p=0}^r [a_1 | \cdots | a_p] \otimes [a_{p+1} | \cdots | a_r]$$

makes $\text{Bar}_X^\perp(A)$ a conilpotent coassociative coalgebra in $\text{FactD}(X)$. The bar differential is a coderivation because every multiplication contracts a consecutive subword.

198.3 INTERVAL FACTORISATION HOMOLOGY

Theorem 198.2 (Bar as interval amplitude). *Let the endpoints of $[0, 1]$ carry the augmentation module. Then*

$$\text{Bar}_X^\perp(A) \simeq \mathbf{i} \otimes_A^\mathbb{L} \mathbf{i} \simeq \int_{([0,1], \partial[0,1])} A.$$

More generally, for a right module R and left module L ,

$$\text{Bar}(R, A, L) \simeq R \otimes_A^\mathbb{L} L \simeq \int_{([0,1], \{0,1\})} (A; R, L).$$

Proof. Cover the interval by an ordered chain of small subintervals and endpoint collars. The factorisation Cech diagram is the two-sided simplicial bar. Excision identifies its colimit with factorisation homology and with the derived relative tensor product. Refinement of the cover is cofinal. $\square$

This is the exact physical meaning of a bar word: an ordered sequence of intermediate boundary operations. The differential fuses adjacent operations. The chiral coefficient accompanies each letter and each multiplication map, but the bar degree is transverse.

No-go. *The transverse bar differential has source $(s\bar{A})^{\otimes r}$ and target $(s\bar{A})^{\otimes (r-1)}$, and it is induced by $m_\perp$. A collision residue changes the normal degree of a distribution on X^I and is induced by restriction to a diagonal. Their source and target gradings differ. They may appear as anticommuting differentials in a bicomplex, but one cannot do the work of the other.*

198.4 ASSOCIAHEDRA AND THE MONADIC BAR

The simplicial bar presents the derived tensor product. The Stasheff associahedron resolves the multiplication used to evaluate it. If A is an algebra over a cofibrant A_∞ -operad, the operadic bar has a cellular filtration with terms

$$\bigoplus_{r \geq 0} C_*(K_r) \otimes (s\bar{A})^{\otimes r}.$$

Rectification gives a zigzag of quasi-isomorphisms to the simplicial bar of a strict model. The simplex and associahedron therefore have different universal properties and must both be retained when the input multiplication is homotopy coherent.

198.5 COMPLETION

The direct-sum bar is conilpotent. In formal or filtered field theory one often needs

$$\widehat{\text{Bar}}^\perp(A) = \prod_{r \geq 0} (\mathfrak{s}\bar{A})^{\otimes r}.$$

Let $F^N A$ be a complete decreasing filtration such that every m_k raises total filtration by at least $k - 1$.

Proposition 198.3 (Completed-bar convergence). *Assume that each quotient $A/F^N A$ has finite bar length in a fixed total weight and that the cohomology tower satisfies Mittag-Leffler. Then*

$$H^*(\widehat{\text{Bar}}^\perp(A)) \cong \varprojlim_N H^*(\text{Bar}^\perp(A/F^N A)).$$

The bar differential is continuous and the completion is separated.

Proof. The completed bar is the inverse limit of the finite quotient bars. The Mittag-Leffler hypothesis kills $\varprojlim^1$, and the filtration-growth assumption makes every coderivation continuous. $\square$

The associative chiral bar and the theorem of two bars

199.1 OPERADIC CHIRAL BAR

Let C be an augmented algebra over Ass_X^{ch} . The operadic bar is a conilpotent coalgebra over the bar cooperad $B\text{Ass}_X^{\text{ch}}$:

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(C) = \bigoplus_T B\text{Ass}_X^{\text{ch}}(T) \otimes_{\text{Aut } T} ({}^s\overline{C})^{\otimes \text{in}(T)}.$$

Here T runs over ordered chiral composition trees. The differential has three sources:

- (i) the internal differential of C ;
- (ii) the cooperadic differential of the chiral resolution, including screen chains when present;
- (iii) contraction of an internal chiral edge by the associative chiral multiplication.

The entire principal-part distribution remains attached to each vertex and edge. A residue is one component of this geometric coefficient, not the universal differential.

On the formal disc, this bar resolves iterated nonlocal fields. Its length is the number of chiral composition vertices, and its relations are weak associativity together with all higher homotopies. When an exchange operator is present, the chiral trees are braided and the differential uses the corresponding transport.

199.2 CHIRAL CHEVALLEY CHAINS ARE A THIRD CONSTRUCTION

A chiral Lie algebra $\mathfrak{g}$ has Chevalley chains

$$\text{CE}_*^{\text{ch}}(\mathfrak{g}) = (\text{Cofree}_{\text{Com}^c}^{\text{ch}}({}^s\mathfrak{g}), d_{\text{int}} + d_{[-,-]_{\text{ch}}}).$$

This is the construction appearing in Francis–Gaitsgory chiral Koszul duality. It is not the associative chiral bar and not the transverse interval bar. The three constructions are related only after one supplies an enveloping map, a commutator map, or another explicit monoidal comparison.

Theorem 199.1 (The theorem of two associative bars). *Let A be a doubly associative chiral algebra in the sense of definition 193.7. Then:*

- (i) $\text{Bar}^{\perp}(A)$ is a conilpotent coalgebra for transverse deconcatenation and remains an associative chiral coefficient whenever the mixed locality maps are compatible with the bar faces;

- (ii) $\text{Bar}_{\text{Ass}}^{\text{ch}}(A)$ is a conilpotent chiral cooperadic object and remains a transverse E_1 -algebra whenever the same mixed data are compatible with chiral edge contraction;
- (iii) the iterated objects $\text{Bar}^\perp \text{Bar}_{\text{Ass}}^{\text{ch}}(A)$ and $\text{Bar}_{\text{Ass}}^{\text{ch}} \text{Bar}^\perp(A)$ are linked by a canonical comparison on the aligned-decomposition sector;
- (iv) if the mixed operad is a genuine distributive law and both realizations preserve the relevant homotopy colimits, the comparison is an equivalence and the total differential is

$$d_{\text{tot}} = d_{\text{int}} + d_\perp + d_X, \quad d_\perp d_X + d_X d_\perp = 0.$$

Proof. The first two statements apply each bar functor internally to the algebra category for the other operation. This is possible precisely because the mixed cells say that every face map is a morphism for the remaining structure. The aligned projector supplies the comparison between the two orders of common decomposition. Under a distributive law, the two-sided bar construction is the geometric realization of one bisimplicial/cooperadic object; Fubini for homotopy colimits gives the equivalence. The anticommutator is the mixed edge identity of theorem 197.2. $\square$

No-go. An equality $\text{Bar}^\perp(A) = \text{Bar}_{\text{Ass}}^{\text{ch}}(A)$ has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. Even their underlying gradings count different geometric events. A similarity of Euler characteristics is not a comparison map.

199.3 UNIVERSAL ENVELOPING COMPARISON

Suppose $\mathfrak{g}$ is a chiral Lie algebra and $U^{\text{ch}}(\mathfrak{g})$ is an associative chiral enveloping algebra satisfying PBW. The universal twisting map induces

$$\text{CE}_*^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{Bar}_{\text{Ass}}^{\text{ch}}(U^{\text{ch}}(\mathfrak{g})).$$

Under the chiral PBW and connectivity hypotheses, the associated graded map is the ordinary Koszul comparison between symmetric coalgebras and the bar of a tensor quotient, hence a quasi-isomorphism. This is the correct setting in which Lie-type and associative-chiral dualities coincide.

For a transversely ordered enveloping algebra $U(\mathfrak{g})$ in the coefficient category, the analogous comparison is

$$\text{CE}_*(\mathfrak{g}) \xrightarrow{\sim} \text{Bar}^\perp(U(\mathfrak{g})),$$

which is the classical Cartan–Eilenberg theorem internal to the coefficient category. The two statements share a proof pattern but have different multiplications.

Bar–cobar reconstruction and the completion locus

200.1 COBAR

Let C be a coaugmented conilpotent coassociative coalgebra in a stable symmetric monoidal category $\mathcal{V}$, and write $\overline{C}$ for its coaugmentation coideal. The cobar construction is

$$\mathrm{Cobar}(C) = (T(s^{-1}\overline{C}), d_C + d_\Delta),$$

where d_Δ is the derivation induced by the reduced coproduct. The universal twisting morphisms give the adjunction

$$\mathrm{Bar} : \mathrm{Alg}_{\mathcal{E}_1}^{\mathrm{aug}}(\mathcal{V}) \rightleftarrows \mathrm{Coalg}_{\mathcal{BE}_1}^{\mathrm{coaug}}(\mathcal{V}) : \mathrm{Cobar}.$$

The notation on the coalgebra side includes the divided-power or norm data required by the ambient category; in ordinary characteristic-zero chain complexes it reduces to the familiar conilpotent coassociative coalgebra.

200.2 NILCOMPLETION AND CONILCOMPLETION

An augmented algebra has an operadic nilpotent tower; its inverse limit is the nilcompletion $A_{\mathrm{nil}}^\wedge$. A coalgebra has a coradical tower; its inverse limit is the conilcompletion $C_{\mathrm{conil}}^\wedge$.

Theorem 200.1 (Internal associative Koszul duality). *Let $\mathcal{V}$ be a stable presentable symmetric monoidal ∞ -category whose tensor product preserves colimits separately, and use the correct bar-cooperad coalgebras in $\mathcal{V}$. Bar and cobar restrict to inverse equivalences*

$$\mathrm{Alg}_{\mathcal{E}_1}^{\mathrm{aug}}(\mathcal{V})_{\mathrm{nilcomp}} \simeq \mathrm{Coalg}_{\mathcal{BE}_1}^{\mathrm{coaug}}(\mathcal{V})_{\mathrm{conilcomp}}.$$

Applied to $(\mathrm{FactD}(X), \otimes_!)$, this is transverse ordered chiral bar–cobar duality. Applied to the associative chiral operad, under its admissibility and continuity hypotheses, it is associative chiral bar–cobar duality.

Proof. The bar–cobar unit and counit are equivalences on finite nilpotent/coradical stages by the operadic filtration and the Koszul self-duality of the associative operad up to suspension. Passing to the inverse limits gives the stated equivalence on complete objects. Stability and separate colimit preservation ensure the operadic constructions exist and commute with the finite stages. $\square$

The theorem is an equivalence on a completion locus, not on all augmented algebras. This is the correct unconditional statement: every hypothesis appears in the object type.

200.3 POSITIVE-WEIGHT PROOF

Let

$$A = \mathbf{1} \oplus \prod_{w \geq 1} A_w, \quad A_u \otimes_l A_v \rightarrow A_{u+v},$$

be connected and complete, and suppose every weight has finitely many decompositions into positive weights.

Theorem 200.2 (Positive reconstruction). *The counit*

$$\mathrm{Cobar} \, \mathrm{Bar}(A) \longrightarrow A$$

is a filtered quasi-isomorphism. Dually, for a conilpotent weight-complete coalgebra with finite decomposition in every weight, the unit $C \rightarrow \mathrm{Bar} \, \mathrm{Cobar}(C)$ is a filtered quasi-isomorphism.

Proof. Fix total weight w . Only finitely many subdivided words occur. Filter $\mathrm{Cobar} \, \mathrm{Bar}(A)$ by the number of surviving cuts. The associated graded of each fixed underlying word is the augmented cellular chain complex of the simplex of its subdivisions. The extra degeneracy joining the first cut contracts this complex onto the unsplit word. Thus the associated-graded counit is a quasi-isomorphism in every weight. Completeness promotes it to the filtered equivalence. The coalgebra proof uses the finite coradical filtration in each weight. $\square$

200.4 MITTAG-LEFFLER LIMITS

For a complete filtered algebra $A \simeq \varprojlim A/F^N$, reconstruction is inherited from the finite quotients if the towers of bar and cobar cohomology satisfy Mittag-Leffler and the tensor product is continuous. Without these conditions, $\varprojlim^1$ can create classes and completion can fail to commute with geometric realization.

No-go. *The canonical map $\mathrm{Cobar} \, \mathrm{Bar}(A) \rightarrow A$ need not be an equivalence for an arbitrary augmented algebra. It computes a nilpotent or augmentation completion. Any theorem writing an unqualified equality has omitted the principal conclusion of the double-centralizer calculation.*

200.5 DEFORMATION COMPLEXES

The convolution complex

$$\mathrm{Conv}(BE_1, \mathrm{End}_A)$$

controls deformations of the associative multiplication. Its Maurer–Cartan elements are homotopy-coherent products, and its gauge equivalences are operadic homotopies. In the two-direction theory, the complete deformation complex is the total convolution algebra of the bicoloured cooperad. Its differential splits into internal, transverse, chiral, and mixed components. Curvature can be inserted only after its arity and degree are specified.

Koszul–Morita theory, relative bases, and the centre

201.1 THE AUGMENTATION BOUNDARY

For an augmented algebra A , the unit is a left and right module. The augmentation-boundary algebra is

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

When bar terms are dualisable, or continuous internal Hom is used,

$$A^! \simeq \mathrm{RHom}_{\mathcal{V}}(\mathrm{Bar}(A), \mathbf{1}).$$

This is the boundary Koszul dual. It is not automatically the quadratic dual, Verdier dual, or chiral dual.

Theorem 201.1 (Double centralizer). *Assume A is connective, derived augmentation completion is computed by the cobar totalization of the Amitsur complex, and $\mathbf{1}$ is compact in the completed module category. Then*

$$A^{!!} := \mathrm{RHom}_{A^!}(\mathbf{1}, \mathbf{1}) \simeq A_\varepsilon^\wedge.$$

On the nilcomplete locus this is A .

Proof. The bar resolution identifies the thick subcategory generated by $\mathbf{1}$ with perfect $A^!$ -modules. The endomorphisms of the corresponding Yoneda generator form the double centralizer. The universal algebra acting on the augmentation-generated subcategory is derived completion along $\ker \varepsilon$, computed by the assumed Amitsur totalization. $\square$

201.2 ONE-SIDED AND TWO-SIDED TWISTING

Put $C = \mathrm{Bar}(A)$ and let $\tau : C \rightarrow A$ be the universal twisting cochain. For an A -module M , define

$$K_A(M) = C \otimes_l M$$

with the sum of the internal differential and the twisting term obtained from Δ_C , τ , and the module action. The first factor makes it a C -comodule. The reverse functor is $L_C(N) = A \otimes_l N$ with the dual twisting differential.

Theorem 201.2 (Positive-complete module–comodule duality). *For a connected positive-complete algebra with finite weight decompositions, the twisting functors are inverse equivalences between positive-complete modules and conilcomplete comodules. The two-sided version gives an equivalence between complete (A, B) -bimodules and conilcomplete $(\text{Bar } A, \text{Bar } B)$ -bicomodules. It sends the regular diagonal bimodule to the regular diagonal bicomodule.*

Proof. Filter the composites by bar cuts and coradical length. In each fixed weight, the associated graded is the subdivision complex of an ordered word with an extra degeneracy, hence contractible onto the unsplit module or comodule. Finite decomposition gives convergence; completion passes from finite windows to the full object. $\square$

For composable bimodules,

$$K_{A,D}^{\text{bi}}(M \otimes_B^{\mathbb{L}} N) \simeq K_{A,B}^{\text{bi}}(M) \square_{\text{Bar } B}^{\mathbb{R}} K_{B,D}^{\text{bi}}(N).$$

Thus tensor composition becomes cotensor composition. Mapping objects identify Hochschild cochains with coHochschild cochains, and Hochschild chains with coHochschild chains, in the complete categories.

201.3 RELATIVE AUGMENTATION BASES

Many important noncommutative algebras are not augmented over $\mathbb{k}$ but are augmented over a separable or semisimple base S .

Definition 201.3 (Relative bar). For an S -algebra A equipped with a retraction $A \rightarrow S$, the relative reduced bar is

$$\text{Bar}_S(A) = S \otimes_A^{\mathbb{L}} S \simeq T_S^c(s\bar{A}_S)$$

with tensors and multiplication taken over S .

For a quiver, $S = \bigoplus_i \mathbb{k}e_i$ is the vertex-idempotent algebra. Tensoring over S automatically retains only composable paths. This is the natural chain complex; choosing an arbitrary scalar augmentation destroys the object decomposition and can change the homology.

201.4 THE INTERNAL DERIVED CENTRE

The derived centre is

$$\mathbf{Z}(A) = \text{RHom}_{A^e}(A, A), \quad A^e = A \otimes_! A^{\text{op}}.$$

The brace operations make $\mathbf{Z}(A)$ an E_2 -algebra internal to the coefficient category. It is universal among E_2 -objects acting centrally on A . Under Morita equivalence, it is invariant.

A physical bulk theory $\mathcal{B}$ coupled to a boundary condition produces a bulk-to-boundary-centre map

$$\mathcal{B} \longrightarrow \mathbf{Z}(A_b).$$

An open–closed theorem can prove this map is an equivalence for a selected sector. The centre alone does not construct the decoupled bulk theory, its stress tensor, its global state space, or its anomaly data.

No-go. *There is no universal inverse to the bulk-to-centre map. Different bulk theories can induce the same central action on a chosen boundary sector, and a boundary category may admit no anomaly-free local bulk realization. The equation $\mathcal{B} \simeq \mathbf{Z}(A_b)$ is a theorem only after generation, locality, and open–closed comparison hypotheses are verified.*

Quadratic, PBW, curved, and Verdier comparisons

202.1 QUADRATIC RECOGNITION

Let V be dualisable and

$$A = T(V)/(R), \quad R \hookrightarrow V \otimes_! V.$$

The quadratic dual algebra is

$$A^{q!} = T(V^*[-1])/(R^\perp),$$

and the quadratic dual coalgebra is the largest subcoalgebra of $T^c(sV)$ whose quadratic core lies in $s^2 R$.

Theorem 202.1 (Quadratic recognition). *Assume every weight component is dualisable and the diagonal Koszul complex is exact in positive homological degree. Then the quadratic dual coalgebra includes quasi-isomorphically into $\text{Bar}(A)$, and continuous duality gives*

$$A^! \simeq A^{q!}.$$

Proof. Filter the bar by tensor weight. Its linear strand is the quadratic Koszul complex. The exactness hypothesis collapses the spectral sequence onto the quadratic dual coalgebra. Continuous linear duality turns the coalgebra into the stated algebra. $\square$

No-go. *The orthogonal-relations algebra $A^{q!}$ records only the quadratic strand. If the Koszul complex has higher homology, $A^{q!}$ is not quasi-isomorphic to $\text{RHom}_A(\mathbf{1}, \mathbf{1})$. Equality requires the Koszulness theorem, not merely a quadratic presentation.*

202.2 PBW DEFORMATIONS AND CURVATURE

A nonhomogeneous relation has the form

$$r - \alpha(r) - \beta(r), \quad \alpha : R \rightarrow V, \quad \beta : R \rightarrow \mathbf{1}.$$

The linear term becomes a differential on the quadratic dual, and the constant term becomes curvature.

Theorem 202.2 (Curved PBW duality). *If the Braverman–Gaitsgory overlap conditions hold, then the filtered algebra has associated graded $T(V)/(R)$, and its dual is a curved dg algebra*

$$(A^{\mathfrak{q}}, d_\alpha, h_\beta), \quad d_\alpha^2 = [h_\beta, -], \quad d_\alpha h_\beta = 0.$$

Conversely, these curved identities are equivalent to resolution of the degree-three overlap ambiguities.

Proof. Evaluate the ambiguity space $(R \otimes V) \cap (V \otimes R)$. Its filtration-two component gives the derivation condition for d_α ; the filtration-one and filtration-zero components give $d_\alpha^2 = [h_\beta, -]$ and $d_\alpha h_\beta = 0$. The Rees spectral sequence then identifies the associated graded algebra. $\square$

This is the correct home of the Weyl relation $yx - xy = 1$: the constant is curvature, not an augmentation-preserving quadratic relation.

202.3 VERDIER DUALITY

Verdier duality acts on the factorisation D -module coefficient and reverses $!$ - and $*$ -extensions. Under holonomicity and finiteness hypotheses it can carry factorisation algebras to factorisation coalgebras. A comparison with a Koszul dual requires an explicit pairing between the bar resolution and the Verdier-dual coefficient. The operations agree in self-dual examples but are not definitionally identical.

No-go. *The interval bar $\mathbf{I} \otimes_A^{\mathbb{L}} \mathbf{I}$, boundary dual $\mathrm{RHom}_A(\mathbf{I}, \mathbf{I})$, quadratic dual $T(V^*[-1])/(R^\perp)$, and Verdier dual $\mathbb{D}A$ have different variances and hypotheses. They can be joined by comparison maps in a finite Koszul self-dual model. Replacing the maps by equality erases the theorem.*

Free and square-zero ordered chiral algebras

203.1 TENSORING A COEFFICIENT BY AN ALGEBRA

Let $F \in \text{FactD}(X)$ be a commutative coefficient object and let B be an augmented associative dg algebra. Then

$$A = B \otimes F$$

is a transversely ordered chiral algebra: the coefficient factorises on X , while the transverse multiplication is $m_B \otimes m_F$. If F also carries an associative chiral product, the two operations act tensor-factorwise and give the first strict doubly associative examples.

This construction is elementary but decisive. It proves that the chiral OPE does not determine the transverse bar: varying B while fixing F changes $\text{Tor}^B(\mathbb{k}, \mathbb{k})$ and leaves every chiral singular coefficient unchanged.

203.2 THE FREE ALGEBRA

Let $B = T(V)$ with the standard augmentation.

Theorem 203.1 (Bar of a free algebra). *There is a natural deformation retract*

$$\text{Bar}(T(V)) \simeq \mathbb{k} \oplus sV.$$

Equivalently,

$$\text{Tor}_0^{T(V)}(\mathbb{k}, \mathbb{k}) = \mathbb{k}, \quad \text{Tor}_1^{T(V)}(\mathbb{k}, \mathbb{k}) = V, \quad \text{Tor}_r^{T(V)}(\mathbb{k}, \mathbb{k}) = 0 \quad (r \geq 2).$$

The same statement holds internally after tensoring with a factorisable coefficient.

Proof. Fix a word $v_1 \cdots v_n$ in the generators. The normalized bar subcomplex spanned by all ways of inserting bars between its letters is the augmented cellular chain complex of the $(n-1)$ -simplex of cuts. If $n > 1$, the extra degeneracy that removes the first cut contracts this complex. If $n = 1$, one suspended generator remains. Summing over all words gives the retract. $\square$

The theorem supplies a strong diagnostic: any table claiming large higher bar homology for a free transverse algebra is counting chain states, configuration coefficients, or a different differential.

203.3 THE SQUARE-ZERO ALGEBRA

Let $B = \mathbb{k} \oplus M$ with $M^2 = 0$.

Proposition 203.2 (Bar of a square-zero algebra). *The multiplication part of the normalized bar differential vanishes, and*

$$\text{Bar}(B) = T^c(\mathfrak{s}M)$$

with only the internal differential. If M is a d -dimensional vector space in degree zero, then

$$\dim \text{Tor}_r^B(\mathbb{k}, \mathbb{k}) = d^r, \quad P_B(t) = \frac{1}{1 - dt}.$$

Proof. Every adjacent multiplication in the augmentation ideal is zero, so every normalized word is a cycle and no nonzero word is a bar boundary. $\square$

Free and square-zero algebras have the same generator space and maximally different bars. This is the cleanest proof of multiplication dependence.

203.4 A FIRST NONLOCAL FACTOR

Let $T_{\text{nv}}(M_X)$ be the free nonlocal vertex algebra of recognition Theorem 196.4. Then

$$T(V_{\perp}) \otimes T_{\text{nv}}(M_X)$$

is free in both associative directions. Its transverse bar contracts to $\mathbb{k} \oplus \mathfrak{s}V_{\perp}$ while its associative chiral bar retains the nonlocal composition trees of M_X . Replacing $T(V_{\perp})$ by $\mathbb{k} \oplus V_{\perp}$ changes the first bar to a tensor coalgebra and leaves the second unchanged. This is the basic calculational laboratory for the theorem of two bars.

The quantum and Jordan planes

204.1 THE QUANTUM PLANE

Fix $q \in \mathbb{k}^\times$ and set

$$A_q = \mathbb{k}\langle x, y \rangle / (yx - qxy), \quad \varepsilon(x) = \varepsilon(y) = 0.$$

Orient the relation as $yx \mapsto qxy$. Every word reduces uniquely to $x^a y^b$, so these monomials form a PBW basis and

$$H_{A_q}(z) = \frac{1}{(1-z)^2}.$$

Theorem 204.1 (Quantum-plane Koszul resolution). *The complex of free left modules*

$$0 \longrightarrow A_q(-2) \xrightarrow{d_2} A_q(-1)^{\oplus 2} \xrightarrow{d_1} A_q \xrightarrow{\varepsilon} \mathbb{k} \longrightarrow 0,$$

with

$$d_1(a, b) = ax + by, \quad d_2(c) = (cy, -cqy),$$

is exact. Consequently

$$\mathrm{Tor}_*^{A_q}(\mathbb{k}, \mathbb{k}) \cong \mathbb{k} \oplus \mathbb{k}\{sx, sy\} \oplus \mathbb{k}\{\omega_q\},$$

where $|\omega_q| = 2$, and a normalized bar representative is

$$\omega_q = [y|x] - q[x|y].$$

The Poincare polynomial is $1 + 2t + t^2$.

Proof. The identity $d_1 d_2(c) = c(yx - qxy) = 0$ proves that this is a complex. The PBW basis and Hilbert-series identity

$$H_{A_q}(z)(1 - 2z + z^2) = 1$$

show that the graded Euler characteristic equals that of $\mathbb{k}$. The leading monomials in the two maps are compatible with the one-relation Grobner basis, so no hidden syzygy occurs; equivalently, the single quadratic relation has the displayed linear syzygy and the complex is the standard quadratic Koszul complex. Tensoring with $\mathbb{k}$ kills both differentials. In the bar complex, $b([y|x] - q[x|y]) = yx - qxy = 0$. $\square$

The quadratic dual is

$$A_q^! = \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

The ordered parameter q survives in the Yoneda product even though the ungraded Betti numbers agree with the commutative plane.

204.2 THE JORDAN PLANE

Set

$$A_J = \mathbb{k}\langle x, y \rangle / (yx - xy - x^2), \quad \varepsilon(x) = \varepsilon(y) = 0.$$

The rule $yx \mapsto xy + x^2$ terminates and has no unresolved critical ambiguity, so $x^a y^b$ again form a PBW basis.

Theorem 204.2 (Jordan-plane Koszul resolution). *The complex*

$$0 \longrightarrow A_J(-2) \xrightarrow{d_2} A_J(-1)^{\oplus 2} \xrightarrow{d_1} A_J \xrightarrow{\varepsilon} \mathbb{k} \longrightarrow 0,$$

with

$$d_1(a, b) = ax + by, \quad d_2(c) = (c(y - x), -cx),$$

is exact. Thus

$$P_{\text{Tor}^{A_J}(\mathbb{k}, \mathbb{k})}(t) = 1 + 2t + t^2,$$

and the degree-two bar class is represented by

$$\omega_J = [y|x] - [x|y] - [x|x].$$

Proof. The composite is $c((y - x)x - xy) = c(yx - xy - x^2) = 0$. The quadratic rewriting system gives the same Hilbert series as the polynomial plane and the displayed complex is its Grobner–Koszul resolution. Tensoring with the augmentation module makes both differentials zero. The bar boundary of ω_J is the defining relation. $\square$

The quantum and Jordan planes have identical Betti numbers but inequivalent products, normal forms, quadratic duals, and module categories. Numerical dimensions alone cannot classify the algebra.

204.3 FINITE CHAIN-MATRIX CALCULATION

For each total polynomial weight $w \leq 5$, take the normalized bar basis formed by words in nonunit monomials and grade it by word length. The alternating adjacent-multiplication differential gives an exact rational matrix. For the quantum plane set $q = 2$; its ranks are generic away from the finite determinantal exceptional locus. The resulting kernel/image dimensions are:

algebra	weight	degree	chain dimension	outgoing rank	homology dimension
A_q	1	1	2	0	2
A_q	2	1	3	0	0
A_q	2	2	4	3	1
A_q	3	1	4	0	0
A_q	3	2	12	4	0
A_q	3	3	8	8	0
A_q	4	1	5	0	0
A_q	4	2	25	5	0
A_q	4	3	36	20	0
A_q	4	4	16	16	0
A_q	5	1	6	0	0
A_q	5	2	44	6	0
A_q	5	3	102	38	0
A_q	5	4	96	64	0
A_q	5	5	32	32	0
A_J	1	1	2	0	2
A_J	2	1	3	0	0
A_J	2	2	4	3	1
A_J	3	1	4	0	0
A_J	3	2	12	4	0
A_J	3	3	8	8	0
A_J	4	1	5	0	0
A_J	4	2	25	5	0
A_J	4	3	36	20	0
A_J	4	4	16	16	0
A_J	5	1	6	0	0
A_J	5	2	44	6	0
A_J	5	3	102	38	0
A_J	5	4	96	64	0
A_J	5	5	32	32	0

All weights four and five have zero homology. Direct application of the degree-two differential sends each displayed cycle to zero.

204.4 INTERNAL CHIRAL REALIZATION

Let F be any augmented commutative factorisation coefficient. Replacing x, y by generator objects $x \otimes F, y \otimes F$ and interpreting the relation as a morphism in $\text{FactD}(X)$ gives quantum- and Jordan-plane ordered chiral algebras. Their transverse bars are the preceding Koszul coalgebras tensored with the coefficient powers, while every collision map of F remains present. If F is an associative chiral/nonlocal coefficient, these become strict doubly associative examples.

Augmentation obstructions, Weyl curvature, and matrix sectors

205.1 THE WEYL OBSTRUCTION

Let

$$W_1 = \mathbb{k}\langle x, y \rangle / (yx - xy - 1).$$

Theorem 205.1 (No scalar augmentation for the Weyl algebra). *There is no unital algebra map $W_1 \rightarrow \mathbb{k}$. Hence $\mathbb{k} \otimes_{W_1}^{\mathbb{L}} \mathbb{k}$ with the alleged “zero-generator augmentation” is not defined.*

Proof. Applying a map to the defining relation gives $\varepsilon(y)\varepsilon(x) - \varepsilon(x)\varepsilon(y) = 1$. The left side is zero in a commutative field, a contradiction. $\square$

The correct replacements are typed.

(i) The homogenized/Rees algebra

$$W_{\hbar} = \mathbb{k}[\hbar]\langle x, y \rangle / (yx - xy - \hbar),$$

with $\hbar$ central, has an augmentation as a $\mathbb{k}$ -algebra only if $\hbar \mapsto 0$. Its fibre at $\hbar = 0$ is the commutative plane, while the fibre at $\hbar = 1$ is not augmented.

- (ii) The nonhomogeneous quadratic dual is curved. The constant term produces a curvature element h with $d^2 = [h, -]$, as in [theorem 202.2](#).
- (iii) A relative bar can be formed for a specified module or boundary idempotent without inventing a scalar augmentation.

Exact calculation. *The quadratic part is the polynomial plane, whose dual is the exterior algebra $\Lambda(\xi, \eta)$. The constant functional on the relation becomes, up to the suspension convention, the curvature*

$$h = \xi\eta.$$

There is no differential in the homogeneous two-generator case, and the curved identity is $d^2 = [h, -] = 0$ because h is central in the graded exterior algebra. Curvature records exactly why the ordinary augmentation and ordinary cohomology differential are unavailable after setting $\hbar = 1$.

205.2 THE MATRIX OBSTRUCTION

Theorem 205.2 (No scalar augmentation for a full matrix algebra). *For $n > 1$, there is no unital algebra map $M_n(\mathbb{k}) \rightarrow \mathbb{k}$.*

Proof. The kernel of a unital homomorphism is a proper two-sided ideal. Since $M_n(\mathbb{k})$ is simple, the kernel is zero, so the map would be injective. Its image lies in a commutative algebra, forcing all matrix commutators to vanish, a contradiction. $\square$

Matrix-valued fields are perfectly valid ordered chiral algebras, but their bar must be relative to a module, an idempotent, or a larger augmented algebra. Morita invariance is the right language; a fictitious trace augmentation is not. The ordinary matrix trace is linear and cyclic but not multiplicative.

205.3 AN AUGMENTED NONCOMMUTATIVE FROBENIUS ALGEBRA

Let $D = \mathbb{k}[\epsilon]/(\epsilon^2)$ and

$$B_n = D \times M_n(\mathbb{k}), \quad \varepsilon_B(a + b\epsilon, M) = a.$$

This algebra is noncommutative for $n > 1$ and genuinely augmented. Give it the symmetric Frobenius trace

$$\tau_B(a + b\epsilon, M) = b + \text{Tr}(M).$$

The pairing $\langle u, v \rangle = \tau_B(uv)$ is nondegenerate: the dual-number block pairs 1 with ϵ , and the matrix block has the trace pairing.

Proposition 205.3 (Bar and handle element of B_n). *As an augmented algebra,*

$$\mathbb{k} \otimes_{B_n}^{\mathbb{L}} \mathbb{k} \simeq \mathbb{k} \otimes_D^{\mathbb{L}} \mathbb{k} \simeq T^c(s\mathbb{k}\epsilon).$$

Thus $\dim \text{Tor}_r^{B_n}(\mathbb{k}, \mathbb{k}) = 1$ for every $r \geq 0$. The Frobenius handle element is

$$e_{B_n} = (2\epsilon, nI_n).$$

Consequently

$$\tau_B(e_{B_n}) = 2 + n^2, \quad \tau_B(e_{B_n}^g) = n^{g+1} \quad (g \geq 2).$$

Proof. Modules over a direct product decompose. The augmentation module is $(\mathbb{k}, 0)$, so its derived tensor product over $D \times M_n$ is the derived tensor product over D . Since the augmentation ideal of D is one-dimensional square-zero, proposition 203.2 gives the tensor coalgebra. For the handle element, the basis $(1, \epsilon)$ of D has dual basis $(\epsilon, 1)$, yielding 2ϵ . The matrix units E_{ij} have duals E_{ji} , and $\sum_{ij} E_{ij}E_{ji} = nI_n$. Powers are computed blockwise. $\square$

This single algebra simultaneously supplies noncommutativity, a legitimate augmentation, a nontrivial infinite bar, and an exact cyclic trace. It will realize the transverse genus in chapter 218.

No-go. *A cyclic trace $\tau(ab) = \tau(ba)$ is generally not multiplicative. Replacing an augmentation by a trace changes a module-valued derived tensor product into a scalar pairing and invalidates the bar faces. Matrix traces, path-integral expectation values, and conformal-block traces must be used in their native cyclic constructions.*

Relative quiver bars and boundary conditions

206.1 THE NATURAL BASE

Let Q be a finite quiver, let

$$S = \bigoplus_{i \in Q_0} \mathbb{k}e_i, \quad M = \bigoplus_{a:i \rightarrow j} e_j M e_i,$$

and let $T_S(M)$ be the path algebra. The map killing all arrows and fixing vertex idempotents is an augmentation to S . Relations generate an S -subbimodule ideal.

The relative bar

$$\text{Bar}_S(\mathcal{A}) = S \otimes_{\mathcal{A}}^{\mathbb{L}} S$$

retains object labels and only allows tensors of composable arrows. It is the algebraic form of a category of boundary conditions: vertices are boundaries, arrows are boundary-changing operators, and paths are ordered compositions.

206.2 A COMPLETE QUADRATIC EXAMPLE

Consider

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3, \quad \mathcal{A} = \mathbb{k}Q/(ba).$$

The relative augmentation ideal is spanned by a, b . The only composable length-two tensor is $b \otimes_S a$, and the defining relation makes its boundary zero.

Theorem 206.1 (Relative bar of the relation quiver). *As an S -bimodule,*

$$\text{Tor}_0^{\mathcal{A}}(S, S) = S, \quad \text{Tor}_1^{\mathcal{A}}(S, S) = \mathbb{k}a \oplus \mathbb{k}b, \quad \text{Tor}_2^{\mathcal{A}}(S, S) = \mathbb{k}[b|a],$$

and $\text{Tor}_r^{\mathcal{A}}(S, S) = 0$ for $r \geq 3$.

Proof. The normalized relative bar has $C_1 = \mathbb{k}a \oplus \mathbb{k}b$, $C_2 = \mathbb{k}[b|a]$, and no composable words of length three. Its only possible differential sends $[b|a]$ to $ba = 0$. There are no incoming boundaries. Adjoining $C_0 = S$ gives the result. $\square$

For the free path algebra with no relation, the same differential sends $[b|a]$ to the nonzero path ba , and the degree-two homology disappears. The relative bar therefore detects the relation itself.

206.3 SIMPLE MODULES AND EXT

Let $S_i = e_i S$ be the simple at vertex i . Cutting the relative bar by vertex idempotents computes derived boundary-changing fields:

$$e_j \operatorname{Tor}_r^A(S, S) e_i \cong \operatorname{Tor}_r^A(S_j, S_i).$$

For the example,

$$\operatorname{Tor}_1^A(S_2, S_1) = \mathbb{k}a, \quad \operatorname{Tor}_1^A(S_3, S_2) = \mathbb{k}b, \quad \operatorname{Tor}_2^A(S_3, S_1) = \mathbb{k}[b|a].$$

The degree-two class is a derived composite whose strict product vanishes. It is the simplest explicit instance of a relation surviving as a higher boundary field.

206.4 FACTORISATION-FAMILY VERSION

Let the arrow spaces be factorisable coefficient objects on X , and form the internal tensor algebra over the vertex coefficient S . If the relation morphisms are compatible with disjoint restriction and diagonal extension, the relative path algebra is a transversely ordered chiral algebra. Marked block-Ran sections recover its module categories. A second associative chiral operation on each arrow coefficient produces a doubly associative quiver algebra, with mixed relations imposed by the derived presentation of chapter 197.

No-go. *A scalar augmentation chooses a single linear combination of vertices and forgets the object decomposition. The canonical bar of a quiver category is $S \otimes_A^{\mathbb{L}} S$, not $\mathbb{k} \otimes_A^{\mathbb{L}} \mathbb{k}$, unless a separate scalar boundary object has been specified.*

Enveloping algebras and planar-sequence obstructions

207.1 CARTAN–EILENBERG COMPARISON

Let $\mathfrak{g}$ be a dg Lie algebra object in the coefficient category and $U(\mathfrak{g})$ its augmented enveloping algebra.

Theorem 207.1 (Enveloping bar). *The antisymmetrisation twisting map induces a natural quasi-isomorphism*

$$CE_*(\mathfrak{g}) = (\mathrm{Sym}^c(s\mathfrak{g}), d_{\mathfrak{g}} + d_{[-,-]}) \xrightarrow{\sim} \mathrm{Bar}^+(U(\mathfrak{g})).$$

Proof. Filter both complexes by PBW degree. The associated graded of $U(\mathfrak{g})$ is $\mathrm{Sym}(\mathfrak{g})$. The comparison becomes the standard Koszul quasi-isomorphism from the exterior/symmetric coalgebra to the bar of a polynomial algebra. The complete, bounded-below filtration spectral sequence lifts the equivalence. $\square$

For a finite-dimensional semisimple Lie algebra over $\mathbb{C}$,

$$H^*(\mathfrak{g}; \mathbb{C}) \cong \Lambda(\zeta_{2m_1+1}, \dots, \zeta_{2m_r+1}),$$

where m_i are the exponents. By duality the same Poincaré polynomial describes homology.

Exact calculation. Take $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. In bases

$$C_1 = (e, f, h), \quad C_2 = (e \wedge f, e \wedge h, f \wedge h),$$

the Chevalley differential is

$$d_2 = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 2 \\ 1 & 0 & 0 \end{pmatrix}, \quad d_3 = 0.$$

Thus d_2 has rank three and

$$H_0 \cong \mathbb{C}, \quad H_1 = H_2 = 0, \quad H_3 \cong \mathbb{C}, \quad P(t) = 1 + t^3.$$

The standard exact values are

$$\begin{aligned} U(\mathfrak{sl}_2) &: 1 + t^3, \\ U(\mathfrak{sl}_3) &: (1 + t^3)(1 + t^5) = 1 + t^3 + t^5 + t^8, \\ U(\mathfrak{g}_2) &: (1 + t^3)(1 + t^{11}) = 1 + t^3 + t^{11} + t^{14}. \end{aligned}$$

207.2 THREE CHAIN COMPLEXES

Three objects must be distinguished:

- (i) the interval bar $\text{Bar}^\perp(\mathcal{A})$, whose differential uses transverse multiplication;
- (ii) chiral Chevalley chains, whose differential uses a diagonal-supported chiral Lie bracket;
- (iii) a screen-enhanced bicomplex

$$B_{\text{scr}}^{p,q}(\mathcal{A}) = (s\bar{\mathcal{A}})^{\otimes_1 p} \otimes C_{\text{scr}}^q$$

with total differential $b_{\mathcal{A}} + d_{\mathcal{A}} + d_{\text{scr}} + \delta_{\text{int}}$.

Only the third can contain both bar words and configuration-space cohomology, and only after the interaction differential is supplied and shown to square to zero.

Filtering by screen degree gives an E_1 -page

$$E_1^{p,q} = H^q(C_{\text{scr}}) \otimes B_p^\perp(\mathcal{A}),$$

while filtering by bar length gives

$${}'_E_1^{p,q} = \text{Tor}_p^{\mathcal{A}}(\mathbf{I}, \mathbf{I}) \otimes C_{\text{scr}}^q.$$

A product of dimensions on an E_1 -page is not automatically the dimension of total homology.

207.3 THE PLANAR SEQUENCES

The Motzkin generating function is

$$M(z) = 1 + zM(z) + z^2M(z)^2 = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}.$$

For $a_n = M_{n+1} - M_n$,

$$\sum_{n \geq 1} a_n z^n = \frac{4z}{(1 - z + \sqrt{1 - 2z - 3z^2})^2},$$

with first values

$$1, 2, 5, 12, 30, 76, 196, 512, 1353, 3610.$$

The Riordan series is

$$R(z) = \frac{1 + z - \sqrt{1 - 2z - 3z^2}}{2z(1 + z)}.$$

Both have dominant singularity $z = 1/3$ and exponential growth rate three. These are correct combinatorial facts.

Theorem 207.2 (Non-identification of Motzkin and Riordan sequences with ordered bar homology). *The displayed sequences are not the ordered bar homology of $U(\mathfrak{sl}_2)$, and a Virasoro chiral coefficient does not determine a transverse bar whose dimensions are Motzkin differences. The former is contradicted by theorem 207.1; the latter fails because the same Virasoro coefficient can be tensored with free, square-zero, enveloping, or quantum-plane transverse algebras having different bars.*

Proof. For $U(\mathfrak{sl}_2)$, the exact Chevalley calculation gives only degrees zero and three. For the Virasoro statement, choose a fixed Virasoro factorisation coefficient $F_{\text{vir},c}$ and form $B \otimes F_{\text{vir},c}$ for the four transverse algebras listed. The coefficient and central charge are unchanged while the interval bar varies with B . $\square$

Theorem 207.2 refutes the $\mathfrak{sl}_2$ claim by computing the answer and the Virasoro claim by non-determination. The Virasoro row admits the stronger statement: in its transverse enveloping presentation the bar homology is known exactly, and it leaves no room for a Motzkin difference.

207.4 THE VIRASORO ROW COMPUTED

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{k} e_n, \quad [e_i, e_j] = (j - i)e_{i+j},$$

to which theorem 207.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 207.3 (Pentagonal rigidity). *The bar homology of $U(L_1)$ satisfies*

$$\dim H^n(\text{Bar}^\perp U(L_1)) = 2 \quad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q(\text{Bar}^\perp U(L_1)) = \prod_{n \geq 1} (1 - q^n),$$

and (207.3) is accounted for degree by degree with no residual freedom.

Proof. By theorem 207.1 the bar of $U(L_1)$ is $\text{CE}_*(L_1)$, whose cohomology is Goncharova's [36]: two dimensional in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (207.3) is theorem 220.1 applied to L_1 , where every root space is one dimensional and even, so $\text{sdim}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_w^n$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

Status. *Finite calculation. Exact rational elimination on the weight-graded Chevalley–Eilenberg complex of L_1 gives the cohomology for $n = 1, 2, 3, 4$ with $d^2 = 0$ checked on every cochain space, occupying the weights $(1, 2), (5, 7), (12, 15), (22, 26)$, one dimension in each. The nonzero coefficients of $\prod_{n \geq 1} (1 - q^n)$ occupy $0, 1, 2, 5, 7, 12, 15, 22, 26$ with signs $+, -, -, +, +, -, -, +, +$. This finite calculation is evidence for, but is not used in place of, the all-degree argument from Goncharova's theorem and Euler's identity.*

Proposition 207.4 (The two sequences are one sequence). *For all $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \quad M(n+1) - M(n) = R(n+2) - R(n).$$

The Virasoro sequence is therefore $R(n+2) - R(n)$ and the $\mathfrak{sl}_2$ sequence is $R(n+3)$: both are elementary transforms of one sequence, and their generating functions are rational in the single algebraic function $\sqrt{1 - 2z - 3z^2}$ for that reason alone. No Drinfeld–Sokolov input is used or needed to explain the shared discriminant.

Proof. Riordan numbers count Motzkin paths with no level step at height zero. Splitting a Motzkin path of length n according to whether its first step is level gives $M(n) = R(n) + R(n+1)$; subtracting consecutive instances gives the second identity. Both were checked termwise for $n < 24$. $\square$

Remark 207.5 (Where the Virasoro formula appears to hold). On degrees $n = 1, 2, 3$ the true dimensions are $2, 2, 2$ by theorem 207.3 while the asserted values are $1, 2, 5$. The two agree at $n = 2$ and nowhere else. A formula tested at that degree alone reproduces the truth, which is the mechanism by which a constant sequence is mistaken for a Motzkin difference.

Remark 207.6 (The type error beneath the numerical one). The bar homology of $U(L_1)$ is bigraded by cohomological degree and weight, and is finite in each degree only because [theorem 207.3](#) concentrates it in two weights. A formula assigning one integer to each degree, with no weight truncation named, does not specify an invariant of a graded object with infinite-dimensional graded pieces.

No-go. *A sequence of integers becomes bar homology only after a chain complex, differential, grading convention, and quasi-isomorphism or rank computation are specified. Matching initial values, discriminants, or growth rates does not construct such a map.*

Zamolodchikov–Faddeev algebras as doubly noncommutative fields

208.1 EXCHANGE PRESENTATION

Let V be finite dimensional and

$$R(u) \in \text{End}(V \otimes V) \otimes \mathbb{k}(u)$$

be invertible away from a specified pole divisor. Introduce generators $Z_i(u)$ and relations

$$Z_i(u)Z_j(v) = \sum_{k,l} R_{ij}^{kl}(u-v)Z_l(v)Z_k(u).$$

The spectral variable belongs to the chiral/meromorphic direction; the word belongs to the ordered associative direction. The relation reverses order and transports the coefficient.

Fix finitely many distinct spectral labels $u_1, \dots, u_m$, choose a total order, and orient every relation toward increasing labels.

Theorem 208.1 (Confluence is Yang–Baxter). *Assume*

$$R(u)R_{21}(-u) = 1.$$

The oriented exchange relations are terminating and confluent exactly when

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

When these equations hold, ordered spectral words form a basis over the localized spectral field.

Proof. The inversion number of a spectral word decreases under each oriented exchange, so reduction terminates. Every irreducible critical overlap has three adjacent labels. The two reduced expressions of the longest permutation in S_3 produce the two sides of the Yang–Baxter equation. A relation followed by its reverse gives the length-two ambiguity, resolved by unitarity. Bergman’s diamond lemma gives unique normal forms. $\square$

This theorem is an exact generators-and-relations construction of a doubly noncommutative algebra. It is not a recognition by analogy: the complete basis and the three-body obstruction are explicit.

208.2 THE RATIONAL MATRIX

Let $P(x \otimes y) = y \otimes x$ and

$$R(u) = \mathbf{1} - \frac{\hbar P}{u}.$$

The permutation identities

$$P_{12}P_{13} = P_{23}P_{12}, \quad P_{13}P_{23} = P_{12}P_{13}, \quad P_{12}P_{23} = P_{13}P_{12}$$

verify Yang–Baxter after clearing denominators. Since

$$R(u)R_{21}(-u) = \mathbf{1} - \frac{\hbar^2}{u^2},$$

a scalar factor $f(u)$ with

$$f(u)f(-u) = \left(\mathbf{1} - \frac{\hbar^2}{u^2} \right)^{-1}$$

produces a unitary normalization.

208.3 A SCALAR BRAIDED EXTERIOR BAR

If V is one-dimensional and the exchange coefficient for labels $a < b$ is a nonzero scalar q_{ab} , the finite-label algebra is multiparameter quantum affine space:

$$z_b z_a = q_{ab} z_a z_b.$$

It is a quadratic PBW algebra. Its Koszul dual is the braided exterior algebra generated by ξ_a with

$$\xi_a^2 = 0, \quad \xi_b \xi_a = -q_{ab}^{-1} \xi_a \xi_b.$$

Hence the bar homology is completely explicit: one basis class for every subset of spectral labels, with the order and braiding recorded in the product. This is the simplest calculational bridge from ZF exchange to chiral Koszul duality.

No-go. *Pairwise rewrite rules can be written for any invertible $R(u)$, but a three-letter word has two reduction paths. Their equality is exactly Yang–Baxter. Without it, the proposed generators-and-relations algebra has unresolved critical pairs and cannot define a coherent quantum vertex or braided factorisation structure.*

RTT Yangians and the bar filtration

209.1 RTT PRESENTATION

Set

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \quad T^{(r)} = (t_{ij}^{(r)})_{1 \leq i, j \leq n}.$$

The Rees RTT Yangian $Y_{\hbar}^{\text{RTT}}(\mathfrak{gl}_n)$ is generated over $\mathbb{k}[\hbar]$ by the coefficients subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v), \quad R(u) = 1 - \frac{\hbar P}{u}.$$

Equivalently, with $t_{ij}^{(\circ)} = \delta_{ij}$,

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar(t_{kj}^{(r)} t_{il}^{(s)} - t_{kl}^{(s)} t_{ij}^{(r)}).$$

209.2 HOPF STRUCTURE

The formulas

$$\Delta T(u) = T_{13}(u)T_{23}(u), \quad \varepsilon(T(u)) = 1, \quad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra. The proof moves auxiliary R -matrices through the product using Yang–Baxter. The counit gives a genuine augmentation, so the ordered bar is defined.

For $n = 2$,

$$\text{qdet } T(u) = t_{11}(u)t_{22}(u - \hbar) - t_{12}(u)t_{21}(u - \hbar)$$

is central and group-like. The quotient $\text{qdet } T(u) = 1$ is the $\mathfrak{sl}_2$ RTT Yangian.

209.3 PBW AND CURRENT SHADOW

Give $t_{ij}^{(r)}$ loop degree $r - 1$. The PBW theorem gives

$$\text{gr}_{\text{loop}} Y_{\hbar}^{\text{RTT}}(\mathfrak{gl}_n) \cong U_{\hbar}(\mathfrak{gl}_n[t]),$$

and ordered monomials in the $t_{ij}^{(r)}$ form a $\mathbb{k}[\hbar]$ -basis. The $\hbar$ -adic completion is topologically free with the same basis.

Theorem 209.1 (Yangian bar spectral sequence). *Filter the normalized bar of the augmented RTT Yangian by total loop degree. Under the usual bounded-below and completion hypotheses, there is a multiplicative spectral sequence*

$$E^1 \cong H_*(\mathfrak{gl}_n[t]; \mathbb{k})[[\hbar]] \implies \mathrm{Tor}_*^{Y_{\hbar}^{\mathrm{RTT}}(\mathfrak{gl}_n)}(\mathbb{k}, \mathbb{k}).$$

For a finite current truncation, the same construction is an ordinary finite-filtration spectral sequence.

Proof. The PBW filtration on the algebra induces a filtration on every bar length. The associated graded bar is the bar of the current Rees enveloping algebra. By theorem 207.1, its homology is the current Lie algebra's Chevalley homology. Standard convergence of the complete filtered bar gives the abutment. $\square$

No general collapse is asserted. The differentials encode the difference between the Yangian and its current shadow.

209.4 THE EXACTLY CALCULABLE ABELIAN CASE

For $\mathfrak{gl}_1$, RTT is commutative and

$$Y(\mathfrak{gl}_1) \cong \mathbb{k}[t^{(1)}, t^{(2)}, \dots].$$

Thus

$$\mathrm{Tor}_*^{Y(\mathfrak{gl}_1)}(\mathbb{k}, \mathbb{k}) \cong \Lambda(st^{(1)}, st^{(2)}, \dots).$$

For the subalgebra on the first N coefficients, the ungraded Poincare polynomial is $(1+t)^N$. This is a genuine Yangian bar calculation, not a recognition slogan.

209.5 ORDERED CHIRAL INTERPRETATION

The coefficients $T_{ij}(u)$ are meromorphic fields on the spectral curve. RTT is their ordered multiplication, the coproduct acts on tensor products of line representations, and the rational R -matrix exchanges parallel lines. The Yangian therefore realizes all three noncommutative layers: ordered word, meromorphic field coefficient, and braid transport. A full associative-chiral bar requires the formal-disc/nonlocal structure maps in addition to the Hopf algebra presentation.

The $\mathcal{A}_2$ Hall algebra in complete detail

210.1 THE REPRESENTATION CATEGORY

Let $Q : 1 \rightarrow 2$ over $\mathbb{F}_q$. Its indecomposables are

$$S_1 = (\mathbb{F}_q \rightarrow 0), \quad S_2 = (0 \rightarrow \mathbb{F}_q), \quad M = (\mathbb{F}_q \xrightarrow{1} \mathbb{F}_q),$$

and there is one nonsplit extension

$$0 \rightarrow S_2 \rightarrow M \rightarrow S_1 \rightarrow 0.$$

The Euler form is

$$\langle (a, b), (c, d) \rangle = ac + bd - ad.$$

Let $[A] * [B]$ count subobjects isomorphic to B with quotient A , and put $e_i = [S_i]$.

210.2 BINARY AND TRIPLE PRODUCTS

The binary products are

$$e_1 * e_2 = [S_1 \oplus S_2] + [M], \quad e_2 * e_1 = [S_1 \oplus S_2], \quad e_1 * e_1 = (q + 1)[2S_1].$$

In dimension vector $(2, 1)$, let

$$X = 2S_1 \oplus S_2, \quad Y = S_1 \oplus M.$$

Flag counting gives

$$\begin{aligned} e_1^2 e_2 &= (q + 1)([X] + [Y]), \\ e_1 e_2 e_1 &= (q + 1)[X] + [Y], \\ e_2 e_1^2 &= (q + 1)[X]. \end{aligned}$$

Theorem 210.1 (Untwisted Hall–Serre relation).

$$e_1^2 e_2 - (q + 1)e_1 e_2 e_1 + q e_2 e_1^2 = 0.$$

The companion relation is

$$q e_2^2 e_1 - (q + 1)e_2 e_1 e_2 + e_1 e_2^2 = 0.$$

Proof. For the first relation, the coefficient of $[Y]$ is $(q+1) - (q+1) = 0$, and the coefficient of $[X]$ is

$$(q+1) - (q+1)^2 + q(q+1) = 0.$$

The second follows from the symmetric flag count in dimension vector $(1, 2)$. □

At $q = 2$, evaluation of the first relation in the basis $([X], [Y])$ gives the zero vector in $\mathbb{Q}^2$.

210.3 TWIST AND QUANTUM GROUP

Let $v^2 = q$ and

$$[A] \diamond [B] = v^{\langle \dim A, \dim B \rangle} [A] * [B].$$

The total twisting factors in the first three triple products are $v^{-1}, 1, v$. Therefore

$$e_1 \diamond e_1 \diamond e_2 - (v + v^{-1})e_1 \diamond e_2 \diamond e_1 + e_2 \diamond e_1 \diamond e_1 = 0,$$

with the companion relation obtained by exchanging 1, 2.

Define

$$e_{12} = e_1 \diamond e_2 - v^{-1}e_2 \diamond e_1 = v^{-1}[M].$$

Every representation is uniquely $S_2^a \oplus M^b \oplus S_1^c$. Ordered divided-power monomials

$$e_2^{(a)} e_{12}^{(b)} e_1^{(c)}$$

are triangular with nonzero diagonal in the isomorphism-class basis ordered by arrow rank. They form a PBW basis, identifying the generated algebra with $U_v^+(\mathfrak{sl}_3)$.

210.4 ORDERED CHIRAL HALL COEFFICIENTS

Let the representation categories vary in a factorisation family over X . Extension correspondences at fixed support define the transverse Hall multiplication; disjoint support gives the chiral coefficient. If pull–push preserves the factorisation system and orientations, the Hall algebra is a transversely ordered chiral algebra. If the sheaf theory itself carries ordered chiral convolution, the derived extension algebra becomes a doubly associative chiral object. The finite-field calculation above gives the exact local structure constants.

Necklace Hamiltonians and the mixed trace algebra

2II.1 THE DERIVED COMMUTING SCHEME

Let

$$R = \mathbb{C}[[z_1, z_2]], \quad \{f, g\} = \partial_{z_1} f \partial_{z_2} g - \partial_{z_2} f \partial_{z_1} g,$$

and let $\mathfrak{h} = R/\mathbb{C} \cdot 1$. The shifted cotangent Lie algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^{\vee}[1]$$

controls formal closed Hamiltonian BF observables.

For N branes, take even matrices $X, Y \in \mathfrak{gl}_N$ and an odd matrix $\psi \in \mathfrak{gl}_N[1]$ with

$$QX = QY = 0, \quad Q\psi = [X, Y].$$

Adding the conjugation ghost gives functions on the derived quotient of the commuting equation. Gauge-invariant open observables are

$$J_N(f) = \text{Tr } f(X, Y).$$

2II.2 NECKLACE BRACKET

Let $F = \mathbb{C}\langle x, y \rangle$ and $F_{\text{cyc}} = F/[F, F]$. The cyclic derivative cuts a cyclic word after every occurrence of the indicated letter. Define

$$\partial_f(x) = -\partial_y^{\text{cyc}} f, \quad \partial_f(y) = \partial_x^{\text{cyc}} f,$$

and

$$\{f, g\}_{\text{neck}} = \left[\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g \right]_{\text{cyc}}.$$

The cyclic Euler identity

$$[x, \partial_x^{\text{cyc}} f] + [y, \partial_y^{\text{cyc}} f] = 0$$

implies $\partial_f([x, y]) = 0$, so the action descends to the derived commuting equation.

Theorem 211.1 (Necklace Lie algebra and trace action). *The bracket makes $F_{\text{cyc}}/\mathbb{C}$ a Lie algebra and*

$$[\partial_f, \partial_g] = \partial_{\{f, g\}_{\text{neck}}}.$$

Moreover,

$$\partial_f J_N(g) = J_N(\{f, g\}_{\text{neck}}).$$

Proof. Hamiltonian derivations preserve the noncommutative symplectic form $dx dy$, so their commutator is Hamiltonian. Evaluating on x, y gives the displayed bracket and Jacobi modulo commutators. Cyclic differentiation of $\text{Tr } g(X, Y)$, followed by cyclicity of trace, gives the second identity. $\square$

This theorem is an exact finite- N open–closed operation. It constructs a coordinate map into the boundary action; it does not yet prove a quasi-isomorphism from all closed observables to the entire derived centre.

211.3 STABLE TRACE POISSON ALGEBRA

On the semisimple commuting locus, write

$$X = \text{diag}(x_1, \dots, x_N), \quad Y = \text{diag}(y_1, \dots, y_N),$$

and define

$$p_{ab} = \sum_{i=1}^N x_i^a y_i^b, \quad a, b \geq 0, \quad a + b > 0.$$

The stable multisymmetric invariant algebra is

$$\Lambda_2 = \mathbb{C}[p_{ab} : a + b > 0].$$

Theorem 211.2 (Stable trace bracket). *For $\{x_i, y_j\} = \delta_{ij}$,*

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1, b+d-1}.$$

This is the commutative reduction of the necklace bracket.

Proof. Different particles Poisson commute, and for one particle

$$\{x^a y^b, x^c y^d\} = (ad - bc)x^{a+c-1}y^{b+d-1}.$$

Sum over particles. Simultaneous diagonalization identifies cyclic traces with power sums. $\square$

For the finite label set $0 \leq a, b \leq 3$, $(a, b) \neq (0, 0)$, all $15^3 = 3375$ ordered Jacobi sums have zero coefficient in the free $\mathbb{Q}$ -module on the resulting labels.

211.4 TWO BRANES AND THE $\mathcal{A}_1$ CONE

For $N = 2$, remove centre-of-mass variables and set $\{x, y\} = 2$. The invariants

$$A = \frac{x^2}{2}, \quad B = \frac{xy}{2}, \quad C = \frac{y^2}{2}$$

satisfy

$$B^2 = AC, \quad \{A, B\} = 2A, \quad \{B, C\} = 2C, \quad \{A, C\} = 4B.$$

After rescaling, this is the Kirillov–Kostant bracket on the nilpotent cone of $\mathfrak{sl}_2$. The first nontrivial finite-rank sector already contains a singular symplectic surface and a nonabelian bracket.

No-go. *The map $f \mapsto J_N(f)$ intertwines Hamiltonian brackets, but it need not be essentially surjective on derived central observables and need not detect non-trace or finite-rank syzygies. A full bulk-centre theorem requires a chain-level map, a filtration, and a proof that its cone is acyclic.*

Modules, orbifolds, and higher E_n directions

212.1 POINTED BARS AND INTERFACES

For a right module M_R and left module M_L ,

$$\text{Bar}_A(M_R, M_L) = |[r] \mapsto M_R \otimes! \bar{A}^{\otimes!r} \otimes! M_L| \simeq M_R \otimes_A^{\mathbb{L}} M_L.$$

A marked chiral insertion gives left and right coactions of $\text{Bar}(A)$ by cutting the interval on either side. Interfaces are therefore bimodules and compose by relative tensor product; after Koszul transform, they compose by cotensor product.

212.2 CROSSED PRODUCTS

Let a finite group G act by augmented automorphisms of A . Then

$$\text{Mod}_{A \rtimes G} \simeq (\text{Mod}_A)^G.$$

The crossed-product bar is the diagonal totalization of the group bar and the A -bar. In characteristic zero, averaging contracts positive group-homology rows. The fixed algebra A^G and crossed product $A \rtimes G$ remain different: the first describes invariant operators, the second equivariant modules and symmetry defects. A Galois or freeness theorem is needed for Morita equivalence.

212.3 SYMMETRIC PRODUCTS AND DMVV

For a graded vector space V with character $f(q)$,

$$\sum_{N \geq 0} p^N \text{ch Sym}^N(V) = \exp\left(\sum_{r \geq 1} \frac{p^r}{r} f(q^r)\right).$$

For a suitable seed elliptic genus $\sum c(m, \ell) q^m y^\ell$, the symmetric-product orbifold trace is

$$\sum_{N \geq 0} p^N \text{Ell}_{\text{orb}}(\text{Sym}^N Y) = \prod_{n \geq 1, m \geq 0, \ell \in \mathbb{Z}} (1 - p^n q^m y^\ell)^{-c(nm, \ell)}.$$

The proof is cycle decomposition in symmetric groups. This is a trace identity. The orbifold operator algebra additionally requires twisted sectors, defect composition, and OPE maps. It must not be identified with a bar complex from the product formula alone.

212.4 HIGHER TRANSVERSE DIMENSIONS

Dunn additivity gives

$$\mathrm{Alg}_{E_{n+1}}(\mathcal{V}) \simeq \mathrm{Alg}_{E_1}(\mathrm{Alg}_{E_n}(\mathcal{V})).$$

An E_n -ordered chiral algebra has n locally constant transverse directions while retaining its chiral coefficient. Its n -fold bar is factorisation homology of the pointed n -disk and produces an E_n -coalgebra under suitable positivity conditions. The derived centre is an E_{n+1} -object.

Independently, one can form E_n -associative chiral algebras for the chiral product. The same theorem-of-two-bars warning persists: “higher E_n ” must specify which monoidal direction is being iterated.

No-go. *A list of graded dimensions, even if it matches a root multiplicity or partition function, contains no coproduct, antipode, associator, or R -matrix. Tannaka reconstruction requires a tensor category and a fibre functor; Hall reconstruction requires extension correspondences and orientations. Scalar agreement is a shadow, not the structured object.*

Perturbative BV realization of the ordered chiral object

213.1 CLASSICAL DATA

A perturbative classical field theory on a manifold $\mathcal{M}$ is described locally by an elliptic L_∞ -algebra of fields $\mathcal{E}[-1]$, an invariant pairing of degree -1 , and an action functional S solving the classical master equation

$$dS + \frac{1}{2}\{S, S\} = 0.$$

Classical observables on an open set are Chevalley cochains of compactly supported fields, with differential induced by the L_∞ -structure. Disjoint support gives factorisation multiplication.

On $X^{\text{an}} \times \mathbb{R}$, assume the kinetic complex is Dolbeault in X and de Rham or locally constant in the real direction. A boundary condition on $X \times \mathbb{R}_{\geq 0}$ restricts fields to a factorisation algebra on $X \times \mathbb{R}$. The ordered interval direction then supplies a transverse E_1 -product.

213.2 RENORMALISED QUANTUM OBSERVABLES

A gauge fixing produces a propagator P_ϵ^L between ultraviolet and infrared scales. Effective interactions $I[L]$ solve the homotopy renormalisation-group equation and the quantum master equation

$$(Q + \hbar\Delta_L)e^{I[L]/\hbar} = 0.$$

The quantum observable differential is

$$Q_L = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

The QME is exactly $Q_L^2 = 0$.

Feynman weights are differential forms on configuration spaces of vertices. Renormalisation extends them to compactified collision spaces after local counterterms are added. Restriction to a boundary face factorises through contraction of the corresponding subgraph.

Theorem 213.1 (Feynman-weight morphism). *Suppose:*

- (i) *renormalised graph weights extend to the chosen two-direction compactifications;*
- (ii) *restriction to every face agrees with graph contraction and the relevant local counterterm;*

- (iii) *the renormalised QME holds;*
- (iv) *boundary observables are holomorphically algebraizable on X and locally constant in the ordered direction.*

Then the graph weights define a morphism from the geometric two-coloured forest resolution to the endomorphism operad of boundary observables. Hence the boundary observables form a homotopy-coherent transversely ordered chiral algebra. If ordered chiral collision vertices are also present, they form the corresponding doubly associative object.

Proof. Face factorisation is operadic composition. Stokes' theorem for the extended weights identifies the de Rham boundary with the sum of graph contractions. The QME cancels the full codimension-one boundary, including BV and counterterm faces, so the assignment is a chain map. Algebraisation and local constancy identify the resulting operations with the coefficient and transverse structures. $\square$

No-go. *A classical factorisation algebra can fail to admit a quantum lift because the one-loop obstruction class is nonzero. Gauge fixing can fail to preserve the chosen boundary condition, and the analytic observables can fail to algebraize as regular holonomic D -modules. The quantum ordered chiral object exists only after these obstructions vanish or are cancelled by an invertible anomaly theory.*

213.3 FREE HOLOMORPHIC-TOPOLOGICAL BF THEORY

Let V be finite dimensional and take fields

$$A \in \Omega^{0,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes V[1], \quad B \in \Omega^{1,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes V^*[-1]$$

with action

$$S_{\text{BF}} = \int_{X \times \mathbb{R}} \langle B, (\bar{\partial} + d_t) A \rangle.$$

The theory is Gaussian. The de Rham contraction on an interval proves local constancy in t ; the Dolbeault complex gives the holomorphic coefficient. Complementary linear boundary conditions select a polarization. Quantization of paired boundary fields produces chiral Weyl or Clifford systems according to parity. The central constant in the Weyl relation is handled by the curved duality of [chapter 205](#), not by a nonexistent scalar augmentation.

BRST descent and Drinfeld–Sokolov reduction

214.1 THE BAR–BRST BICOMPLEX

Let $Q : A \rightarrow A[1]$ be a square-zero derivation of the transverse multiplication, augmentation, and every factorisation map. On a bar word define

$$Q_B[a_1 | \cdots | a_r] = \sum_{i=1}^r (-1)^{\sum_{j<i} (|a_j|+1)} [a_1 | \cdots | Qa_i | \cdots | a_r].$$

Proposition 214.1 (BRST compatibility).

$$Q_B^2 = 0, \quad Q_B d_B + d_B Q_B = 0.$$

Thus $(\text{Bar}(A), d_B, Q_B)$ is a bicomplex in factorisation coefficients.

Proof. The first identity is the tensor differential calculation. For the second, compare applying Q before and after an interior face. The two terms agree by the derivation rule and acquire opposite totalisation signs. Outer faces agree because the augmentation is Q -closed. $\square$

Theorem 214.2 (Bar–BRST comparison). *Assume tensor products preserve Q -quasi-isomorphisms and the combined filtration is complete and bounded below in every total weight. Then*

$$E_1 \cong \text{Bar}(H_Q(A)) \implies H^* \text{Tot}(\text{Bar}(A), d_B + Q_B).$$

If $H_Q(A)$ is concentrated in degree zero, the edge map

$$\text{Bar}(H_Q^0(A)) \xrightarrow{\sim} \text{Tot}(\text{Bar}(A), d_B + Q_B)$$

is a quasi-isomorphism.

Proof. Filter first by BRST degree and then by bar length. The vertical complex is a tensor product of copies of (A, Q) . The Kunnet hypothesis identifies its cohomology with tensor powers of $H_Q(A)$, and the induced horizontal differential is the bar differential of the cohomology algebra. Completeness gives convergence. Concentration leaves one row. $\square$

214.2 PRINCIPAL DRINFELD–SOKOLOV REDUCTION

For an affine vertex algebra $V_k(\mathfrak{g})$, choose a good grading for a nilpotent element f , introduce charged and neutral ghosts, and form the BRST current from the moment map, cubic ghost term, character, and neutral correction. Its residue is a derivation Q with $Q^2 = 0$. The reduced algebra is

$$\mathcal{W}_k(\mathfrak{g}, f) = H_Q^0(V_k(\mathfrak{g}) \otimes \text{Ghosts}).$$

For principal f at generic noncritical level, the standard concentration theorem places cohomology in degree zero. Then [theorem 214.2](#) identifies the ordered bar of the principal $\mathcal{W}$ -algebra with BRST cohomology of the affine-ghost bar.

214.3 THE $\mathfrak{sl}_2$ CALCULATION

For affine currents e, h, f at level k , introduce fermionic ghosts b, c and

$$j_{\text{BRST}}(z) = (e(z) - 1)c(z), \quad Q = \text{Res } j_{\text{BRST}}(z).$$

The e - e and c - c OPEs are regular, hence $Q^2 = 0$. Improving the Sugawara stress tensor and adding the ghost stress tensor gives a Q -closed class with central charge

$$c_{\text{DS}}(k) = 13 - 6(k + 2) - \frac{6}{k + 2} = 1 - \frac{6(k + 1)^2}{k + 2}.$$

A filtration by f - and b -modes pairs the constrained currents into contractible quartets. The surviving Heisenberg field has the Feigin–Fuchs background charge, and its invariant Miura subalgebra is Virasoro. This realizes principal $\mathfrak{sl}_2$ reduction in degree zero for generic $k \neq -2$.

Warning 214.3. At special levels, higher BRST cohomology and nonsemisimple phenomena can occur. The concentration hypothesis must be checked in the representation category being used.

From order to exchange: KZ, Tannaka, and four-dimensional Chern–Simons

215.1 INFINITESIMAL BRAID RELATIONS

Let $\mathfrak{g}$ be a quadratic Lie algebra and $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ the inverse invariant form. On $V_1 \otimes \cdots \otimes V_n$, write Ω_{ij} for its action in factors i, j . Invariance gives

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{for disjoint pairs,} \quad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0.$$

These are the infinitesimal braid relations.

The KZ connection on $\text{Conf}_n(\mathbb{C})$ is

$$\nabla_{\text{KZ}} = d - \kappa \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Proposition 215.1 (KZ flatness). *The KZ connection is flat.*

Proof. The logarithmic one-forms are closed, so curvature is the wedge-square of the connection form. Terms on four distinct labels vanish by disjoint commutativity. On three labels, the coefficient is an infinitesimal braid commutator multiplied by Arnold’s relation

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

Both factors vanish in the required combination. □

Monodromy gives a braid-group representation. The Drinfeld–Kohno theorem identifies the formal monodromy category with the braided representation category of the corresponding quantum group after an associator and parameter identification are chosen. The associator is part of the comparison; it is not produced by the ordered line.

215.2 MEROMORPHIC TANNAKA RECONSTRUCTION

Let $\mathcal{C}$ be a rigid meromorphic tensor category of line operators and let $\omega : \mathcal{C} \rightarrow \text{Vect}_{\mathbb{K}((u))}$ be a faithful exact strong monoidal fibre functor. The coend

$$H = \int^{V \in \mathcal{C}} \omega(V)^* \otimes \omega(V)$$

reconstructs a bialgebra. Rigidity gives an antipode. Braided exchange gives a coquasitriangular structure. A topological completion may be required when spectral expansions are infinite.

No-go. *Tannaka reconstruction outputs an ordinary or topological Hopf algebra only after a suitable fibre functor is specified. Without it, the intrinsic output is the tensor category itself, or a Hopf object in a different base category. “The quantum group” is not a scalar extracted from braid matrices.*

215.3 FOUR-DIMENSIONAL CHERN–SIMONS THEORY

Let Σ be an oriented real surface, C a complex curve with meromorphic one-form ω , and consider the partial connection of four-dimensional Chern–Simons theory on $\Sigma \times C$:

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma \times C} \omega \wedge \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

The theory is topological on Σ and holomorphic on C . A Wilson line in Σ at spectral point $z \in C$ is a line object. Parallel fusion is ordered; exchange in Σ produces an R -matrix. In the rational case, the tree-level exchange is

$$R(z_1 - z_2) = 1 + \hbar \frac{\Omega}{z_1 - z_2} + O(\hbar^2).$$

Gauge invariance at the first nontrivial order imposes the classical Yang–Baxter equation. Quantisation produces the rational Yangian-type exchange algebra under the boundary, anomaly, and representation hypotheses of the model.

The geometry explains why ZF/RTT relations and Yang–Baxter occur. The complete identification with a specific Yangian remains a theorem about the line category and its generators, not a consequence of dimensional analysis.

Cyclic traces, modular sewing, and the two-edge master equation

216.1 CLOSING THE TRANSVERSE INTERVAL

Let $\mathcal{A}$ be a proper cyclic A_∞ -algebra object in the coefficient category. A degree- d Calabi–Yau structure is a negative-cyclic class inducing an equivalence

$$\mathcal{A} \xrightarrow{\sim} \mathcal{A}^\vee[d].$$

On a finite strict model it is represented by a cyclic trace with perfect pairing $\langle a, b \rangle = \text{Tr}_{\mathcal{A}}(ab)$. The circle value is

$$\int_{S^1} \mathcal{A} \simeq \text{HH}_*(\mathcal{A}).$$

A ribbon graph is evaluated by cyclic higher multiplications at vertices and the inverse pairing at edges.

Theorem 216.1 (Ribbon-graph action). *A finite perfect cyclic A_∞ -algebra defines a dg PROP morphism from the positive-boundary ribbon-graph complex to endomorphisms of its Hochschild chains. Edge contraction is sent to composition of cyclic operations, and gluing boundary circles is sent to insertion of the coevaluation pairing.*

Proof. The graph evaluation is the tensor contraction determined by cyclic order at vertices. The ribbon differential sums internal-edge contractions. The cyclic A_∞ -relations are exactly the assertion that this sum intertwines the graph and Hochschild differentials. Disjoint union and gluing agree with tensor product and coevaluation. $\square$

This produces a transverse ribbon genus $g_\perp$.

216.2 CHIRAL SEWING OVER STABLE CURVES

A modular chiral coefficient assigns state spaces or complexes to pointed curves and contraction maps to separating and nonseparating clutching morphisms

$$\begin{aligned} \overline{\mathcal{M}}_{g_1, n_1+1} \times \overline{\mathcal{M}}_{g_2, n_2+1} &\xrightarrow{o} \overline{\mathcal{M}}_{g_1+g_2, n_1+n_2}, \\ \overline{\mathcal{M}}_{g-1, n+2} &\rightarrow \overline{\mathcal{M}}_{g, n}. \end{aligned}$$

A perfect chiral pairing contracts the two flags of a node. The resulting genus g_X is the genus of the algebraic curve and has no reason to equal $g_\perp$.

216.3 BICOLOURED STABLE GRAPHS

Give every internal edge a colour X or $\perp$. An X -edge is a chiral clutching node; a $\perp$ -edge is a ribbon/trace contraction. Orient a graph by

$$\det E_X \otimes \det E_\perp.$$

Exchanging the two edge colours changes sign. The graph brackets and loop operators are

$$[-, -]_X, \quad [-, -]_\perp, \quad \Delta_X, \quad \Delta_\perp.$$

Recognition Theorem 216.2 (Two-edge sewing theorem). *Suppose a state object carries:*

- (i) *cyclic transverse gluing and a perfect transverse pairing;*
- (ii) *chiral clutching and a perfect chiral sewing pairing;*
- (iii) *common genus-zero corollas;*
- (iv) *the one-colour associativity/modular identities;*
- (v) *a mixed interchange square for one ribbon and one chiral edge, with the product determinant sign.*

Then there is a unique action of the bicoloured stable-graph transform extending the two one-colour actions. Its master element satisfies

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_X + \frac{1}{2}[\Theta, \Theta]_\perp + \hbar_X \Delta_X \Theta + \hbar_\perp \Delta_\perp \Theta = 0.$$

In particular,

$$\Delta_X \Delta_\perp + \Delta_\perp \Delta_X = 0.$$

Proof. Every bicoloured graph is built by gluing common corollas along coloured edges. The one-colour axioms identify all decompositions involving only one colour. The mixed square identifies adjacent contractions of different colours. These moves connect any two edge-orderings, giving existence and uniqueness. The codimension-two boundary of the bicoloured graph complex is the sum of the two brackets, two loop contractions, and the internal differential. Its square is zero by the determinant orientation, yielding the master equation. $\square$

No-go. *The ribbon genus counts handles in the surface thickening of a cyclic associative graph. The chiral genus counts handles of the algebraic curve in Deligne–Mumford moduli. They are gradings from different edge colours. A concrete duality may relate selected amplitudes, but there is no natural transformation identifying g_X with $g_\perp$ for all ordered chiral algebras.*

Prime forms, plumbing, and the analytic chiral genus

217.1 THE NORMALIZED BIDIFFERENTIAL

Let Σ_g be a compact Riemann surface with symplectic homology basis and period matrix Ω . The prime form $E(p, q)$ is a $(-1/2, -1/2)$ -form with a simple zero on the diagonal. The normalized fundamental bidifferential has local behaviour

$$B(p, q) = \left(\frac{1}{(z(p) - z(q))^2} + O(1) \right) dz(p) dz(q)$$

and vanishing A -periods. It may be expressed through logarithmic derivatives of the prime form with the normalization correction fixed by periods.

For a rank-one Heisenberg current,

$$\langle J(p)J(q) \rangle_{\Sigma_g} = kB(p, q).$$

Higher current correlators are sums over pairings. This is an exact Gaussian realization of the chiral coefficient at every genus once the determinant line and zero modes are included.

217.2 PLUMBING A NODE

For local coordinates z, w at two marked points, plumbing identifies

$$zw = q, \quad |q| \ll 1.$$

A separating node joins two components; a nonseparating node joins two points on one component. Formal sewing inserts a complete set of states:

$$\mathcal{F}_{\text{sew}}(q) = \sum_{\alpha} q^{L_0(\alpha) - c/24} \mathcal{F}_1(\dots, v_{\alpha}) \mathcal{F}_2(v^{\alpha}, \dots).$$

The expression is formal in q . Analytic sewing additionally requires convergence, control of the completion, and compatibility of the chosen local coordinates.

No-go. *An identity in $V[[q]]$ can hold while the corresponding series has zero radius of convergence in a chosen topology. Trace-class estimates, energy bounds, or a rational/CFT finiteness theorem are separate analytic inputs. Formal modular-operad structure and analytic conformal blocks must be stated separately.*

217.3 GENUS ONE

For a positive-energy module M , the torus character is

$$\chi_M(\tau) = \text{Tr}_M q^{L_0 - c/24}, \quad q = e^{2\pi i \tau}.$$

For a rank-one Heisenberg algebra,

$$\chi_{\text{Heis}}(\tau) = \eta(\tau)^{-1}.$$

For an even lattice L ,

$$\chi_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^{\text{rk } L}}.$$

The theta numerator records zero modes; the eta denominator is the oscillator determinant. Their modular transformations are sections of the anomaly line rather than ordinary invariant functions until the determinant factor is included.

217.4 HIGHER GENUS AND SIEGEL THETA SERIES

The lattice zero-mode sum becomes

$$\Theta_L^{(g)}(\Omega) = \sum_{\lambda_1, \dots, \lambda_g \in L} \exp\left(\pi i \sum_{a,b} (\lambda_a, \lambda_b) \Omega_{ab}\right).$$

The oscillator factor is the corresponding chiral determinant. Separating degeneration factorises the theta series into the products of lower-genus theta series; nonseparating degeneration is a Fourier–Jacobi expansion. These are genuine analytic data that survive the correction from one genus to two independent genera: they belong to g_X , not to the ribbon loop count.

A concrete two-genus model

218.1 THE TRANSVERSE FROBENIUS ALGEBRA

Take the augmented noncommutative Frobenius algebra $B_n = D \times M_n(\mathbb{k})$ of proposition 205.3. Its cyclic trace produces a ribbon theory. Let

$$R_h(n) = \tau_B(e_{B_n}^h) = \begin{cases} n, & h = 0, \\ 2 + n^2, & h = 1, \\ n^{h+1}, & h \geq 2. \end{cases}$$

This is an exact closed-surface amplitude in the associated two-dimensional topological field theory convention.

218.2 A RANK-ONE DELIGNE–MUMFORD THEORY

Let $C_\lambda = \mathbb{k}\mathbf{1}$ be the one-dimensional commutative Frobenius algebra with

$$\mathbf{1} \cdot \mathbf{1} = \mathbf{1}, \quad \eta(\mathbf{1}, \mathbf{1}) = \lambda \in \mathbb{k}^\times.$$

The inverse metric is λ^{-1} , and the handle element is $\lambda^{-1}\mathbf{1}$. The associated rank-one cohomological field theory assigns the constant class

$$\Omega_{g,r} = \lambda^{1-g} \in H^0(\overline{\mathcal{M}}_{g,r}; \mathbb{k})$$

to every stable pointed curve.

Proposition 218.1 (Clutching). *The classes $\Omega_{g,r}$ satisfy both separating and nonseparating CohFT gluing equations.*

Proof. A separating gluing contracts two classes with the inverse metric and gives

$$\lambda^{1-g_1} \lambda^{1-g_2} \lambda^{-1} = \lambda^{1-(g_1+g_2)}.$$

A nonseparating gluing gives

$$\lambda^{1-(g-1)} \lambda^{-1} = \lambda^{1-g}.$$

The unit axiom is immediate. □

218.3 STRICT MIXED INTERCHANGE

Form the tensor-product state object

$$\mathcal{H} = B_n \otimes C_\lambda.$$

Ribbon contractions act on the first factor; Deligne–Mumford clutching acts on the second. Common corollas are tensor products of the two multiplications. Operators on different tensor factors commute before the determinant-line sign is inserted.

Theorem 218.2 (Realized two-edge recognition). *The state object $\mathcal{H}$ satisfies all five hypotheses of recognition Theorem 216.2. Its bivariate closed amplitude is*

$$Z_{g_X, g_\perp}(n, \lambda) = \lambda^{1-g_X} R_{g_\perp}(n).$$

Thus

$$Z_{g_X, 0} = n\lambda^{1-g_X}, \quad Z_{g_X, 1} = (2 + n^2)\lambda^{1-g_X}, \quad Z_{g_X, h} = n^{h+1}\lambda^{1-g_X} \quad (h \geq 2).$$

The two genus variables are independent and the mixed interchange square is strict.

Proof. Cyclicity and perfection for B_n give the transverse sewing. proposition 218.1 gives the chiral modular sewing. Tensor-product corollas satisfy each one-colour associativity law. A contraction on B_n commutes with a contraction on C_λ , and exchanging their odd edge orientations supplies the required sign. The recognition theorem applies. The amplitude factorises into the two exact one-colour amplitudes. $\square$

This is the missing realized invocation of the recognition theorem. It is deliberately elementary enough that every hypothesis and scalar can be checked. It proves the formal two-genus architecture unconditionally. Tensoring the rank-one CohFT with a rational vertex-algebra modular functor gives richer analytic examples once convergence and anomaly-line hypotheses are supplied.

218.4 WHAT THE MODEL PROVES AND WHAT IT DOES NOT

The example proves existence of a noncommutative transverse sector, a genuine augmentation and bar, two independent modular contractions, and a mixed graph action. It does not claim that every holomorphic–topological gauge theory factorizes as a tensor product, or that its two loop parameters are physically unrelated. Interacting theories can have mixed vertices; then the mixed square is a substantive calculation rather than strict tensor-factor commutation.

Typed anomalies and invertible theories

219.1 FIVE DOMAINS

The word “anomaly” names several obstruction classes.

Anomaly type	Native object
Curved Koszul term	arity-zero element h in a curved dg algebra, with $d^2 = [h, -]$.
BRST anomaly	failure of the quantum BRST operator to square to zero, a class in the BRST deformation complex.
BV anomaly	obstruction to the QME, usually a degree-one local functional modulo exact counterterms.
Conformal/determinant anomaly	curvature or first Chern class of a determinant line over moduli.
Framing anomaly	invertible field theory or central extension measuring dependence on a framing/trivialization.

A family index map, transgression, boundary map, or open–closed comparison can carry one class to another in a concrete theory. The map is the theorem.

Theorem 219.1 (Anomaly cancellation by an inverse theory). *Let $\mathcal{L}$ be the invertible anomaly theory of a quantum field theory $\mathcal{T}$. If an inverse $\mathcal{L}^{-1}$ and a trivialization of $\mathcal{L} \otimes \mathcal{L}^{-1}$ are chosen compatibly with gluing, then $\mathcal{T} \otimes \mathcal{L}^{-1}$ has an honest, rather than projective, state assignment on the corresponding bordism/moduli category.*

Proof. The anomalous theory is a relative natural transformation valued in the line assigned by $\mathcal{L}$. Tensoring with the inverse line makes the value canonically scalar after the chosen trivialization. Compatibility with gluing is exactly naturality of the trivialization. $\square$

No-go. *There is no number that simultaneously represents curved Koszul data, BV obstruction, determinant curvature, BRST failure, and framing dependence. They live in different cohomology theories and can have different degrees. A scalar appears only after evaluating a specified class on a cycle, trace, index, or background.*

No-go. *The Virasoro central charge is one coefficient of one OPE. Distinct vertex algebras, module categories, fusion rules, modular data, and transverse products can share the same central charge. Therefore no classification of ordered or associative chiral algebras can be based on c alone.*

Bar Euler products and Borcherds denominators

220.1 ROOT-GRADED LIE SUPERALGEBRAS

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha, \quad \mathfrak{n}_+ = \bigoplus_{\alpha \in \Gamma_+} \mathfrak{g}_\alpha,$$

where every root space is finite dimensional and each degree has finitely many decompositions in the chosen completion. PBW and theorem 207.1 give

$$\text{Bar}^\perp(U(\mathfrak{n}_+)) \simeq \text{CE}_*(\mathfrak{n}_+) = \text{Sym}(\mathfrak{n}_+[1]).$$

Theorem 220.1 (Bar denominator product). *The completed graded Euler supercharacter is*

$$\chi_{\Gamma} \text{Bar}^\perp(U(\mathfrak{n}_+)) = \prod_{\alpha \in \Gamma_+} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_\alpha}.$$

If $\mathfrak{g}$ is a Borcherds–Kac–Moody superalgebra with Weyl vector ρ , multiplying by $e^{-\rho}$ gives the product side of its denominator identity.

Proof. An even root vector becomes odd after suspension and contributes $1 - e^{-\alpha}$ to the Euler supercharacter. An odd root vector becomes even and contributes its inverse. Multiplication over a homogeneous basis gives the superdimension exponent. Finite decomposition makes each coefficient of the completed product finite. $\square$

The theorem is an object-to-character map: the Lie bracket controls the Chevalley differential, while the Euler character remembers only signed root multiplicities.

220.2 THE IGUSA PRODUCT

Let

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n y^\ell$$

be the normalized weak Jacobi form of weight zero and index one. In the standard chamber, the monic Borcherds product is

$$D_5(Z) = q^{1/2} y^{1/2} p^{1/2} \prod_{(n, \ell, m) > 0} (1 - q^n y^\ell p^m)^{f(nm, \ell)}.$$

For the Gritsenko–Nikulin Borchers algebra, the signed root multiplicities are $f(nm, \ell)$. With the theta-constant normalization $\Delta_5 = 64D_5$, the denominator convention gives

$$e^{-\rho} \chi_{\Gamma} \text{Bar}^{\perp}(U(\mathfrak{n}_{\Delta_5, +})) = D_5(2Z) = 64^{-1} \Delta_5(2Z).$$

This equality is a typed Euler-character identity. It does not identify the entire bar complex with an automorphic function.

The scalar 64 is not a convention. Writing $\mathfrak{E}$ for the ten even genus-two characteristics and $\Delta_5^{\theta}(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \mathfrak{F}[\mathfrak{m}](Z, \mathfrak{o})$, the cusp leading terms of the ten factors group by first half-characteristic: four contribute the scalar 1, and the remaining six each contribute a factor 2 from a pair of shortest lattice representatives exchanged by sign. The exponents sum to $(1/2, 1/2, 1/2)$, so

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^{\theta} = 2^6 = 64,$$

which fixes $\Delta_5^{\theta} = 2^6 \Delta_5^{\text{mon}}$ and with it the normalization used above. The derivation is carried out in `Volume_III_Igusa_Borchers_Theory.tex` of the `platonic-integrated spine`, deployed at `igusa-cusp-form/platonic/` and `calabi-yau-quantum-groups/platonic/`; the weight is $f(\mathfrak{o}, \mathfrak{o})/2 = 5$, consistent with the normalization $f(\mathfrak{o}, \mathfrak{o}) = 10$ fixed above.

220.3 FINITE POSITIVE WINDOWS

Let $h : \Gamma_+ \rightarrow \mathbb{N}_{>0}$ be proper and additive. The quotient

$$\mathfrak{n}_+(H) = \mathfrak{n}_+ / \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\alpha}$$

is finite-dimensional nilpotent. Its level-zero current vertex algebra has PBW character, and the inverse limit over H is separated and complete. Coefficientwise finiteness follows because a fixed root and oscillator degree sees only finitely many positive roots and modes.

220.4 POLARIZED GRAM TRANSFORM

Let $\Pi : \Gamma_{\text{phys}} \rightarrow \Lambda$ be quadratic with polarization

$$B(\gamma, \delta) = \Pi(\gamma + \delta) - \Pi(\gamma) - \Pi(\delta).$$

For a charge-graded algebra $\mathcal{A} = \bigoplus_{\gamma} \mathcal{A}_{\gamma}$, define

$$(a_{\gamma} u^{\lambda}) \star_B (b_{\delta} u^{\mu}) = a_{\gamma} b_{\delta} u^{\lambda + \mu + B(\gamma, \delta)}.$$

Biadditivity gives the cocycle identity and hence associativity. The Gram embedding

$$\iota_{\Pi}(a_{\gamma}) = a_{\gamma} u^{\Pi(\gamma)}$$

is an algebra homomorphism for $\star_B$. This is the exact device that turns quadratic Fourier indices into an additive grading without pretending that Π itself is additive.

Warning 220.2. A compact Hall realization of the Igusa root algebra is a further theorem. It must construct the moduli stack, orientation data, extension correspondence, PBW comparison, and map to the Borchers algebra. The denominator product alone supplies none of these maps.

Higher-dimensional sources and factorisation pushforward

221.1 PUSHFORWARD ALONG AN ACTUAL MAP

Let $f : M \rightarrow X$ be a map and $\mathcal{O}$ a factorisation algebra on M . Under properness/support and descent conditions, factorisation pushforward is

$$(f_*^{\text{Fact}} \mathcal{O})(U) = \mathcal{O}(f^{-1}U).$$

Disjoint opens in X have disjoint inverse images, so the factorisation maps descend. Holomorphic or topological reduction to a curve is therefore a functor only after an actual map and support theory are specified.

221.2 CALABI–YAU CATEGORIES AND HALL PRODUCTS

A Calabi–Yau category of branes can produce:

- (i) an E_1 -algebra of endomorphisms of a chosen brane;
- (ii) extension correspondences on moduli of objects, yielding Hall or cohomological Hall multiplication;
- (iii) cyclic pairings and open-string traces;
- (iv) closed-string maps into Hochschild invariants.

Shifted symplectic geometry can orient the correspondences and control quantization. Each arrow needs its own finiteness and orientation theorem. A slogan such as “Calabi–Yau source” is not yet an ordered chiral algebra.

221.3 COULOMB BRANCHES AND SHIFTED YANGIANS

In suitable supersymmetric gauge theories, convolution on affine Grassmannian-type spaces produces quantized Coulomb branches, while line defects organize modules over shifted Yangians. To place such an algebra in the present theory one must exhibit:

- (i) the curve-valued factorisation coefficient;
- (ii) the ordered line multiplication;
- (iii) the augmentation or relative base used by the bar;

- (iv) the exchange geometry or R -matrix;
- (v) the comparison to the RTT/Drinfeld presentation.

Once these data are present, bar, centre, and BRST constructions are functorial. Without them, name matching is not a proof.

22I.4 SOURCE-TO-CHIRAL COMPARISON

The correct general pattern is a diagram

$$\begin{array}{ccc}
 \text{higher-dimensional geometric category} & \xrightarrow{\text{pushforward/restriction}} & \text{factorisation category on } X \\
 \text{Hall or convolution} \downarrow & & \downarrow \text{RHom}(b,b) \\
 \text{geometric algebra} & \xrightarrow{\text{comparison}} & \text{Alg}_{E_1}(\text{FactD}(X)).
 \end{array}$$

Every square is a theorem about correspondences, supports, and descent. The diagram is the disciplined replacement for a universal “shadow tower.”

The no-go constitution

The theory must distinguish a genuine comparison map from a coincidence of shadows. Each prohibition below names the datum whose absence blocks the proposed identification and the construction that would make a comparison possible.

222.1 CARRIER PROHIBITIONS

- N1. Residue is not the transverse bar differential.** Residue changes normal/collision degree; the bar differential contracts adjacent transverse letters. They can anticommute in a bicomplex.
- N2. A logarithmic form does not extract all principal parts.** Higher poles require the complete normal-jet filtration or a chosen connection.
- N3. Incidence does not contract screen cohomology.** A terminal corolla can make an incidence nerve contractible while the geometric screen has nontrivial cohomology.
- N4. Order does not produce exchange.** An ordered chamber on a line is contractible and has no braid generator.
- N5. A trace is not an augmentation.** Cyclic linear functionals do not define multiplicative bar faces.
- N6. Graph topology does not determine coefficients.** A stable graph specifies possible contractions; a pairing and state assignment specify their values.

222.2 MONOIDAL AND OPERADIC PROHIBITIONS

- N7. Hadamard and Cauchy products are not globally strongly duoidal.** The exact relation is a split inclusion/projection with aligned idempotent.
- N8. Transverse E_1 does not imply associative chiral E_1 .** The two products have different carriers and operads.
- N9. Associative order does not imply a quantum vertex $\mathcal{S}$ -operator.** Unitarity, Yang–Baxter, shift, and braided locality are extra equations.
- N10. Two pairwise exchange rules do not imply coherent three-field exchange.** The critical-pair obstruction is Yang–Baxter.
- N11. The interval bar, associative chiral bar, and chiral Chevalley complex are not one complex.** A comparison requires an enveloping or distributive-law map.
- N12. A quadratic dual is not automatically the derived Koszul dual.** Exactness of the quadratic Koszul complex is the missing theorem.

- N13. Bar–cobar is not an equivalence without completion.** The output is nilcompletion/conilcompletion in general.
- N14. Verdier, boundary, quadratic, and bar duals are not definitionally identical.** Finite self-dual examples supply comparison maps.

222.3 ALGEBRA AND CALCULATION PROHIBITIONS

- N15. The Weyl algebra has no scalar augmentation.** Use a Rees degeneration, a curved dual, or a relative module.
- N16. A full matrix algebra has no scalar augmentation.** Use Morita/relative bar data or embed it in an honestly augmented algebra.
- N17. A multi-idempotent quiver is naturally relative over its vertex algebra.** Scalar augmentation forgets the boundary objects.
- N18. A chain-space count is not a homology dimension.** The differential and its rank are part of the invariant.
- N19. A matching integer sequence is not a quasi-isomorphism.** Motzkin and Riordan arithmetic survives only in its actual planar combinatorial role.
- N20. Central charge does not classify a chiral algebra.** OPEs, modules, fusion, modular data, and transverse multiplication remain independent.
- N21. A character does not reconstruct a Hopf algebra.** Coproduct, antipode, associator, and exchange are structured maps.
- N22. A denominator product does not construct a Hall realization.** Moduli, orientations, extension correspondences, and PBW comparison are separate.

222.4 PHYSICS AND SEWING PROHIBITIONS

- N23. A boundary centre does not determine an arbitrary decoupled bulk.** An open–closed equivalence is a theorem with generation and locality hypotheses.
- N24. The two genera are not universally equal.** Ribbon and Deligne–Mumford handles are different edge colours.
- N25. Formal sewing does not imply analytic convergence.** Energy bounds or trace-class estimates are additional inputs.
- N26. Different anomaly classes are not one scalar.** Curved, BV, BRST, determinant, and framing anomalies live in different complexes.
- N27. Classical factorisation does not guarantee quantum factorisation.** The QME obstruction and boundary anomaly must vanish.
- N28. A physical analogy does not supply a functor.** Every comparison between gauge theory, Hall geometry, vertex algebra, and automorphic form must specify source, target, and map.

222.5 POSITIVE REPLACEMENTS

The constitution is not merely restrictive. Each no-go theorem has a constructive replacement:

False shortcut	Correct construction
residue as bar differential	total complex with typed collision and bar degrees
strong duoidal equivalence	aligned-decomposition retract and locality squares
“nonlocal” from transverse order	independent associative chiral operad and formal-disc recognition
ordinary quotient by relations	homotopy pushout of free cofibrant algebras
bar of a Weyl/matrix algebra	curved, Rees, relative, or Morita bar
sequence recognition	explicit chain matrices and quasi-isomorphism
boundary centre equals bulk	bulk-to-centre map plus open–closed theorem
one genus	bicoloured stable graphs and mixed recognition
one anomaly number	comparison maps between typed deformation complexes
automorphic product as algebra	bar character theorem plus separate Hall realization

Typed obstruction and finite-calculation ledger

223.1 CARRIER-TO-CONSTRUCTION ATLAS

The mathematical claims separate according to their carriers.

Carrier	Available construction	Additional datum required
factorisable D -modules	aligned pointwise/chiral locality	a mixed distributive law off the aligned sector
augmented transverse algebra	interval bar and its word differential	a separate chiral multiplication
associative chiral algebra	collision-tree operadic bar	an enveloping or distributive-law comparison with the interval bar
relative quiver algebra over S	$S \otimes_A^{\mathbb{L}} S$	no scalar augmentation is inferred
exchange algebra	KZ, ZF, or RTT transport	flatness, unitarity, Yang–Baxter, and a fibre functor as applicable
cyclic or modular object	trace and sewing operations	pairing, convergence, and anomaly data
root-graded bar complex	Euler supercharacter denominator	a separate Hall correspondence and PBW comparison

223.2 FINITE CALCULATIONS AND THEIR BOUNDARIES

For the quantum plane at $q = 2$ and the Jordan plane, normalized bar words are graded by total polynomial weight $w \leq 5$ and word length. The alternating adjacent-multiplication matrices are rational. Their nonzero homology in this window is

$$H_{1,1} \cong \mathbb{k}^2, \quad H_{2,2} \cong \mathbb{k},$$

with all other groups zero; Appendix 226 gives the bases, matrix ranks, and cycles. For $U(\mathfrak{sl}_2)$, the Chevalley–Eilenberg differential has $\det d_2 = -4$, $d_3 = 0$, and

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (1, 0, 0, 1).$$

These calculations refute universal Motzkin or Riordan formulas for the corresponding ordered bar homology. A scalar sequence, chain-space count, or finite window does not establish a quasi-isomorphism outside its stated carrier and range.

223.3 THE EXACT EVIDENTIARY STANDARD

A worked example is complete only when it contains:

- (i) the coefficient object and its diagonal extension;
- (ii) the associative product and augmentation or relative base;
- (iii) generators, relations, and a normal-form or PBW theorem;
- (iv) the actual bar or Koszul differential;
- (v) cycles, boundaries, and homology, not only chain dimensions;
- (vi) any exchange, trace, modular, or physical realization as a named extra map;
- (vii) an exact finite-window calculation, with its parameter range, when constants or signs are nontrivial.

The quantum plane, Jordan plane, relation quiver, $\mathfrak{sl}_2$, ZF system, $\mathcal{A}_2$ Hall algebra, stable trace bracket, and two-genus Frobenius/CohFT model meet this standard in the present volume.

The definitive structure theorem

224.1 THE PRIMITIVE DATUM

Let X be a smooth complex curve. The complete local object is the tuple

$$\mathbb{A} = (F, m_{\perp}, m_X, \lambda, \varepsilon; \text{ completions}),$$

where:

- (i) F is a factorisable D -module carrying every diagonal singularity and regular product;
- (ii) $m_{\perp}$ is an augmented transverse E_1 -multiplication for $\otimes$;
- (iii) m_X is an augmented associative chiral E_1 -multiplication for $\star_{\text{ch}}$ or Ass_X^{ch} ;
- (iv) λ is the aligned mixed distributive law, including all higher homotopies;
- (v) ε is a scalar or relative augmentation appropriate to the algebra;
- (vi) the nilpotent, weight, pole, coradical, and analytic completions are specified separately.

Any subset of this tuple defines the corresponding partial theory. No missing component is inferred from the others.

Theorem 224.1 (Definitive structure theorem). *For a complete datum $\mathbb{A}$ as above, the following constructions are functorial and type preserving.*

- (i) $\text{Bar}^{\perp}(\mathbb{A})$ is interval factorisation homology and a conilpotent transverse coalgebra retaining the associative chiral coefficient.
- (ii) $\text{Bar}_{\text{Ass}}^{\text{ch}}(\mathbb{A})$ is the operadic associative chiral bar retaining the transverse algebra.
- (iii) The mixed distributive law gives a total two-direction bar with anticommuting edge differentials and an aligned Fubini comparison.
- (iv) On the nilcomplete/conilcomplete locus, the corresponding bar and cobar functors are inverse.
- (v) The augmentation boundary, relative module categories, two-sided twisting transform, and derived centre form the Koszul–Morita atlas.
- (vi) An E_2 -lift or flat meromorphic transport adds exchange; a fibre functor reconstructs the corresponding quantum symmetry.
- (vii) A compatible BRST operator forms a bicomplex and descends all commuting structures to cohomology.
- (viii) A cyclic transverse pairing and a modular chiral pairing satisfying the mixed square produce the two-edge master equation with independent genus variables.

- (ix) *A perturbative BV realization gives the preceding object when renormalised graph weights extend, the QME holds, and the observables algebraize.*
- (x) *Root-graded Euler character, trace, period, index, and monodromy maps produce scalar shadows only after the structured object has been constructed.*

Conversely, every construction assembled from the displayed carriers factors through one or more of these functors. A proposed construction fails whenever it omits one of the required data or hypotheses.

Proof. Items 1–3 are theorems 198.2 and 199.1; item 4 is theorems 200.1 and 200.2; item 5 is theorems 201.1 and 201.2; item 6 follows from proposition 215.1 and theorem 208.1 and Tannaka reconstruction; item 7 is theorem 214.2; item 8 is recognition Theorem 216.2, realized by theorem 218.2; item 9 is theorem 213.1; item 10 is exemplified by theorems 211.2 and 220.1. Conversely, each operation factors by type as collision, ordered composition, exchange, trace, sewing, reduction, pushforward, or scalar evaluation, and each appears in the atlas. The no-go constitution records the missing input in every forbidden identification. $\square$

224.2 THE SINGLE INVARIANT SENTENCE

The complete statement is:

A genuinely noncommutative chiral field theory on a curve may carry two independent associative structures: ordered fusion in a transverse topological direction and associative, possibly nonlocal, chiral composition on the curve. Their coherent coexistence is an algebra over a two-direction cofibrant operad, with locality supported on the aligned-decomposition sector. Bar, centre, exchange, trace, sewing, reduction, and arithmetic are functorial constructions from this typed datum, never substitutes for it.

224.3 CONSEQUENCES OF THE TYPED DATUM

The typed datum has four consequences.

- (i) The aligned projector is the organizing law of mixed locality.
- (ii) It identifies the missing associative chiral/nonlocal E_1 as a second operadic direction rather than a synonym for transverse order.
- (iii) It turns generators and relations into a derived, calculable presentation theory over the two-coloured cofibrant operad.
- (iv) The examples are chain-level calculations that can falsify a proposed identification as well as illustrate it.

224.4 CODA: THE GEOMETRY REMEMBERS

The diagonal remembers singularity. The interval remembers order. The formal disc remembers associative chiral iteration. The plane remembers exchange. The circle remembers trace. The stable curve remembers sewing. The compactified boundary remembers the equations. A scalar remembers only the value of a chosen functional. The typed theory keeps these memories separate until a theorem joins them.

Suspensions, orientations, and low-arity signs

22.1 COHOMOLOGICAL SUSPENSION

For homogeneous a , $|sa| = |a| + 1$. The Koszul rule for exchanging homogeneous factors u, v is

$$u \otimes v \mapsto (-1)^{|u||v|} v \otimes u.$$

The binary bar coderivation is fixed by

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

On a word, moving this degree-one coderivation past the preceding suspended letters gives

$$b_2[a_1 | \cdots | a_r] = \sum_{i=1}^{r-1} (-1)^{\sigma_i} [a_1 | \cdots | a_i a_{i+1} | \cdots | a_r],$$

$$\sigma_i = \sum_{j < i} (|a_j| + 1) + |a_i|.$$

For degree-zero letters the signs alternate. Any convention differing by a global suspension conjugation produces an isomorphic complex; mixing conventions within one calculation does not.

22.2 FOUR-LETTER BAR WORD

For ungraded letters and the alternating convention,

$$b[a|b|c|d] = [ab|c|d] - [a|bc|d] + [a|b|cd].$$

Applying b again gives

$$\begin{aligned} b^2[a|b|c|d] &= [(ab)c|d] - [ab|cd] \\ &\quad - [a(bc)|d] + [a|b(cd)] \\ &\quad + [ab|cd] - [a|(bc)d]. \end{aligned}$$

The middle terms cancel, and associativity cancels the remaining two pairs. This is the cellular boundary of the two-dimensional associahedron in its simplest word model.

225.3 TWO EDGE COLOURS

For a bicoloured graph G , choose

$$\text{or}(G) = \det E_X(G) \otimes \det E_\perp(G).$$

Contracting an X -edge removes a basis vector from the first determinant; contracting a $\perp$ -edge removes one from the second. Reversing the contraction order exchanges two odd normal directions and contributes -1 . Hence

$$d_X d_\perp + d_\perp d_X = 0.$$

The same convention controls the two loop operators in the master equation.

225.4 ARNOLD RELATION AND KZ CURVATURE

For three points,

$$\omega_{ij} = d \log(z_i - z_j)$$

satisfy

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

The arity-three KZ curvature is

$$\begin{aligned} F_\nabla = \kappa^2 & \left([\Omega_{12}, \Omega_{13}] \omega_{12} \wedge \omega_{13} + [\Omega_{12}, \Omega_{23}] \omega_{12} \wedge \omega_{23} \right. \\ & \left. + [\Omega_{13}, \Omega_{23}] \omega_{13} \wedge \omega_{23} \right). \end{aligned}$$

The infinitesimal braid relation rewrites the three commutators as one common commutator with the coefficients in Arnold's relation, so $F_\nabla = 0$.

225.5 RATIONAL YANG-BAXTER EXPANSION

For $R_{ij}(u) = 1 - \hbar P_{ij}/u$, expand both sides of Yang-Baxter. The order- $\hbar$ terms agree trivially. The order- $\hbar^2$ identity reduces, after a common denominator, to the permutation identities

$$P_{12}P_{13} = P_{23}P_{12}, \quad P_{13}P_{23} = P_{12}P_{13}, \quad P_{12}P_{23} = P_{13}P_{12}.$$

The cubic term agrees because both sides represent the same longest permutation.

Exact finite computations

226.1 FINITE CHAIN-MATRIX CONSTRUCTION

For each total polynomial weight $w \leq 5$ and word length r , let $C_{w,r}$ be the normalized bar space spanned over $\mathbb{Q}$ by words whose nonunit monomial letters have total weight w . The bar differential is the alternating sum of adjacent multiplications. In the quantum plane at the rational parameter $q = 2$, multiplication is

$$(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}.$$

In the Jordan plane it is determined by the exact reduction $yx \mapsto xy + x^2$. Thus every differential matrix in the stated window has rational entries; its kernel, image, and rank are taken over $\mathbb{Q}$.

226.2 QUANTUM-PLANE AND JORDAN-PLANE WINDOWS

The complete nonzero homology through weight five is:

$$H_{1,1} \cong \mathbb{k}^2, \quad H_{2,2} \cong \mathbb{k},$$

for both algebras, with every other $H_{w,r}$ zero in the window. The selected chain dimensions and ranks are:

algebra	w	r	$\dim C_r$	$\text{rk } d_r$	$\text{rk } d_{r+1}$	$\dim H_r$
$A_q, q = 2$	1	1	2	0	0	2
$A_q, q = 2$	2	1	3	0	3	0
$A_q, q = 2$	2	2	4	3	0	1
$A_q, q = 2$	3	1	4	0	4	0
$A_q, q = 2$	3	2	12	4	8	0
$A_q, q = 2$	3	3	8	8	0	0
$A_q, q = 2$	4	1	5	0	5	0
$A_q, q = 2$	4	2	25	5	20	0
$A_q, q = 2$	4	3	36	20	16	0
$A_q, q = 2$	4	4	16	16	0	0
$A_q, q = 2$	5	1	6	0	6	0
$A_q, q = 2$	5	2	44	6	38	0

algebra	w	r	$\dim C_r$	$\operatorname{rk} d_r$	$\operatorname{rk} d_{r+1}$	$\dim H_r$
$A_q, q = 2$	5	3	102	38	64	0
$A_q, q = 2$	5	4	96	64	32	0
$A_q, q = 2$	5	5	32	32	0	0
A_J	1	1	2	0	0	2
A_J	2	1	3	0	3	0
A_J	2	2	4	3	0	1
A_J	3	1	4	0	4	0
A_J	3	2	12	4	8	0
A_J	3	3	8	8	0	0
A_J	4	1	5	0	5	0
A_J	4	2	25	5	20	0
A_J	4	3	36	20	16	0
A_J	4	4	16	16	0	0
A_J	5	1	6	0	6	0
A_J	5	2	44	6	38	0
A_J	5	3	102	38	64	0
A_J	5	4	96	64	32	0
A_J	5	5	32	32	0	0

The quantum cycle $[y|x] - 2[x|y]$ and Jordan cycle $[y|x] - [x|y] - [x|x]$ evaluate to the zero vector under the exact degree-two differential.

226.3 THE $\mathfrak{sl}_2$ COMPLEX

The matrix in [chapter 207](#) has determinant -4 , so d_2 is an isomorphism. The top differential is zero by the Jacobi identity and direct calculation. The result is

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (1, 0, 0, 1).$$

226.4 THE A_2 HALL RELATION AT $q = 2$

In the basis $([X], [Y])$,

$$e_1^2 e_2 = (3, 3), \quad e_1 e_2 e_1 = (3, 1), \quad e_2 e_1^2 = (3, 0).$$

Therefore

$$(3, 3) - 3(3, 1) + 2(3, 0) = (0, 0).$$

The symbolic proof in [theorem 210.1](#) holds for arbitrary q ; the finite check catches coefficient transpositions.

226.5 FINITE STABLE-TRACE JACOBI CALCULATION

For labels (a, b) with $0 \leq a, b \leq 3$ and $a + b > 0$, evaluate all ordered triples under

$$[(a, b), (c, d)] = (ad - bc)(a + c - 1, b + d - 1).$$

All $15^3 = 3375$ Jacobi sums vanish exactly in the free $\mathbb{Q}$ -module on the resulting labels. This statement is confined to the displayed finite label set; the all-label Poisson identity is supplied by the separate symbolic proof.

Generators-and-relations templates

227.1 QUADRATIC TRANSVERSE PRESENTATION

Given coefficient generators $V \in \text{FactD}(X)$ and relations $R \rightarrow V \otimes_! V$, the strict quadratic algebra is

$$A = T_{\otimes_!}(V)/(R).$$

The derived presentation is the homotopy pushout

$$\begin{array}{ccc} \text{Free}_{E_!}(R) & \xrightarrow[\partial_!]{\partial_o} & \text{Free}_{E_!}(V) \\ \downarrow & & \downarrow \\ \mathbf{I} & \longrightarrow & A_{\text{der}}. \end{array}$$

A cofibrant replacement adds syzygy cells whenever relations among relations occur.

227.2 ASSOCIATIVE CHIRAL PRESENTATION

For formal-disc generators M and ordered field relations R_X , replace $\text{Free}_{E_!}$ by $\text{Free}_{\text{Ass}_X^{\text{ch}}}$. A relation can be a finite-pole identity

$$(z-w)^N \left(Y(a, z)Y(b, w) - \sum_i f_i(z-w)Y(b_i, w)Y(a_i, z) \right) = 0.$$

A quantum relation includes the $\mathcal{S}$ -matrix coefficients. The derived quotient remembers a null-homotopy of this field identity and its compatibility with nested substitutions.

227.3 DOUBLY ASSOCIATIVE PRESENTATION

Use the two-coloured operad $\mathcal{P}$. The data consist of

$$(V; R_{\perp}, R_X, R_{\text{mix}}),$$

where R_{mix} contains the rectangular locality square and any braided permutation between middle factors. Resolving the presentation automatically adds:

- transverse overlap cells;

- chiral collision overlap cells;
- mixed prisms comparing the two reduction orders;
- higher cells filling the resulting spheres.

This is the promised model-category implementation of homotopically complete generators and relations.

227.4 RELATIVE PRESENTATION

For a semisimple base S , use free S -bimodule algebras and relative tensor products throughout. The quiver template is

$$\mathcal{A} = T_S(\mathcal{M})/(R), \quad \text{Bar}_S(\mathcal{A}) = S \otimes_{\mathcal{A}}^{\mathbb{L}} S.$$

A coloured chiral quiver replaces every $e_j M e_i$ by a coefficient object and imposes the factorisation compatibility on every source-target component.

Logical status of comparison statements

A comparison is used only after its source, target, carrier category, and map have been specified. A finite chain-matrix calculation establishes its stated finite window only. A formal identity carries no analytic convergence claim, and a conditional comparison is invoked only with its named hypothesis. The obstructions in Chapter 222 and the exact calculations in Appendix 226 give the corresponding negative and finite criteria.

Extended comparison table

Geometry	Object	Operation	Required theorem or datum
partial diagonal	factorisable D -module	singular OPE/principal part	diagonal extension, coordinate covariance, chiral identities
ordered interval	E_1 -algebra for $\otimes_l$	transverse fusion	rectangular locality and augmentation/relative base
formal disc	Ass_X^{ch} -algebra	nonlocal field iteration	weak associativity and completion
transverse plane	E_2 -lift or connection	exchange	braid paths, flatness, Yang–Baxter
circle	Hochschild chains	trace/closure	cyclic or Calabi–Yau structure
ribbon surface	cyclic A_∞ -algebra	open TCFT	perfect pairing and ribbon graph action
stable curve	modular chiral object	conformal-block sewing	clutching maps, pairing, convergence/anomaly line
bicoloured graph	two-edge algebra	mixed master equation	common corollas and mixed interchange square
augmentation boundary	module $\mathbf{r}$	boundary Koszul dual	compactness and completion
annulus	Hochschild theory	centre/trace pairing	two-sided Morita transform
quiver	S -algebra	boundary-changing composition	relative base and path relations
ZF spectral line	exchange algebra	ordered normal form	unitarity and Yang–Baxter
RTT monodromy	Hopf algebra	coproduct and evaluation	RTT PBW theorem
Hall stack	correspondence algebra	extension product	proper pull–push, orientations, associativity

Geometry	Object	Operation		Required theorem or datum
derived commuting scheme	necklace algebra	Hamiltonian action	trace	cyclic derivatives and derived quotient
BV configuration	graph-weight complex	quantum	observables	renormalisation and QME
BRST complex	bicomplex	reduction		square-zero derivation, Kunneth, convergence
root lattice	graded bar	denominator character		PBW and finite-decomposition completion
scalar output	field/ring element	trace, period, index		specified functional on a structured object

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